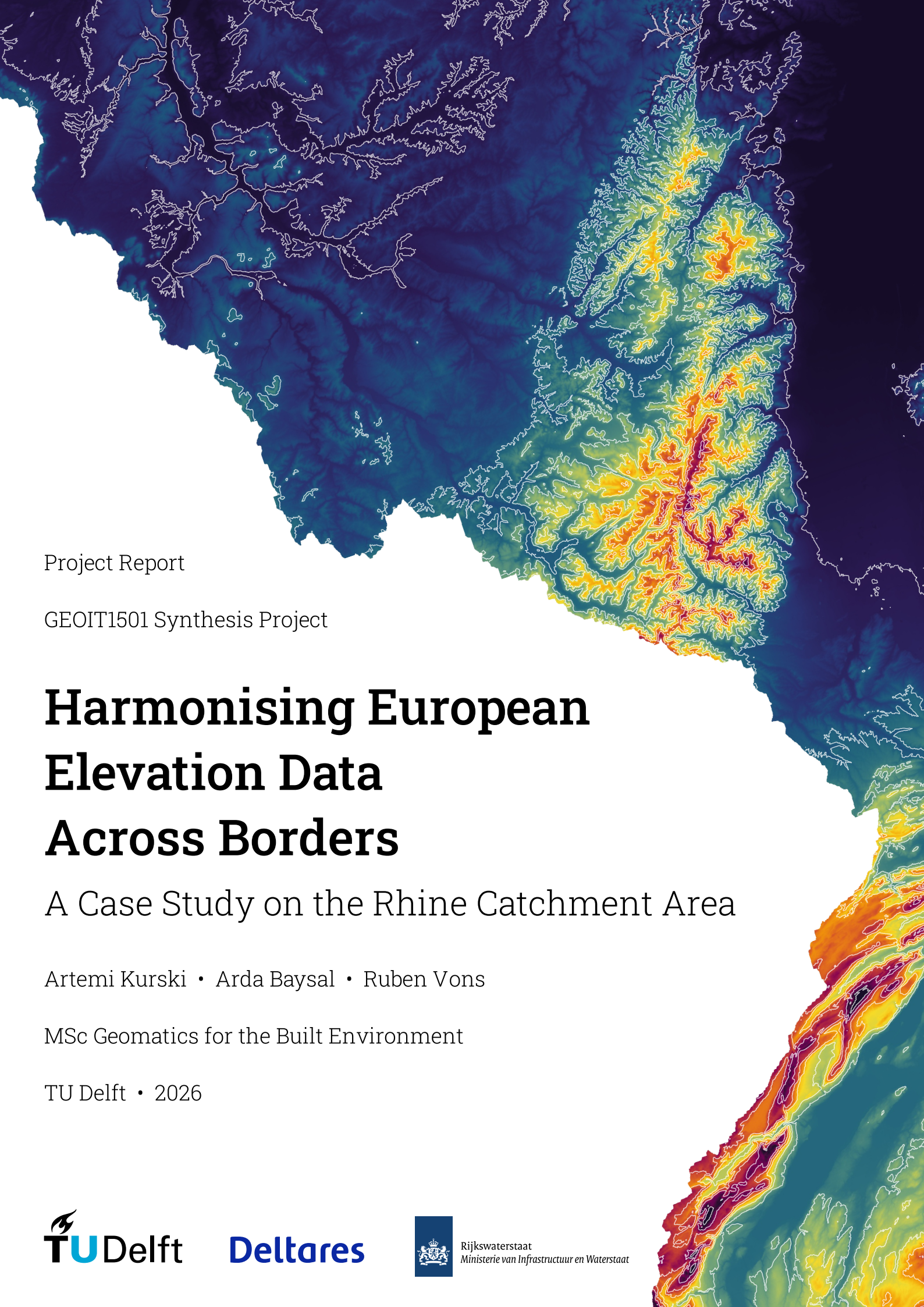

Project Report

GEOIT1501 Synthesis Project

# Harmonising European Elevation Data Across Borders

## A Case Study on the Rhine Catchment Area

Artemi Kurski • Arda Baysal • Ruben Vons

MSc Geomatics for the Built Environment

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TU Delft Deltares Rijkswaterstaat
*Ministerie van Infrastructuur en Waterstaat*

# Harmonising European Elevation Data Across Borders: a Case Study on the Rhine Catchment Area

MSc Geomatics for the Built Environment
GEOIT1501 Synthesis Project Report


**Artemi Kurski**
6195784

**Arda Baysal**
5484987

**Ruben Vons**
6285880

Supervised by:

**Gina Stavropoulou**
TU Delft

**Maarten Pronk**
Deltares

**Daan van der Heide**
Rijkswaterstaat

Faculty of Architecture and the Built Environment,
Delft University of Technology,
The Netherlands


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The cover image shows a fragment of the DTM of the Rhine catchment area covering parts of Switzerland, France, and Germany.


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## Abstract

High-quality digital elevation models (DEMs) are increasingly available across Europe through regional and local mapping and research programmes. However, these datasets are created by many different organisations, which results in differences in CRS, resolution, VCRS, processing approaches, and distribution formats between datasets. These differences pose a challenge to creating one unified elevation dataset spanning across several regions.

This project focuses on creating a harmonised cross-border DEM of the Rhine catchment area in an automated manner. Within this project, an automated pipeline has been developed which handles data harmonisation, processing, and visualisation. We acquire the datasets which are then reprojected, vertically transformed, resampled, and aggregated into a single DEM. Additionally, the pipeline supports interpolation of missing data, generating a high-resolution water mask, and a semi-automated border analysis for neighbouring datasets.

The resulting DTM covers Liechtenstein, Switzerland, France, Luxembourg, the Netherlands, and the German states of Bavaria, Baden-Württemberg, Hessen, Rhineland-Palatinate, Saarland, and Nordrhein-Westfalen. To evaluate the quality of the harmonised dataset, neighbouring DTMs are compared by performing a difference of elevation rasters, statistical analysis, co-registration, and creating a set of diagnostic plots and raster files which are then evaluated against the requirements set out in the INSPIRE specification on elevation. The analysis identifies several recurring sources of disagreement, categorised into interpolation, ground filtering, temporal differences, water mask, and pre- and post-processing artefacts. Of the 17 border pairs within the catchment area, 15 meet the INSPIRE requirements for vertical accuracy (for 15 pairs, a mean RMSE of 0.50 m compared to INSPIRE requirement of 1.67 m at 5 m horizontal resolution) and only two exceed that threshold (RMSE of 3.12 m and 5.36 m). The sources of errors are best explained by differences in acquisition processing by data providers; the pipeline does not introduce significant errors.

The project demonstrates that harmonised cross-border DTM data can be generated automatically from heterogeneous elevation data while retaining the original data without applying methods such as cross-border smoothing, unlike existing cross-border elevation products. We acknowledge that the challenges to harmonising the cross-border DEMs remain, and provide recommendations for improving the interoperability of elevation data.

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# List of Abbreviations

| | |
|---|---|
| **AHN** | Actueel Hoogtebestand Nederland |
| **ALOS** | Advanced Land Observing Satellite |
| **ALS** | Aerial Laser Scanning |
| **AOI** | Area of Interest |
| **CH** | Switzerland |
| **COG** | Cloud-Optimized GeoTIFF |
| **COPC** | Cloud-Optimized Point Cloud |
| **CRS** | Coordinate Reference System |
| **CSF** | Cloth Simulation Filtering |
| **DE** | Germany |
| **DE-BW** | Baden-Württemberg, Germany |
| **DE-BY** | Bavaria, Germany |
| **DE-HE** | Hessen, Germany |
| **DE-NW** | North Rhine-Westphalia, Germany |
| **DE-RP** | Rhineland-Palatinate, Germany |
| **DE-SL** | Saarland, Germany |
| **DEM** | Digital Elevation Model |
| **DHHN2016** | German Main Height Network 2016 |
| **DSM** | Digital Surface Model |
| **DTM** | Digital Terrain Model |
| **EEA** | European Environment Agency |
| **ETRS89** | European Terrestrial Reference System 1989 |
| **ETRS89-GRS80** | ETRS89 using the Geodetic Reference System 1980 ellipsoid |
| **ETRS89-LAEA** | ETRS89 Lambert Azimuthal Equal Area |
| **ETRS89-LCC** | ETRS89 Lambert Conformal Conic |
| **EU** | European Union |
| **EVRS** | European Vertical Reference System |
| **FEE** | Freshwater Ecosystems Explorer |
| **FR** | France |
| **GDAL** | Geospatial Data Abstraction Library |
| **GDEM** | Global Digital Elevation Model |
| **GRS80** | Geodetic Reference System 1980 |
| **GSD** | Ground Sample Distance |
| **ICPR** | International Commission for the Protection of the Rhine |
| **ICSF** | Improved CSF |
| **IDW** | Inverse Distance Weighted to a Power |
| **INSPIRE** | Infrastructure for Spatial Information in the European Community |
| **InSAR** | Interferometric Synthetic-Aperture Radar |
| **LAZ** | LASzip-compressed point-cloud format |
| **LE90** | Linear Error at 90th Percentile |
| **LI** | Liechtenstein |
| **LU** | Luxembourg |
| **MAE** | Mean Absolute Error |
| **MCC** | Multiscale Curvature Classification |
| **MSF** | Maximum Slope Filtering |
| **MoSCoW** | Must, Should, Could, Won't |
| **NAP** | Normaal Amsterdams Peil |
| **NL** | Netherlands |
| **NMAD** | Normalized Median Absolute Deviation |
| **NNI** | Natural Neighbour Interpolation |
| **OSM** | OpenStreetMap |

| | |
|---|---|
| **PC** | Point Cloud |
| **PMF** | Progressive Morphological Filter |
| **PTIN** | Progressive Triangulated Irregular Network |
| **RBF** | Radial Basis Function |
| **RMSE** | Root Mean Square Error |
| **STD** | Standard Deviation |
| **TIN** | Triangulated Irregular Network |
| **UTM** | Universal Transverse Mercator |
| **VCRS** | Vertical Coordinate Reference System |
| **WCS** | Web Coverage Service |
| **WLS** | Weighted Linear Least Squares |
| **lidar** | Light Detection and Ranging |

# 1 Introduction

High-quality elevation data is increasingly available across Europe through national, regional, and local mapping and research programmes. These datasets support a wide range of applications, including hydrological modelling, flood risk assessment, infrastructure planning, and environmental management. At the same time, many of these applications extend beyond administrative boundaries. The Infrastructure for Spatial Information in the European Community (INSPIRE) framework (European Parliament and Council of the European Union, 2007) therefore places emphasis on the interoperability of spatial datasets and on their consistent use across national borders. Elevation is one of the spatial data themes addressed within this framework (TWG-EL, 2024).

The Rhine catchment is a clear example of an area where cross-border spatial information is important. The catchment covers approximately 180,000 km² and extends across several European countries, with around 60 million people living within the basin (Internationale Arbeitsgemeinschaft der Wasserwerke im Rheineinzugsgebiet, no date). The Rhine and its tributaries are important for water supply, navigation, industry, agriculture, ecosystems, and flood management. Management of the river basin is already coordinated internationally through organisations such as the International Commission for the Protection of the Rhine (International Commission for the Protection of the Rhine (ICPR), 2022). Many processes within the catchment, particularly water flow and flooding, depend directly on terrain. The quality and consistency of elevation data can therefore influence analyses performed at catchment scale (Muthusamy *et al.*, 2021).

Although detailed elevation datasets are available for large parts of Europe, they are generally collected and maintained by individual national or regional authorities. As a result, neighbouring datasets can differ in quality of acquired data, spatial resolution, coordinate and height reference systems, classification, and more. Such differences create practical challenges when elevation data from several jurisdictions needs to be combined into a single terrain model. Simply joining the available datasets does not necessarily result in a consistent representation of the terrain. Recent work by Rijkswaterstaat, TU Delft, Het Waterschapshuis, and EuroSDR documents these differences across European point cloud datasets and identifies the absence of a harmonised cross-border point cloud dataset within the European Union (van der Heide *et al.*, 2026). Similarly, we are not aware of a harmonised quality-assured high-resolution cross-border gridded elevation dataset.

The lack of a joint European elevation model forms the main challenge addressed by this project. The aim is to investigate how elevation data covering the Rhine catchment can be processed into a harmonised digital elevation model in an automated manner. The project considers the differences between the available source datasets and the processing steps required to bring them into a common representation. The exact geographic coverage, datasets, and harmonisation aspects considered within the project are defined further through the research questions and subquestions.

The project is defined in Section 2, presented alongside the research questions and subquestions. Section 3 contains the summary of our findings during literature review; in Section 4, we describe the methodology underpinning the implementation of the harmonisation pipeline. The results of the project are presented in Section 5. The report concludes with a discussion of the limitations, findings, and recommendations for future work in Section 6.

# 2 Project Definition

## 2.1 Terrain Models

Ledoux *et al.* (2025) provide the following definition of a terrain model: "A terrain is a representation of the Earth's surface. It gives us the elevation, which is the height above/below a certain reference point (a vertical datum)." In this project, the main representation of terrain is a gridded Digital Elevation Model (DEM), i.e. a raster dataset where the value of each grid cell equals the height of the terrain at its centre — within this report, the gridding is implied whenever a DEM is mentioned. DEMs are further differentiated into Digital Terrain Models (DTMs), which only show the elevation of bare earth without human-made structures or vegetation, and Digital Surface Models (DSMs), which show all objects on the surface, such as trees, buildings, and the ground itself.

Each terrain model is associated with a Coordinate Reference System (CRS). A CRS describes the units of measurement, horizontal and vertical datums, and direction of the axes. Different countries use different reference systems, and conversion between them is needed before different terrain models can be integrated with one another.

## 2.2 The Need for the European DEM

There are currently several products which offer EU-wide elevation data as described in Section 3.3. These products, however, have several shortcomings. Satellite-derived DEMs have coarse resolution and do not fully support extracting the DTM; aggregate DEMs compromise on accuracy in the border regions.

The aim of this project is to develop an automated pipeline for a higher-quality European DEM which would support runoff, flood, and other modelling. We specifically consider the DTM of the Rhine catchment area as the object of the case study.

## 2.3 Research Questions

As described in the introduction, the project focuses on the issues presented by the heterogeneous data within the Rhine catchment area, which we translate to the creation of an automated pipeline and a description of the challenges that come with attempting the creation of a harmonised DEM.

The main research question of this project is thus:

*How to create a harmonised cross-border DEM of the Rhine catchment area in an automated manner?*

The aspects of the main question can be broken down into the following subquestions:

- What is the Rhine catchment area?
- What, if any, existing approaches are there to creating a harmonised DEM from heterogeneous sources?
- What are the existing DEMs at different scales (regional, national, and global)?
- What are the INSPIRE requirements for DEMs?
- What are the issues with cross-border data?
- What are the limitations of an automated harmonisation pipeline?
- How can the accuracy of a reconstructed cross-border DEM be tested?

Answering these subquestions helps answer the main research question. The first four subquestions are evaluated in the research phase of the project and relate mostly to existing literature and available data. Answering these questions informs data acquisition and subsequent software development. The next questions are answered during the software development phase as they are related to testing of the pipeline by evaluating the procedures and the output, and discussing the findings with the client.

# 3 Theory and Context

This chapter covers the reviewed literature. In it, we present the definition for the Rhine catchment area; the summary of the relevant aspects of INSPIRE specification on elevation; existing cross-border digital elevation models; aspects of processing which affect the agreement between elevation products from different providers; and approaches to modelling water body elevation.

## 3.1 Rhine Catchment Area

The first step of the project is determining the area on which the project is going to get carried out. For this, different extents of the Rhine catchment area were considered. The most commonly agreed-upon definition comes from International Association of Waterworks in the Rhine Basin (Internationale Arbeitsgemeinschaft der Wasserwerke im Rheineinzugsgebiet, no date) and International Commission for the Protection of the Rhine (ICPR) (International Commission for the Protection of the Rhine (ICPR), 2013). It excludes the upstream Meuse but includes the rest of the delta, e.g. IJssel and Waal. Another possible definition can be found on Wikimedia (see Figure 1), and it also includes the Meuse.

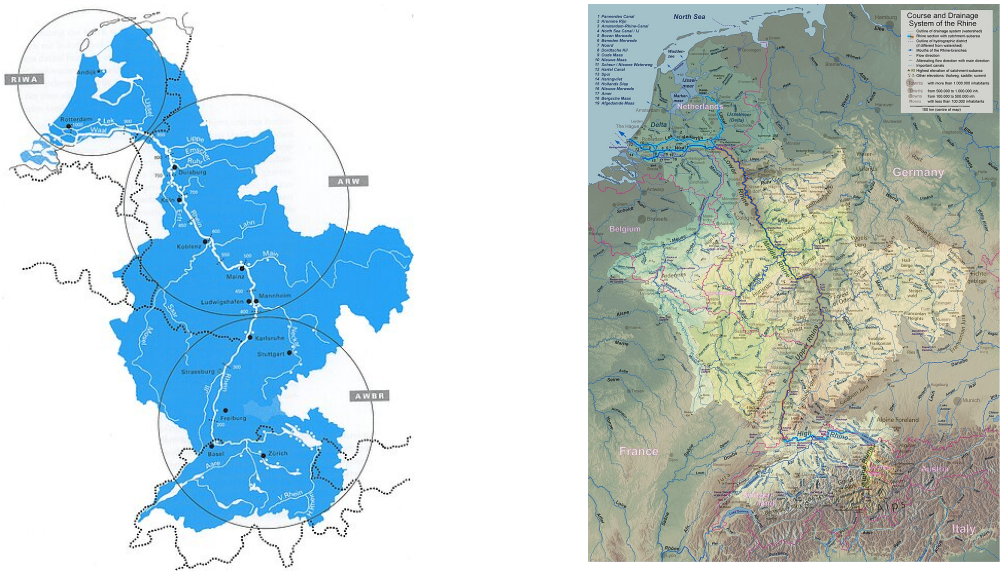



Figure 1: Maps of the Rhine catchment area from ICPR (left, from Internationale Arbeitsgemeinschaft der Wasserwerke im Rheineinzugsgebiet (no date)) and Wikimedia (right, from Wikimedia Commons (2026)).

For this project, the Rhine river definition comes from HydroBASINS (Lehner and Grill, 2013), derived from a global elevation model. The Meuse and IJssel are excluded from the main catchment area of the Rhine in this case — their exclusion has been discussed with the client and was deemed appropriate as the automated pipeline supports expanding the geographic extent of the DEM. Figure 2 shows the map of the area used in the project, coloured by elevation data provider.

The Rhine catchment area nominally covers (parts of) Switzerland, Liechtenstein, Austria, Italy, Germany, France, Luxembourg, Belgium, and The Netherlands. The following regions are explicitly defined to be within the scope of the project: Switzerland (CH), Liechtenstein (LI), France (FR), Luxembourg (LU), Germany (DE) — specifically, states of Baden-Württemberg, Germany (DE-BW), Bavaria, Germany (DE-BY), Hessen, Germany (DE-HE), North Rhine-Westphalia, Germany (DE-NW), Rhineland-Palatinate, Germany (DE-RP), and Saarland, Germany (DE-SL) — and Netherlands (NL). Austria, Italy, and Belgium are excluded due to poor availability of data (van der Heide *et al.*, 2026). Germany's Thüringen has been excluded as only a small part of the state falls within the borders of the catchment area.

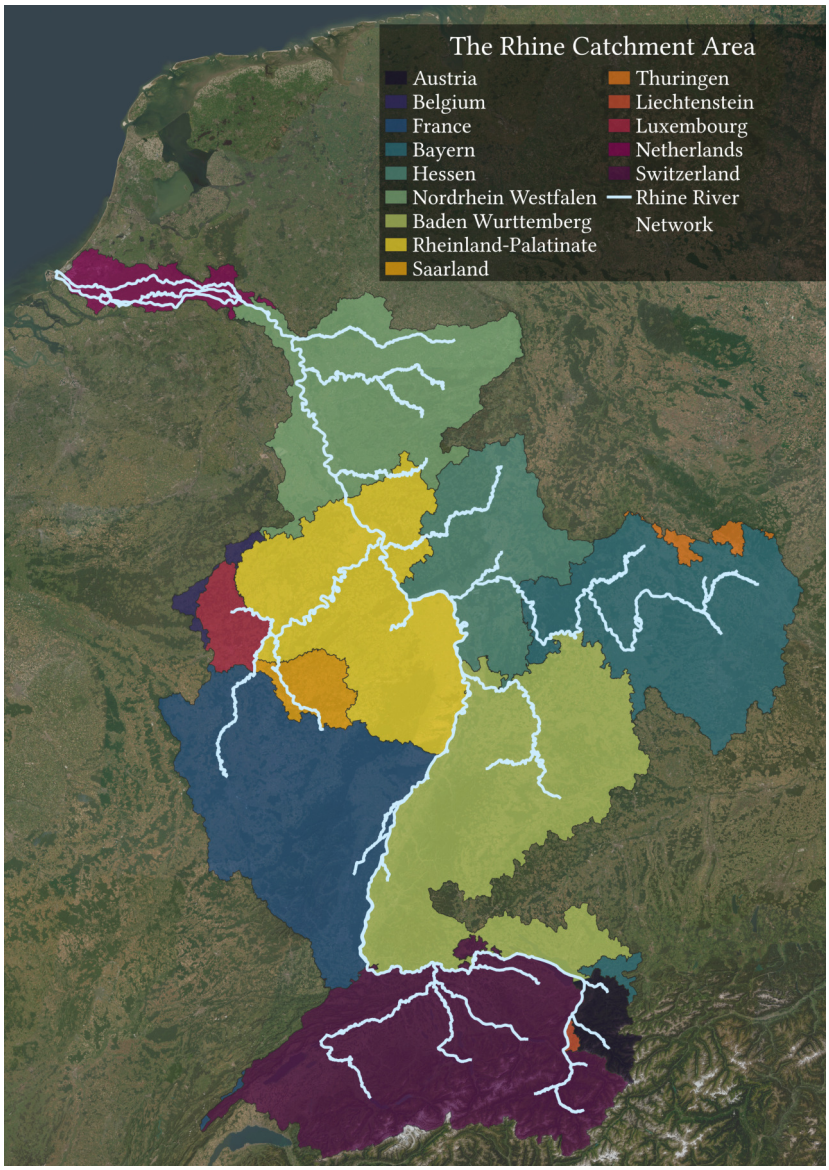



Figure 2: The Rhine catchment area with its regions and main river network. (image by authors).

## 3.2 INSPIRE Specification on Elevation

In the European Union (EU), the INSPIRE Directive (European Parliament and Council of the European Union, 2007) governs the "sharing, access, and use" of spatial data, specifically within the context of environmental policy. INSPIRE defines 34 spatial data themes; elevation data is one of those. Below is a summary of the data specification on elevation (TWG-EL, 2024), which can be considered the practical guidance on the theme. The specification contains necessary contextual information, requirements (as either general implementing rules or elevation-specific technical guidance), and recommendations. Certain implementation-specific details, such as the XML schema or data structures and formats, are excluded from the summary below, while the conceptual details are retained.

### 3.2.1 INSPIRE View on Elevation

Within the scope of the specification, the elevation property — either depth or height — is understood to be the three-dimensional shape of the Earth's surface (TWG-EL, 2024, pp. 40–43). Generally, land-elevation and bathymetry are considered separately; for land-elevation, the elevation property is called height and its positive direction is "upwards", and for bathymetry, it is depth

with positive direction pointing downward. The scope for bathymetry is specifically the sea, inland standing water bodies, and navigable rivers. The specification also considers an integrated land-sea model, where, measured from a given datum, the height is positive and the depth is negative.

The specification considers two possible "shapes" of the terrain (TWG-EL, 2024, pp. 43–44):
- a DTM, which only represents the bare surface of the Earth;
- a DSM, which represents the Earth's surface with all static natural and artificial features (e.g. this includes trees and buildings, but not cars).

"Digital elevation model" (DEM) is used as an umbrella term which encompasses both a DTM and a DSM.

Within the scope of the specification, the surface is considered 2.5-dimensional — for each $(x, y)$ position, only one elevation value is possible — and it can be represented with gridded coverage (raster data), vector data (contour lines and spot elevations), or a Triangulated Irregular Network (TIN). For land-elevation, grid coverage is required and vector representation is optional but recommended (TWG-EL, 2024, pp. 44–45). For bathymetry, either grid coverage or vector representation are required.

### 3.2.2 Gridded Coverage

Following both INSPIRE specification's requirements and the clients' request, only the gridded coverage is considered within the scope of the project at hand. The specification requires that the grid is a two-dimensional regular quadrilateral grid and is geo-rectified, i.e. can be transformed to/from a grid of a CRS via an affine transform (TWG-EL, 2024, p. 52).

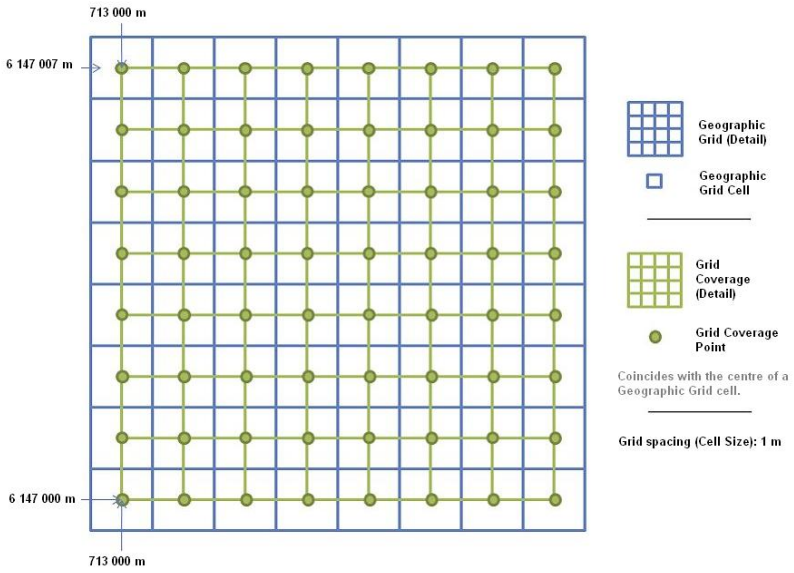



Figure 3: INSPIRE definition of gridded coverage. Image from TWG-EL (2024).

Furthermore, the specification includes guidance on providing tiled datasets — it refers to such coverage as "aggregated coverage", and the process of assembling it is called aggregation. While the illustrations (e.g. on pp. 52 and 53) show that the tiles may be overlapping, the implementation rule (p. 60) requires that:
- the tiles are consistent, i.e. sharing CRS, encoding, and resolution;
- the grids of the tiles are aligned;
- the footprints of the tiles are either adjacent or disjoint — that is, non-overlapping;
- the domain of the aggregated coverage is the union of the tiles' domains.

It is also clarified that the coverage describes the elevation values at the centres of the geographic grid; and the geographic grid cells are the squares rendered when a GeoTIFF is displayed.

### 3.2.3 Coordinate Reference Systems

Generally, INSPIRE demands that the CRS used for datasets of continental Europe is based on European Terrestrial Reference System 1989 (ETRS89), with a matching datum. These include unmodified ETRS89, based on the Geodetic Reference System 1980 (GRS80) ellipsoid, which uses geodetic coordinates; and different metric CRSs, usually ETRS89 Lambert Azimuthal Equal Area (ETRS89-LAEA) or ETRS89 Lambert Conformal Conic (ETRS89-LCC). European Vertical Reference System (EVRS) is prescribed as the vertical reference system.

Annex D, however, provides specific guidance for gridded elevation data. It deems ETRS89-LAEA and ETRS89-LCC unsuitable for the elevation data for a number of reasons, notably that they do not preserve the geometry of the terrain, i.e. planar angles. It instead recommends the use of a zoned geographic grid based on the geodetic coordinates, denoted ETRS89-GRS80zn. Under such a grid, the longitudinal resolution of the grid is different for different zones. Below 50°, the grid is "square": the latitude and longitude are in a 1:1 ratio; between 50° and 70°, the ratio is 1:2; and it further increases in the higher latitudes. Annex D is informative only, i.e. it is not expressly required to follow its guidance.

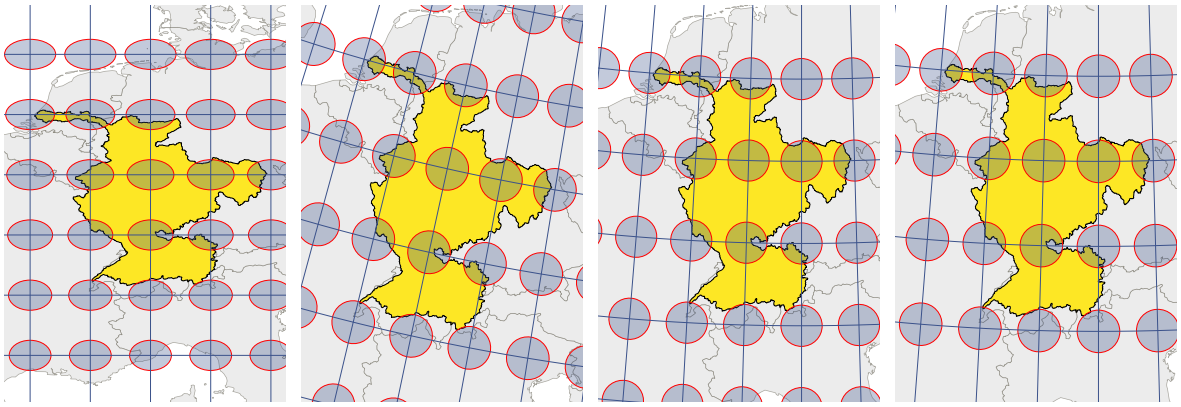
#### Input CRS

Most national data providers supply elevation coverage in the local CRS, such as Amersfoort/ RD New (EPSG code 28992) for the Netherlands. The only exception to this rule is Germany as it uses Universal Transverse Mercator (UTM) zones We found that every dataset uses a national vertical reference system, such as Normaal Amsterdams Peil (NAP) for the Netherlands or German Main Height Network 2016 (DHHN2016) for Germany. For further description of input data, see Section 5.1.

#### Target CRS

Following a discussion with the clients, we determined that ETRS89-LAEA is best suited for the current project for the following reasons:

- **Cell area.** The primary use of the DEM is expected to be hydrological modelling. The cells of ETRS89-LAEA each have the same area, removing the need to recompute it for each cell.
- **Cell shape.** The area of the project is close to the origin of the CRS which minimises the shape distortion, as seen in Figure 4: the Tissot's indicatrices are near-circular, with the semi-major and semi-minor axes being within 0.2% of 1. In practice, this means that the equal spacing of the grid in the X and Y directions will also be near-equal in reality (intuitively, a square cell on a map represents a square parcel of land).
- **Cell orientation.** The angles are distorted somewhat more: the westernmost point is at 4° E, which results in a cell being rotated 6° relative to true North at that point; however, the centroid of the dataset is at 8° E, which means a rotation of only 2°.



Geodetic ETRS89 Equi7 Europe ETRS89-LCC ETRS89-LAEA

Figure 4: Comparison of Tissot indicatrices for different projections.

The preservation of the angles — which is among the key reasons to use geodetic coordinates in the INSPIRE specification — has been determined to be of lowest priority considering the expected use case of the product, while the preservation of area is of highest priority. A qualitative analysis of Tissot indicatrices (Figure 4) shows that the distortions in the Rhine catchment area are relatively minimal with ETRS89-LAEA, whereas ETRS89 using the Geodetic Reference System 1980 ellipsoid (ETRS89-GRS80) noticeably distorts shape and ETRS89-LCC distorts area.

Additionally, the use of Equi7Grid was considered. Equi7Grid has been developed by Vienna University of Technology specifically to support high-resolution raster data by minimising data loss and data inflation (GEO Department, TU Wien, no date). Figure 4 shows the distortion caused by using the Europe subgrid compared to the ETRS89 and ETRS89-derived CRSs. The use of a CRS other than one derived from ETRS89 is explicitly disallowed by the INSPIRE specification, which excludes Equi7Grid from the candidate CRSs for this project. However, we recognise that in other cases, a different CRS may be advantageous. Therefore, the pipeline is designed to support selection of a user-specified target CRS by its EPSG code.

### 3.2.4 Data Quality

The specification defines several criteria to evaluate the data quality against. These are split into three categories: completeness, logical consistency, and positional accuracy (TWG-EL, 2024, pp. 121–138).

Completeness encompasses omission and commission. Omission errors are data which is within the dataset's scope but is absent from the dataset (in practice, these are expressed as `NoData` values); commission errors are data in excess of the dataset's scope, such as duplicates. In both cases, the target error rate is 0%.

Logical consistency includes conceptual consistency, domain consistency, format consistency, and topological consistency. Conceptual consistency describes compliance with the conceptual schema; domain consistency describes compliance with the value domain; format consistency is related to the physical structure of the dataset, i.e. the formats and encodings used. Topological consistency is only applicable to the topology vector data and therefore out of scope for this project. The error rate describes the ratio of non-compliant items and the target is 0%.

Positional accuracy is defined differently for vector and TIN data and for gridded data. For a grid, only the planimetric accuracy is considered. The Root Mean Square Error (RMSE) — here, defined as the radius of a circle around the point, such that the true value of a point lies within that circle with a given probability — is the chosen evaluation metric. The target for the maximum RMSE is $\frac{\mathrm{GSD}}{6}$,

where Ground Sample Distance (GSD) is the ground sample distance, essentially the resolution of the grid.

In the "Data Capture" section (TWG-EL, 2024, p. 171), the resolution is additionally related to the vertical accuracy of the dataset. For flat terrain, the resolution should be between $3 \times \text{RMSE}$ and $20 \times \text{RMSE}$, while for mountainous terrain it should be between $3 \times \text{RMSE}$ and $10 \times \text{RMSE}$. This will be the target when evaluating the agreement between overlapping border areas.

## 3.3 Global Elevation Models

A Global Digital Elevation Model (GDEM) covers most of the land surface of the Earth. Currently, there are two main approaches to creating a GDEM: a conventional approach relies on satellite data, and more recently, there have been efforts to derive a GDEM from national and regional datasets as suggested by INSPIRE. Below is a brief overview of both approaches.

### 3.3.1 Satellite Data-Derived GDEMs

Most GDEMs are constructed using data from space-borne Interferometric Synthetic-Aperture Radar (InSAR) or optical sensors. An overview of the most common existing GDEMs is provided by Ledoux *et al.* (2025, Table 3.2). The majority of satellite GDEMs are obtained using InSAR; only two of those described are derived from optical data. The common GSD is 30 m or 90 m a few, like WorldDEM or Advanced Land Observing Satellite (ALOS) are also available at finer resolutions. The majority of the GDEMs are DSMs; some products offer a "*pseudo*-DTM" where some ground filtering has been performed.

The European data portal lists two official DTM datasets: EU-DEM (Eurostat, no date a) and Copernicus DEM (Eurostat, no date b). Both are based on Copernicus data and have a 30 and 90 m cell resolution respectively. EU-DEM is available in geodetic ETRS89 and ETRS89-LAEA and is described to have a vertical RMSE of 2.9 m (DHI GRAS, 2014) while Copernicus DEM is available only in ETRS89 with a vertical RMSE of 1.68 m (Airbus Defence and Space GmbH, 2020). Copernicus further offer a 10 m dataset titled EEA-10 which is limited to the European Environment Agency (EEA) member states and is only available to select users (Copernicus Data Space Ecosystem, no date). EEA-10 is the only dataset which is available in INSPIRE-compatible format and projected to ETRS89-LAEA for continental Europe whereas 30 m and 90 m global datasets are only supplied in WGS84 (Airbus Defence and Space GmbH, 2022).

Krdžalić *et al.* (2024) evaluate the accuracy of several GDEMs against two sets of reference points in Bosnia and Herzegovina, additionally comparing them with the national DEM. The authors further compare different interpolation methods in their assessment — here, only results achieved with bilinear interpolation are referenced as it is the interpolation method recommended by INSPIRE. For both sets of reference points, national DEM has the lowest RMSE of 1.64 m and 3.02 m respectively; Copernicus DEM has the lowest RMSE of the GDEMs with 2.35 m and 3.11 m respectively.

### 3.3.2 Aggregate Global Elevation Models

An aggregate elevation model serves as an alternative to the conventional GDEM. With this approach, several heterogeneous DEMs are assembled into one finished product — this is also an approach taken in this project. The only example of this approach known to us is Mapterhorn (Wipfli, 2026). Currently, Mapterhorn combines 70 open-access national, regional, and local elevation products alongside Copernicus DEM GLO-30 into one harmonised DEM.

There are several shortcomings with the current implementation of Mapterhorn. The first one is that it is neither a DTM nor a DSM: Copernicus DEM is a *surface* model, and national products can be both a *terrain* and a *surface* model — therefore, Mapterhorn's elevation data is a mix of both. The second one is the approach to resolving elevation conflicts at the border. Mapterhorn

uses a Gaussian blur to blend the elevation data where two datasets overlap, which may introduce elevation values which are not representative of reality for those point samples. Lastly, Mapterhorn enforces that the CRS transform is only horizontal, explicitly ruling out vertical transform.

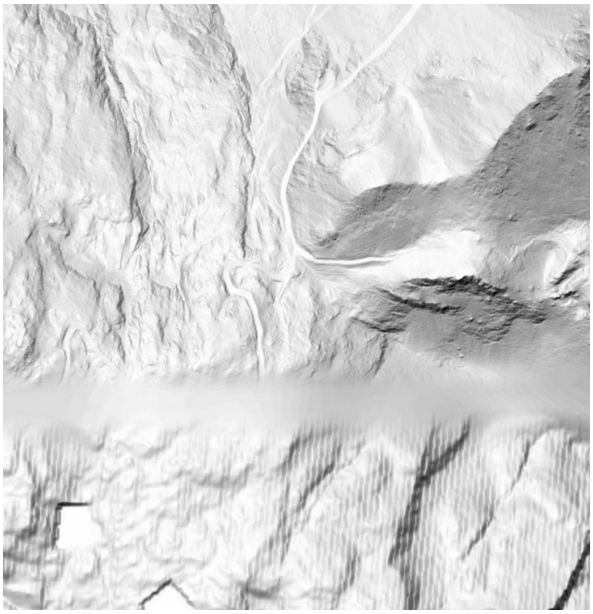


Figure 5: Example of smoothing at the border in Mapterhorn. From Wipfli (2026).

A similar approach was described by Noardo *et al.* (2016) who report on harmonisation of the CRS, orthophoto, semantic information, and the DEM and Biagi and Carcano (2016) who focus on DEM harmonisation specifically. The papers try and harmonise Italian elevation data with France and Switzerland respectively, focusing on the mountainous regions. Both demonstrate presence of errors of high magnitude, exceeding 100 m. To resolve those errors, both sets of authors suggest an approach conceptually similar to that of Mapterhorn — Noardo *et al.* obtain the elevation of the border regions by performing a linear interpolation on the overlapping region while Biagi and Carcano perform bicubic interpolation — which also introduces issues similar to using a Gaussian blur in that the data around the border may not match reality. Noardo *et al.* comment on the potential discrepancy; Biagi and Carcano leave it unaddressed.

## 3.4 Point Cloud Processing

A potential source of disagreement between gridded elevation data from different providers is the approach used to derive it from point clouds.

Ground filtering is one of key steps to extracting a DTM from a point cloud. Silva *et al.* (2018) compare Progressive Morphological Filter (PMF), Multiscale Curvature Classification (MCC), Progressive Triangulated Irregular Network (PTIN), and Weighted Linear Least Squares (WLS). One notable finding from this study is that PMF outputs a set of ground points with constant density whereas the other three algorithms produce a set of ground points with uneven density depending on the type of surface. PMF's set is also approximately 10–15 times smaller than the other algorithms. However, the resulting DTMs are still very close to each other, with highest RMSE of $0.37$ m (for shrubland; when comparing MCC to PMF). Silva *et al.* note that PMF assumes constant slope and therefore struggles in steeper parts of the terrain. Cai *et al.* (2023) carry out further assessment of Maximum Slope Filtering (MSF), PMF, Cloth Simulation Filtering (CSF), and Improved CSF (ICSF). The study provides both a quantitative and a qualitative assessment of the different methods. Quantitatively, omission and commission errors are evaluated: MSF has the highest error rate of up to 20% for complex scenes; PMF and CSF have generally comparable and slightly lower error rates

(ca 14% for the same scene); ICSF performs best, with an error rate of ca 7% for the same scene. Qualitative analysis is illustrated with Figure 6. Compared to the reference (column a), MSF (b) can introduce noise, PMF (c) and CSF (d) tend to smooth out steep slopes — this is especially apparent in Figure 6 (c) — while ICSF is mostly artefact-free.

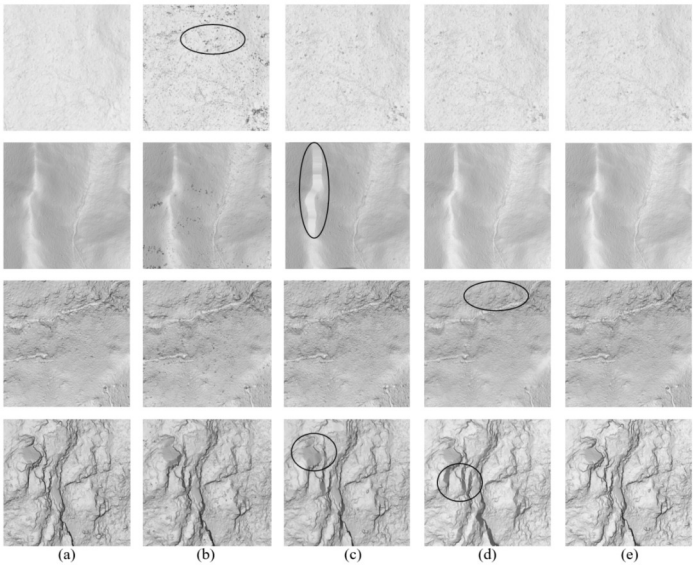



Figure 6: Comparison of ground filtering algorithms. From Cai *et al.* (2023).

Point distribution and interpolation methods can introduce further discrepancies between the raster products from different data providers. It is expected that these discrepancies are minimal for flatter areas or regions with high point density, and higher for steep slopes or regions with low point density. It is also expected that the documentation of the interpolation methods is primarily available in the national language of each data provider, if at all, therefore investigating it was deemed to be out of scope of the project.

## 3.5 Interpolation Methods

Some elevation datasets — within the study area, the Dutch DTM is the only such dataset — are supplied with "gaps" of `NoData`, which warrants filling those gaps. Other data providers fill those gaps themselves by performing interpolation. There are several papers (Hu, 2013; Chen *et al.*, 2014; Gradka and Kwinta, 2018; Raza *et al.*, 2023) comparing different interpolation methods for terrain modelling. The methods compared include Inverse Distance Weighted to a Power (IDW), Radial Basis Function (RBF), Natural Neighbour Interpolation (NNI), and kriging. The first three methods are deterministic as the results are strictly based on the input data and parameters of the function; kriging is stochastic and also requires assumptions about the statistical properties of the dataset. All authors agree that IDW produces less accurate results than IDW produces less accurate results than the alternatives: Chen *et al.* and Gradka and Kwinta find that kriging is most accurate; Hu does not evaluate kriging but recommends that an implementation of RBF produces the best result — these findings are partly corroborated by Gradka and Kwinta who state that results obtained with RBF and kriging closely match; and Raza *et al.* find that NNI works best for their dataset. However, Gradka and Kwinta note that IDW is less computationally expensive than RBF and especially kriging.

For this project, we chose IDW as the interpolation method as it requires minimal tuning compared to the alternatives; it is computationally cheaper; and a ready implementation is available in Geospatial Data Abstraction Library (GDAL). Given only a small part of the study region (the Netherlands) requires interpolation, the compromise is deemed acceptable as interpolation is not the primary focus of the project.

## 3.6 River Elevation Data

The way different DEMs treat water body surfaces can further influence the differences between products.

Currently, the Netherlands assigns a `NoData` value to the water bodies without efforts to estimate either the elevation of the water surface or the bathymetry. All other countries in the region either perform hydro-flattening as described by Zheng *et al.* (2019) which estimates a flat across-channel surface and a continuous along-channel slope; or interpolate the shoreline. Similar approaches have been trialled for the Dutch elevation model (Keurentjes *et al.*, 2020) but are currently not included in the official product. Hydro-flattening may introduce artefacts due to temporal differences between different lidar surveys — these are presented in Section 5.

A third approach is to estimate the riverbed as proposed by Dysarz (2018) which would allow for a mixed elevation/bathymetry model. No national product currently uses this approach; and estimating river elevation was deemed to be out of scope for this project.

# 4 Methodology

This chapter describes the methodology used in the DTM harmonisation pipeline. It provides both a general overview and specifics of our approach to processing heterogeneous elevation data and subsequent analysis of the results. It additionally covers water mask generation alongside the visualisation and comparison methods employed.

## 4.1 Pipeline Overview

Within this project, we create a fully automated pipeline for processing and harmonising heterogeneous gridded elevation data. Raster processing has been chosen as the focus of the project as re-extracting a DTM from scratch is computationally expensive and is deemed unnecessary: as shown in Section 5, most national DTMs are sufficiently accurate to allow near-seamless integration. The pipeline can currently handle data from 6 countries within the Rhine catchment area and is expected to support similar data from other providers. Point cloud processing has been excluded from the pipeline due to the limited time of the project and limited availability of the data; where available, point clouds are used for additional evaluation of the gridded DTM.

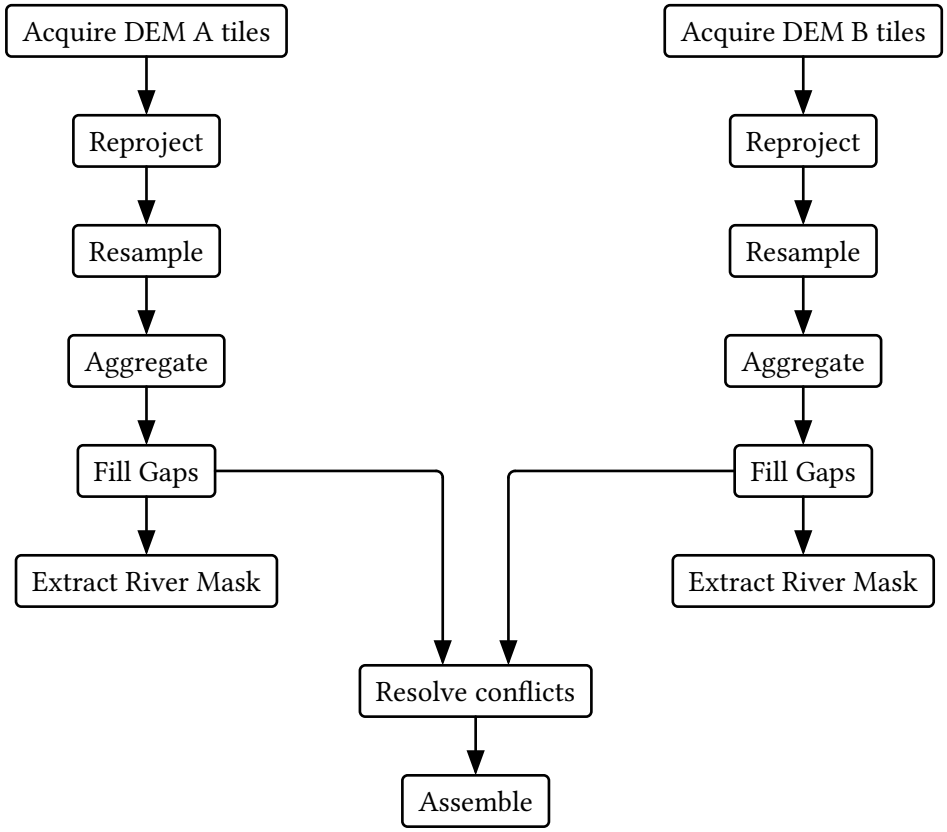



Figure 7: Flowchart of the processing steps of the pipeline.

The pipeline is split into two stages: processing and comparison — the above flowchart only shows the processing stage; comparison is described in Section 4.3. Both stages take a `.json` file as input and write the intermediate and final results to it. The comparison stage reads and evaluates data completed during processing. The user is not expected to interact or pass arguments directly to the pipeline — instead, they fill in the `.json` with input data.

The global settings influence both the final product and the performance of the pipeline. The key raster settings include the target CRS and resolution, region extent, interpolation settings, and path where the output raster should be written to. The performance settings govern the maximum

number of threads which can be used for processing or the number of regions which can be processed at once.

The region settings include the path to vector and raw raster data, information about regions CRS and Vertical Coordinate Reference System (VCRS) in case they are not specified within metadata, and whether interpolation is needed — currently, only Dutch DTM needs interpolation. Additionally, a water mask is passed for the river detection part of the algorithm.

For the comparison stage, the settings include the threshold for flagging errors and the path to region's river or interpolation masks.

A full breakdown of all settings can be found in Appendix C.

### 4.1.1 Data Acquisition

Data acquisition is a crucial step which enables the rest of the workflow despite technically being outside of the fully automated part of the pipeline. Data discovery was performed primarily based on (van der Heide *et al.*, 2026) supported by additional online searches.

We find that national and regional data providers offer different download options: the Netherlands and DE-NW offer a Web Coverage Service (WCS) server for access/download; DE-BY and France offer tiled raster data in a binary (`.xyz` and `.asc` respectively) format; Luxembourg offers a single untiled Cloud-Optimized GeoTIFF (COG); the remaining datasets are available as GeoTIFF (`.tif`) tiles from a catalogue or a download portal. Some providers (DE-HE, France) bundle the tiles into zip archives by administrative subdivision.

The downloads are handled automatically but each region essentially requires a bespoke solution due to the differences in the data portals and the catalogues.

For each region, the data is queried within a specified Area of Interest (AOI) (here understood as the intersection of the region's extent and the Rhine catchment area); where possible, the tiles are downloaded in parallel; where necessary, the downloaded tiles are recursively unzipped. The downloaded data is then validated and checked against a catalogue.

### 4.1.2 Reprojection & Resampling

This is the first step in the raster processing pipeline.

First, all regions without a processed DEM are listed and a queue is built from largest (processed first) to smallest (processed last) region. The regions can be processed in parallel and this order helps avoid situations when one process is performing the reprojection while the remaining processes are sitting idle.

For each region, all tiles are first aggregated into a virtual mosaic (`.vrt`). Each tile is assigned the region's CRS as specified in the `.json` file — this behaviour is necessary as parts of regional datasets (most notably subdivisions of DE-HE and DE-RP) have their CRS either missing or incorrectly specified within the tile's metadata — and is placed into the mosaic according to that CRS. No region has been found to have a mixed CRS; we note, however, that this approach may cause issues if that is the case.

Similarly, for certain regions, a `NoData` value needs to be specified. Parts of DE-BW and DE-HE, again, record `NoData` as 0 but do not state this in metadata, which warrants forcing a `NoData` value.

The output grid is established based on the resolution supplied in the `.json`. The grid is identical for all regions. For the product described in this report, the grid is in EPSG:3035 and at 5 m resolution; the vertical CRS is EVRF2007 (EPSG:5621) as the PROJ library does not yet natively support transformation to EVRF2019.

Then, the rasters are warped (using `gdalwarp`) to the common grid. At this step, the data is transformed from the source CRS to target CRS both horizontally and vertically. After that, the data is resampled to the target grid. By default, the chosen resampling method is averaging all the valid (i.e. not `NoData`) sample points located in the given grid cell, but bilinear interpolation is also possible.

The resampled raster is written to file, without cropping, with a `NoData` value of −9999.

#### 4.1.3 Filling Gaps

If specified by a region's setting in the `.json`, gap filling is applied after resampling using IDW interpolation with a specified maximum search distance and no smoothing. This order targets gaps remaining in the final-resolution surface and preserves all already valid target elevations. Valid surrounding terrain may supply interpolation donors outside the AOI. Alongside the interpolated surface, a binary interpolation mask is written to file to distinguish unchanged cells from interpolated cells.

#### 4.1.4 Resolving Conflicts and Assembling the Output

At this step, we need to decide which data to write to the output DEM where DEMs of several regions overlap. Several approaches were considered, such as evaluating for best temporal match or data quality. The former is not guaranteed by raster metadata — referring to the GPS time of the corresponding point cloud was considered but ruled out due to storage requirements — and the latter is difficult to infer from raster elevation data alone. Instead, we simply write each region's data within its boundaries (as determined by a vector file of that region).

However, for regions where interpolation was performed during our processing, we decide to overwrite the interpolated grid cells with non-interpolated data from the neighbouring region if that data is available. If the agreement between regions is high (like it is in our case, see Section 5) this approach produces fewer and smaller artefacts than accepting interpolation results.

The resulting elevation data is written to an output GeoTIFF, cropped to the specified region extents.

### 4.2 Water Mask Processing

An expected behaviour of the pipeline is retaining the errors introduced by water body interpolation by different providers, which significantly impact the error rate when comparing the datasets. The solution to this problem is to exclude the subset of points which fall within the river area from the error analysis.

This mask could be identified using the classified point clouds. However, the re-extraction of a DEM from point clouds is out of the scope for this project. Therefore, we only rely on the DEM data for segmentation. While river/lake masks do exist, they either provide global low-resolution vector data (Lehner, 2022), or low-resolution raster data (United Nations Environment Programme, no date).

In order to create a high-quality water mask, the Freshwater Ecosystems Explorer (FEE) mask was used as an input. The FEE mask contains a value 1, 2, or 3 to describe the certainty of the underlying value being water — corresponding to "no water", "seasonal water", and "permanent water" respectively — based on historical remote sensing data.

However, the FEE mask is not fine enough to be applied to the elevation map. Instead, we can use the FEE mask's values of 2 and 3 as seed points in a region growing algorithm. Since rivers and water bodies are known to be relatively flat, we perform region growing detecting neighbourhoods with small changes in elevation.

The mask is transformed to the elevation map resolution, CRS and grid. Each grid point with value 2 or 3 raster is taken as a seed for a region growing algorithm. These seed points are added to the region growing queue. Each entry has eight neighbours and for each, the height difference with the given cell is calculated. When these differences are smaller than the set threshold — empirically, it was found that a $0.02\,\mathrm{m}$ threshold works best — the neighbouring point is added to the end of the queue. The result is a breadth-first search which incrementally classifies cells as a water body. The end result is a mask with region IDs, where cell values above 0 denote the existence of a water body. A pseudo-code description of the water masking is found in Algorithm 1.

**River Mask Generation**

**Input:** DTM $D$ with valid cells $V$, extent $E$, water probability raster $C$, seed values $S$, vertical tolerance $\Delta_z$
**Output:** River mask $R$
$M \leftarrow$ empty boolean grid on the grid of $D$
**for each** grid cell $c \in C$ within extents of $D$ where $C[c] \in S$ **do:**
    transform grid point of $c$ to target CRS
    $M[c] \leftarrow 1$
$M \leftarrow M \cap P \cap V$ *discard seeds outside the region or without DEM data*
$R \leftarrow 0$ everywhere; $r \leftarrow 0$
**for each** seed $s$ in $M$, in row-major order **do**
    **if** $L[s] \neq 0$ **do:**
        **continue** *already part of an earlier water body*
    $r \leftarrow r + 1$; $R[s] \leftarrow r$
    $Q \leftarrow [s]$ *FIFO queue: breadth-first growth*
    **while** $Q$ is not empty **do**
    $c \leftarrow$ pop front of $Q$
    **for each** of the 8 neighbours $n$ of $c$ inside the grid **do**
        **if** $n \in V$ **and** $L[n] = 0$ **and** $|E[n] - E[c]| \leq \Delta_z$ **then**
            $L[n] \leftarrow r$
            push $n$ onto $Q$
    $R[\neg V] \leftarrow -1$
    **return** $L$

Algorithm 1: Binary river-mask generation by seeded region growing from hydrographic classes.

## 4.3 DEM Analysis

Once all harmonisation steps are complete, the next step is to pairwise compare neighbouring datasets and evaluate the results. This is first done by determining which regions share borders and have overlapping elevation products. This is followed by computing the elevation difference rasters, performing co-registration, and analysing statistical aspects of said difference such as correlation between error magnitude and slope, or the rate of INSPIRE compliance.

To determine which DTMs overlap one another, we first compute the bounding boxes of each DTM and check if they overlap:
$\max(x_{A\,\min}, x_{B\,\min}) < \min(x_{A\,\max}, x_{B\,\max}) \wedge \max(y_{A\,\min}, y_{B\,\min}) < \min(y_{A\,\max}, y_{B\,\max})$, where $A, B$ denote two regions.

However, the bounding boxes may overlap even when the datasets do not. Therefore, another check is done by cropping the DTM of one country with the other and checking if the cropped dataset is not empty. This is a computationally expensive operation and is thus only performed if an overlap between the bounding boxes is found.

The next step in the analysis process is to calculate the elevation differences between the DTM pairs in the overlapping border regions by subtracting the DTM rasters. The resulting elevation difference rasters contain the cell-wise elevation difference in the region, and constitute the core of the conducted qualitative analysis. Qualitative analysis is also supported by satellite imagery or OpenStreetMap (OSM) vector maps.

Based on this difference, we construct polygons to highlight errors greater than a set threshold. Nearby polygons are merged into one which allows highlighting potential areas with high concentration of disagreements rather than relying on a count of cells in excess of the threshold. We selected the threshold to match the minimum ratio between resolution and vertical RMSE — at 5 m, that threshold is 1.67 m. This threshold is a little lower than Copernicus's 2 m Linear Error at 90th Percentile (LE90) benchmark. For the analysis, the number of cells where the error rate exceeds the threshold is also presented.

The spatial correlation of the elevation differences is also analysed. Overall, it is expected that Tobler's (1970) first law of geography will hold: the close-by errors will have a higher correlation than far-away ones. It is further expected that stochastic errors are uncorrelated, and a high correlation of far away errors likely indicates a systematic error instead. However, the correlation does not account for error magnitude — a small systematic error will result in a higher correlation rate than a large stochastic one.

Another way to evaluate systematic errors is through coregistration. This is a function in XDEM which performs an affine transform (Nuth and Kääb, 2011), which ensures best alignment of the DTMs in $x$, $y$ and $z$ directions. The statistics (RMSE, LE90, mean error etc) are calculated for the border region before and after coregistration. A small affine transform with a high RMSE would indicate stochastic errors.

Several diagnostic plots are generated for each border pair: an elevation-difference histogram, an RMSE versus slope plot using 5-degree slope bins, and a border-effect plot showing RMSE in 50-metre distance bins. The slope plot examines the relationship between terrain slope and elevation differences, while the border-effect plot can help identify errors caused by extrapolation near dataset edges. The latter can also be sensitive to a single large error.

INSPIRE suggests different target ratios between horizontal resolution and vertical accuracy depending on slope angle: 3–20 for flat terrains and 3–10 for steep terrain. To evaluate the implied increase in error rate, we produce a plot showing the relationship between the slope angle and the error rate.

We also plot a histogram of errors, set to logarithmic scale to better highlight the less prevalent errors too. We expect the errors to largely follow a normal distribution with a mean at 0. A mean shifted away from zero may suggest a systematic vertical offset, while secondary peaks may indicate artefacts such as extrapolation errors.

### 4.3.1 Analysis Outputs and Interpretation

The generated `.csv` file with statistics complements the spatial plots with summary statistics for the entire border region. The "before" and "after" describe the differences before and after co-registration. Since we do not apply the affine transform to national DTMs, the "before" rows are more informative. However, the after section does show the effect of vertical transform errors on the total

difference. The “subset” refers to the cells with and without the values interpolated by us, which may be the same number if no interpolation was applied; both figures already include the removed river mask. The “mean” and “median” help describe the systematic errors of the border region, while the Standard Deviation (STD), RMSE, Mean Absolute Error (MAE), and Normalized Median Absolute Deviation (NMAD) describe the magnitude of the outliers (Table 1). The minimum and maximum values indicate the greatest disagreements between the datasets. Furthermore, the “above threshold” shows the amount of cells that do not meet the error threshold, while the “percentage above threshold” describes what percentage of cells that accounts for (Table 2).

The remaining attributes are mainly used to determine the reliability of the comparison, and the effectiveness of the coregistration (Table 3). The “mean $\sigma\Delta$ h” represents the average uncertainty associated with the elevation differences, while $N_{\text{eff}}$ reflects the number of independent observations after accounting for spatial correlation. The “mean $\Delta h/\sigma$” metric describes the magnitude of the differences, relative to the expected random error. Furthermore, the $\rho_{1\text{ km}}$ provides an indication of the spatial correlation of the errors at 1 km distance, with larger values suggesting that cells exhibit similar error behaviour. Finally, the estimated $\Delta X$, $\Delta Y$, and $\Delta Z$ values quantify the horizontal and vertical corrections determined during coregistration and can help identify systematic offsets between neighbouring datasets. Small values of this shift indicate that the vertical and horizontal CRS transformations already properly align the two DTMs.

| Stage | Subset | Valid cells | Mean | Median | Std | RMSE |
|---|---|---|---|---|---|---|
| Before | All | 11,282,135 | −0.0328 | −0.0324 | 0.2574 | 0.2595 |
| Before | Measured | 11,282,135 | −0.0328 | −0.0324 | 0.2574 | 0.2595 |
| After | All | 11,282,135 | −0.0042 | −0.0001 | 0.2518 | 0.2518 |
| After | Measured | 11,282,135 | −0.0042 | −0.0001 | 0.2518 | 0.2518 |

Table 1: Example of general metrics for FR/DE-RP border.

| Stage | Subset | MAE | NMAD | LE90 | Min | Max | Above thr. | % Above |
|---|---|---|---|---|---|---|---|---|
| Before | All | 0.1176 | 0.1129 | 0.2255 | −30.37 | 34.54 | 26,792 | 0.237 |
| Before | Measured | 0.1176 | 0.1129 | 0.2255 | −30.37 | 34.54 | 26,792 | 0.237 |
| After | All | 0.1057 | 0.1054 | 0.1928 | −30.15 | 34.84 | 26,976 | 0.239 |
| After | Measured | 0.1057 | 0.1054 | 0.1928 | −30.15 | 34.84 | 26,976 | 0.239 |

Table 2: Example accuracy and threshold metrics for FR/DE-RP border.

| Stage | Mean σΔh | $N_{\text{eff}}$ | Mean Δh/σ | $\rho_{1\text{ km}}$ | ΔX (m) | ΔY (m) | ΔZ (m) |
|---|---|---|---|---|---|---|---|
| Before | 0.1075 | 4.567 | 0.0503 | 0.2353 | - | - | - |
| After | 0.1075 | 4.567 | 0.0503 | 0.2353 | −0.0056 | 0.2830 | −0.0286 |

Table 3: Example coregistration metrics for FR/DE-RP border.

We find that of these metrics, mean, median, RMSE, NMAD, and LE90 best inform the analysis, and the rest are primarily used for context.

## 4.4 Visualising the DTMs

Next to designing the pipeline and implementing analysis methods, an important aspect of this study was to decide on effective visualisation methods and colour schemes for the catchment-wide and analysis related DTMs.

The catchment-wide DTM product contains elevation values ranging from $-325\,\text{m}$ to $4272\,\text{m}$. The DTM also contains various terrain types from the Netherlands’ low variation in elevation to the Swiss Alps where elevation changes by several thousands of metres. This made visualising the catchment-wide DTM challenging, as with most colour maps used, the Netherlands appeared completely dark while Switzerland became very bright and saturated, or vice versa. The Pride

(Velden, 2026a) colour map resulted in a good balance between having an interpretable DTM, also at the extremes of the range, and a visually striking product (Figure 8).

Figure 8: Pride gradient. From Velden (2026a).

The primary and most informative analysis tools are the elevation difference rasters obtained by subtracting two different DTMs from each other after they have been projected onto the same grid. Various colour gradients were tested to best evaluate the difference rasters. However, those provided by default in QGIS were not satisfactory for both visualisation and analysis purposes. Therefore, Fusion (Velden, 2026b) was used as the primary gradient for elevation difference visuals (Figure 9) as it is divergent and perceptually uniform. Using this gradient allows for the clear illustration of difference extremes. For the purposes of analysis, the colour range is set such that it is symmetrical around 0, e.g. $[-8, 8]$. The exact magnitude differs for different border pairs as some have higher error magnitudes than others.

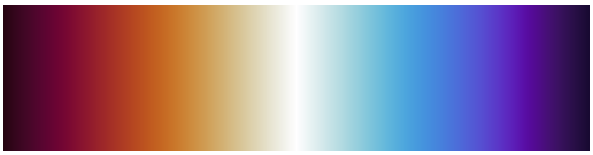


Figure 9: Fusion gradient. From Velden (2026b).

In addition to the gradients mentioned above, the Viridis and Spectral gradients were used in certain occasions, for example when discovering flight strip patterns or inspecting specific areas.

# 5 Results

This chapter discusses the results of the data acquisition, the pipeline, analysis of the border-pairs and general findings about the DTMs. First, the challenges encountered while acquiring the data are described. Then, the results of the reprojection, resampling, regridding and interpolation are presented, alongside the description of our experience executing the pipeline. After, the analysis of the process is discussed and a summary of the common error types is given.

## 5.1 Data Acquisition

The data scraping is not considered a part of the automated processing pipeline due to the variations in scraping methods, accessing differences, and computation time. For the pipeline to be executed, all data should be made available in their raw formats which in most cases are individual DTM tiles. In case the requested data is not available, the acquisition script proceeds to download it from the specified source. Currently, all acquisition of data labelled as a "must" in Appendix E.1 is implemented within the scraping software.

The pipeline acquires elevation data from 11 regions within 6 countries. The data comes in different formats, availability, CRS, and resolution. Table 4 summarises the characteristics of datasets. Some regions offer multiple DTM products that differ in resolution, format, and download accessibility. Selecting a suitable product therefore involved trade-offs between data quality and computational efficiency. For France, two products were reviewed: RGE ALTI and MNT LiDAR HD. Although MNT LiDAR HD provides newer, higher-resolution elevation data, its publicly available GeoTIFF tiles have a resolution of 0.5 m, which would require considerably more storage and computational resources. In contrast, RGE ALTI is available at 5 m resolution, although its ASCII Grid format requires conversion to GeoTIFF. RGE ALTI was therefore selected, prioritising storage and computational efficiency over the potentially higher accuracy and more recent acquisition of MNT LiDAR HD. RGE ALTI is also more complete, although omissions from MNT LiDAR HD are outside of the study area.

| Region | Dataset / provider | Source format | Survey method | Native grid (m) | CRS (EPSG) | Vertical datum |
|---|---|---|---|---|---|---|
| NL | AHN / Rijkswaterstaat | GeoTIFF | ALS | 0.5 | 28992 | NAP |
| DE-BY | DGM5 / LDBV | XYZ Tiles | ALS | 5 | 25832 | DHHN2016 / NHN |
| DE-BW | DGM1 / LGL | XYZ Tiles | ALS | 1 | 25832 | DHHN2016 / NHN |
| DE-HE | ATKIS DGM1 / HVBG | GeoTIFF | ALS | 1 | 25832 | DHHN2016 / NHN |
| DE-NW | DGM1 / Geobasis NRW | GeoTIFF | ALS | 1 | 25832 | DHHN2016 / NHN |
| DE-RP | DGM1 / LVermGeo | GeoTIFF | ALS | 1 | 25832 | DHHN2016 / NHN |
| DE-SL | DGM1 / LVGL | GeoTIFF | ALS | 1 | 25832 | DHHN2016 / NHN |
| FR | RGE ALTI 5 m / IGN | ASCII Grid | Mixed[1] | 5 | 2154 | IGN69 |
| LU | LiDAR 2024 MNT / ACT | GeoTIFF | ALS | 0.5 | 2169 | n/a |
| LI | swissALTI3D / swisstopo | GeoTIFF | Mixed[2] | 2 | 2056 | LN02 |
| CH | swissALTI3D / swisstopo | GeoTIFF | Mixed[2] | 2 | 2056 | LN02 |

Table 4: Characteristics of the source elevation datasets used in the study.

[1] IGN uses ALS in flood-prone areas, coastal zones and large forests combined with radar in mountainous areas (Alps, Pyrenees, Cévennes and Corsica), and aerial image correlation elsewhere. In some areas processed by image correlation, older elevation data fill gaps where vegetation or other surface objects prevent ground measurements.

[2] swissALTI3D combines airborne laser scanning with aerial photogrammetry, including automatic stereo correlation and manual stereoscopic measurements. The source methods vary spatially.

For some regions, however, data acquisition and processing had already been implemented before the target grid resolution of 5 m was established. For the Netherlands, the existing implementation using the 0.5 m AHN product was retained despite the availability of a 5 m product. For most German regions, the standard 1 m DTM products were used, with DE-BY being an exception where a 5 m product was available for download. For Switzerland and Liechtenstein, the nationwide 2 m product was selected to the target grid. In Luxembourg, the more recent 0.5 m nationwide product was preferred over an older 5 m product, as the relatively small study area made processing the higher-resolution data feasible.

The total file size of the source data is measured at 804.46 GB. Notably, France occupies only 23.04 GB or 2.9% of the total compared to the Netherlands' 213.25 GB or 26.5%, clearly demonstrating the trade-off between data quality and storage costs.

## 5.2 Pipeline

During this project, we have created an automated pipeline which can process, aggregate, and compare national gridded elevation products. The code has been executed on a server which has a CPU with 128 threads and a maximum clock speed of 2900 MHz, and 512 GB RAM. The server was shared with other members of the research group; we cannot guarantee that we had access to its full resources at all times. The pipeline has been run utilising 64 threads; in 4 parallel processes for individual DTM rasters, and sequential processing for river mask extraction and border pair comparison. We have attempted parallel processing for river mask and comparison; however, we found that both tasks have high memory demands which would result in some, especially larger, regions being skipped if sufficient memory could not be allocated.

The time required to produce an aggregate DEM was measured at 2:29:22; and time to produce the full comparison was measured at 14:44:39. The river mask extraction was not measured, but is estimated at 4:12:04. Note that these times are for a single run of the full pipeline, starting on 2026-10-25 at 00:16:57 and finishing on 2026-10-25 at 20:43:03; they may not be representative of optimal or typical resource availability.

If the associated `.json` file is configured properly, the code can be executed in full and produce the desired outcome. However, the pipeline and the `.json` needed to be modified several times to account for the unforeseen errors. The pipeline was first tested on the Netherlands and DE-NW to ensure individual components are integrated, and it was subsequently run in full. During the full runs, several issues were discovered and patched.

The first of these was that DE-HE publishes data by administrative unit; for some of these units, `NoData` is represented with a 0 value but the value is not encoded in the metadata. Some of DE-HE's tiles also had no horizontal CRS. Stitching these tiles resulted in regions of flat terrain with elevation value around 20 cm (non-zero due to CRS transform). Similar behaviour was observed in DE-RP, where the CRS was not consistently described in the metadata. This behaviour has been patched by force-assigning a horizontal and vertical CRS and a `NoData` value for each region.

Issues with `NoData` value not declared were also observed in DE-BW. These are mostly resolved except for several tiles overlapping with DE-BY, where the `NoData` issue persists. We acknowledge these issues within the analysis of the border regions, but resolving them was deemed to be of low priority given the time constraints.

During the later runs, we observed that large parts of the Netherlands and France were absent from the final product. We determined that this issue stems from a missing VCRS transform. Previously, a warning was issued and the pipeline continued with no VCRS transform applied, but during the later runs, some of the tiles were skipped. We believe this to be a change on the PROJ server likely

following the PHP2shell vulnerability (Tzadik *et al.*, 2026). Our fix is setting the VCRS for affected regions to `null` such that no transform is attempted, which allows the pipeline to be executed.

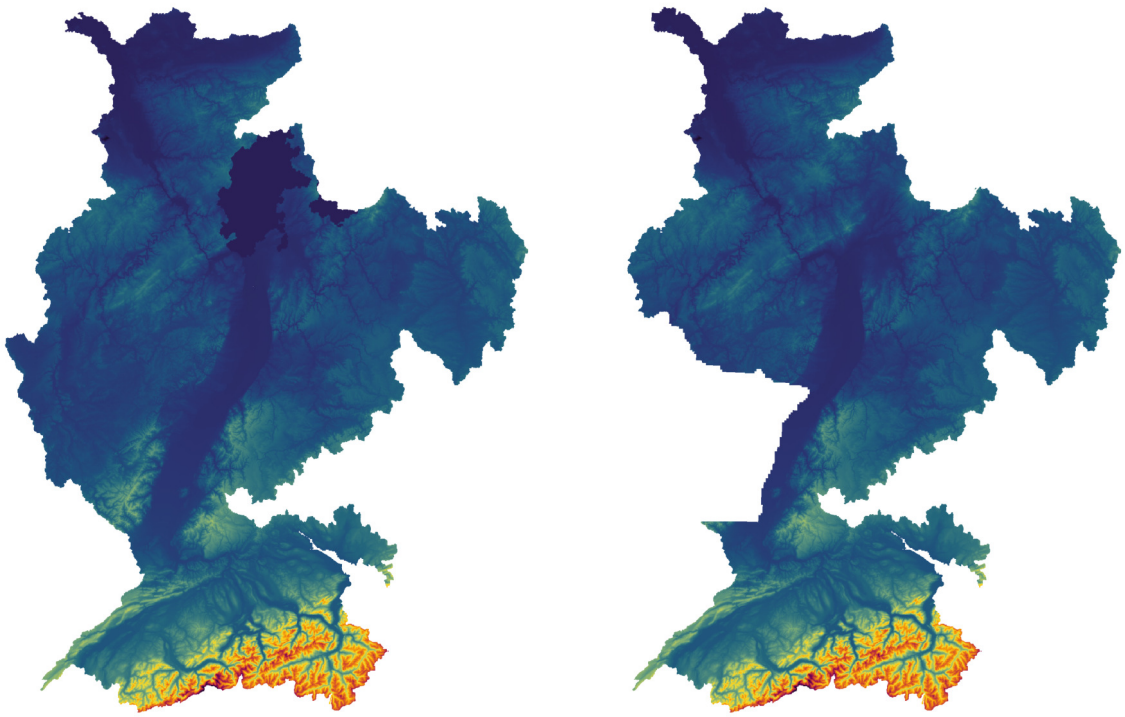


Figure 10: Issues diagnosed with earlier runs of the pipeline: incorrect `NoData` value for DE-HE (left) and missing parts of FR (right).

An additional issue was observed during the border analysis stage. If the data provider cropped the dataset following the border and their definition of the border did not match ours, gaps along the border appeared. These issues are not observed within Germany where full tiles are supplied, but occur along the borders with France and the Netherlands. This issue stems from the coarse vector masks used. These were replaced with masks acquired from OSM which has patched the gaps.

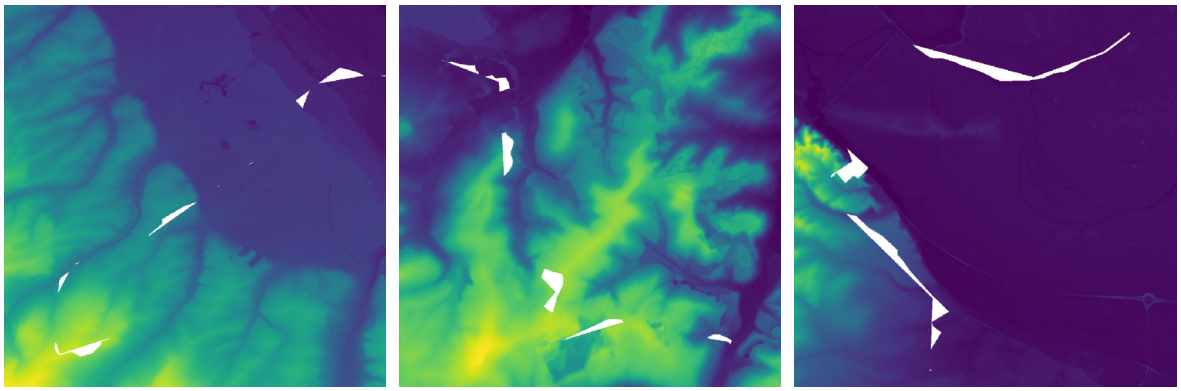


Figure 11: Gaps on the FR/CH, FR/LU, and NL/DE-NW border.

## 5.3 DTM of the Rhine Catchment Area

The pipeline successfully produces a DTM of the Rhine catchment area, shown in Figure 12. For visualisation purposes, the elevation colour scheme was adjusted to improve the visual distinction between terrain features across the catchment. The displayed elevation range was limited to −35 m to 1,100 m, with values outside this range assigned the colours of the respective lower and upper limits. This clipping affects only the visualisation and does not alter the underlying elevation values.

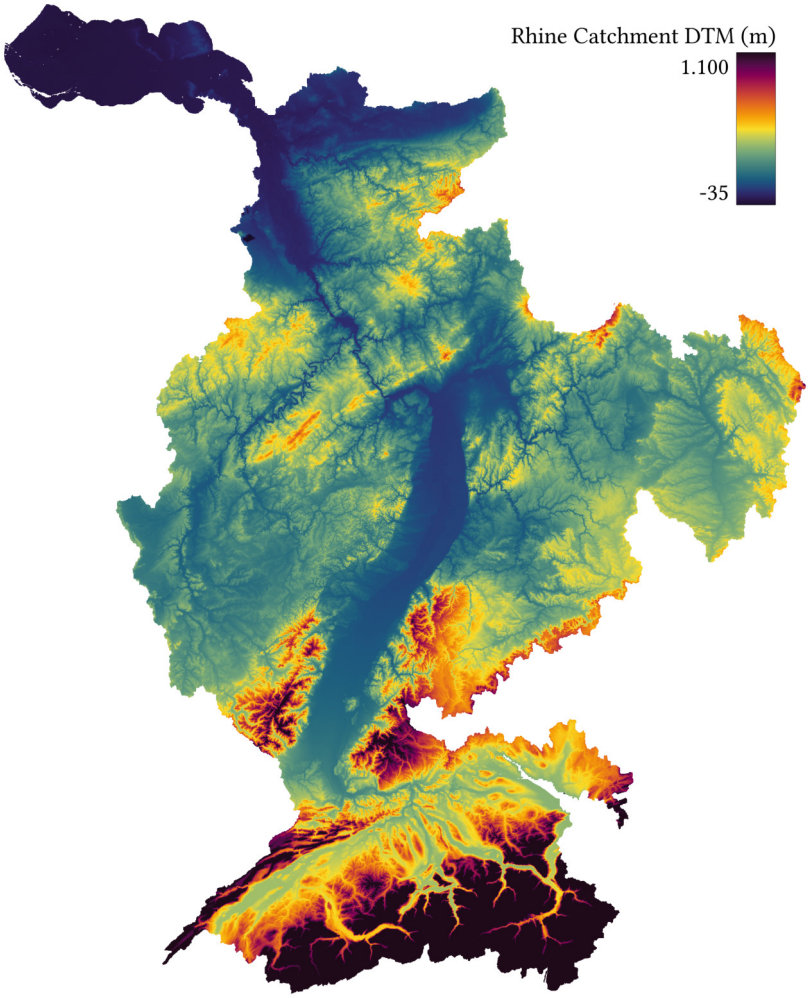



Figure 12: Rhine Catchment DTM.

The final DTM is stored as a Cloud-Optimized GeoTIFF file which occupies 17.5 GB of disk space. The final DTM uses ETRS89 / LAEA Europe (EPSG:3035) as its CRS, with a resolution of 5 m and elevation values stored as Float32. The minimum elevation is −324.90 m, occurring in a quarry in DE-NW, while the maximum of 4,272.68 m occurs in the Swiss Alps. The mean elevation is 452.32 m, and the elevation range is 4,597.58 m.

The DTM covers 97.33% (159284.05 km² or 6371361919 cells out of 163645.59 km² or 6545823685 cells) of the catchment area, for an omission rate of 2.67 %. The omission rate can be attributed exclusively to the missing regions (Austria, Belgium, and Thüringen) and `NoData` values within the Dutch dataset.

The commission rate within the final DTM is zero. However, the overlaps at the border add up to 4140.09 km², resulting in a commission rate of the underlying data of 2.60%.

The seamlessness of the DTM is dependent on the quality of the underlying datasets. Where the agreement is high, the DTM is seamless. This is the case for most of the datasets, illustrated by 3 regions of Germany in Figure 13. Where it is low, such as due to temporal or processing differences, a "step" can be observed. In our case, this happens primarily on the border with France as it is an older and less accurate dataset, as shown in Figure 14.

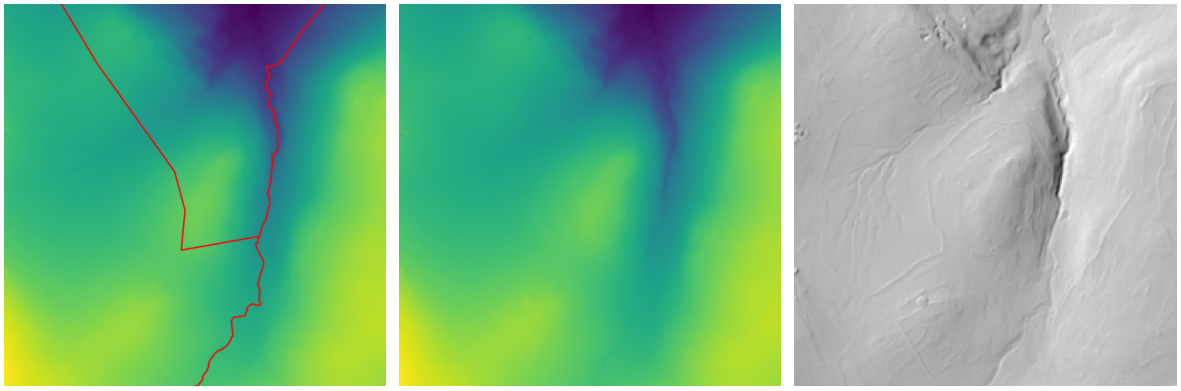


Figure 13: Example of seamless border between DE-HE, DE-NW, and DE-RP.

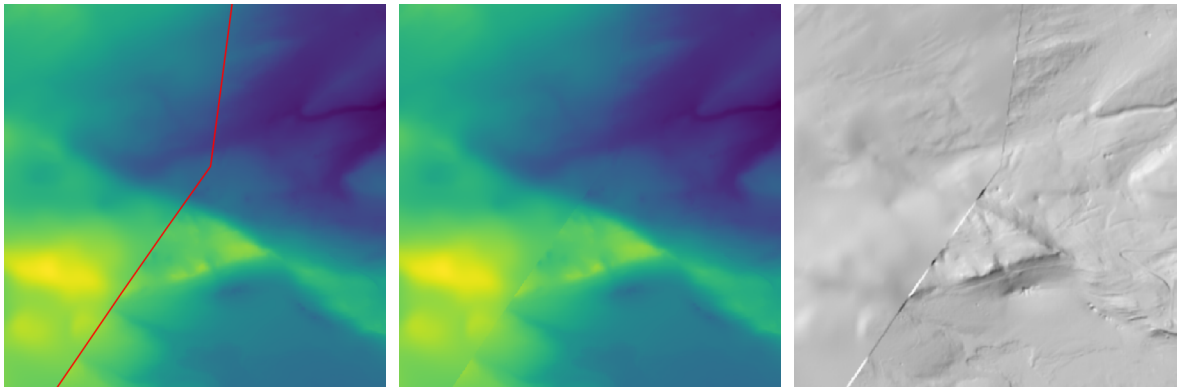


Figure 14: Example of border seam between FR and CH.

Whenever a border passes through a water body, we reliably see "jumps" in elevation. Figure 15 shows that such jumps are present not only at the border but also within the individual DTMs and are likely due to temporal differences between different surveys by the same provider.

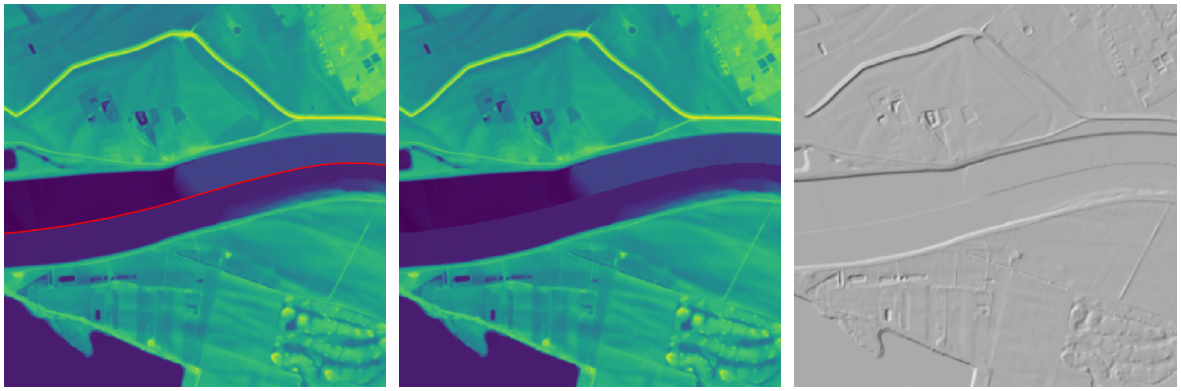


Figure 15: Elevation changes in water, as shown on the border between DE-HE and DE-RP.

### 5.3.1 Water Mask

The elevation differences shown above emphasise the need for masking the rivers — within the scope of this project, primarily for performing more accurate analysis. Due to a lack of an existing high-quality water mask, an existing mask was refined as detailed in Section 4.2. However, due to

the nature of the underlying data ("jumps" in elevation happen even within the same dataset) and the applied algorithm, artefacts still occur, even if their impact on the analysis is minimised. The final result of the river mask can be observed in Figure 16.

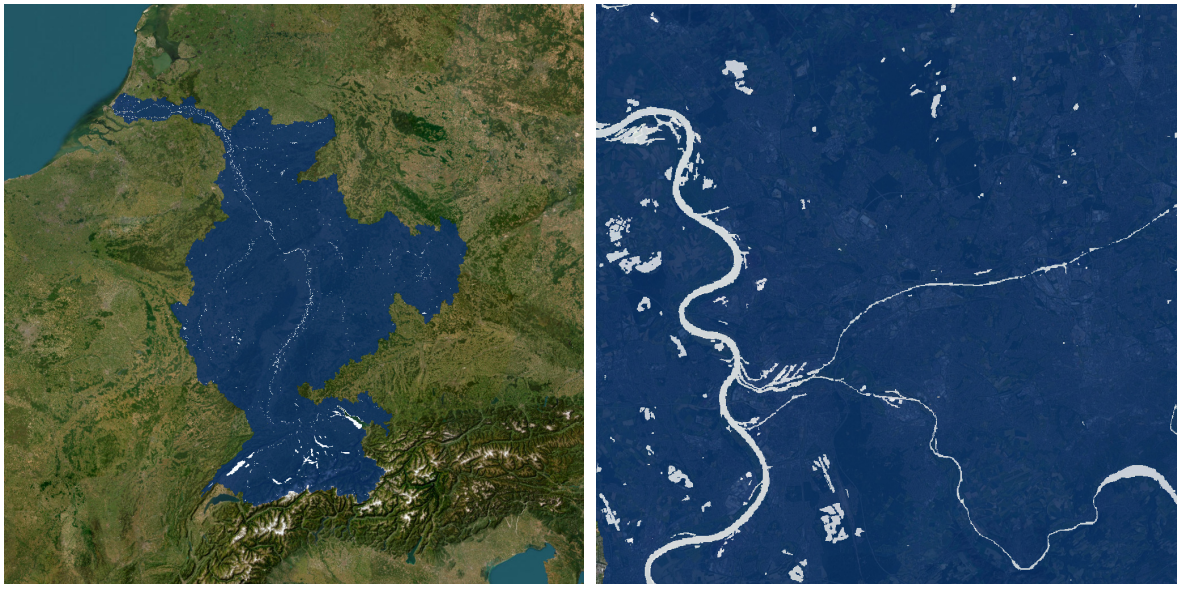
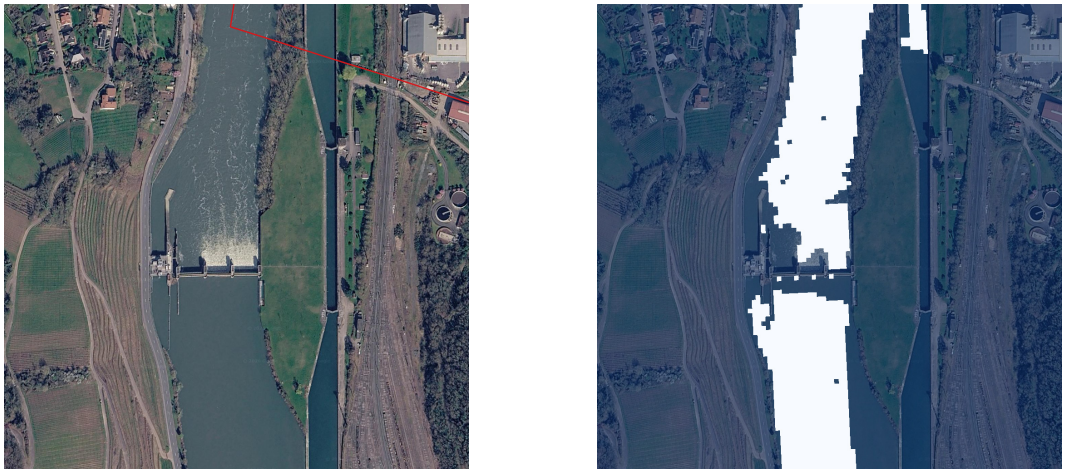


Figure 16: River mask visualised on the entire Rhine Catchment (left), and around the Cologne area in DE (right).

The region growing algorithm relies on the assumption that the water surface is smooth. However, this is not always the case. The first example of this is found in Figure 17. Here a hydroelectric power plant causes a change in the elevation of the water surface. The rough water mask from FEE marks this as an area which is not considered water. Therefore, there is a break in the water mask, and the area surrounding the plant such as the left secondary channel, is not modelled.

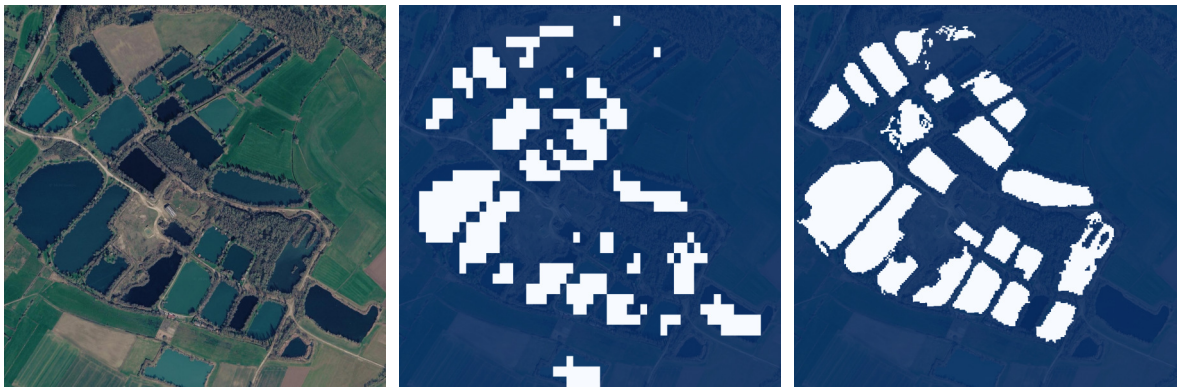


Satellite imagery. From Airbus, Maxar Technologies (2026).

Non-continuous river mask.

Figure 17: River mask failure where the mask is not continuous due to a hydroelectric power plant.

Another similar shortcoming is that the FEE mask is incomplete. Not all lakes contain valid seed points. Furthermore, the reprojection of FEE cell centres onto the empty mask sometimes fails due to invalid row and/or column values which results in runtime errors. These are known limitations of the river mask which could be solved by extracting the mask from the national point clouds of the respective regions.



Satellite imagery. From Airbus, Maxar Technologies (2026). FEE mask. Incomplete river mask.

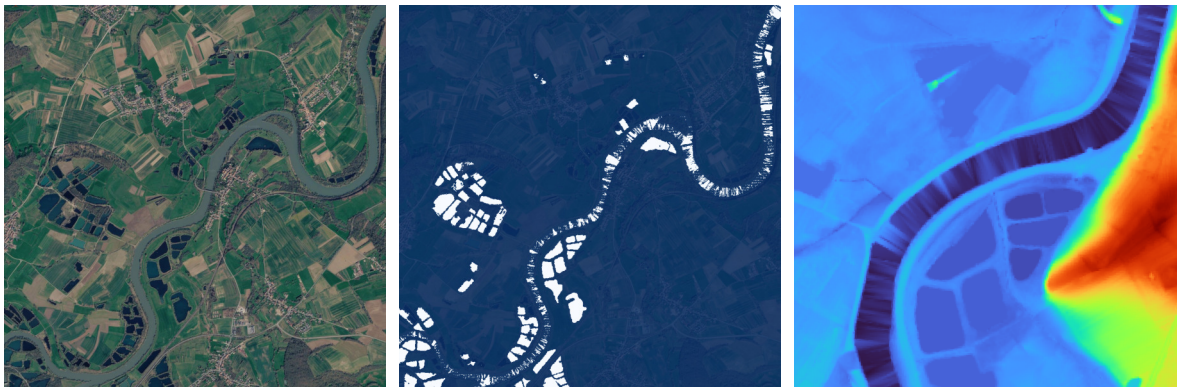
Figure 18: River mask failure due missing water cells in the input raster.

Other issues include areas that are not marked as water within the FEE mask. Examples include swampy areas on NL/DE-NW border. This is shown in Figure 19.

Satellite imagery. From Airbus, Landsat / Copernicus, Maxar Technologies (2026). Incomplete water mask.

Figure 19: Omission of swamp-like areas from the water mask prevents correct detection of water bodies. The marsh located between the upper lake and smaller patched lakes is not properly accounted for.

The last artefact can be observed along several regions but is most pronounced within France. It is caused by using the TIN to approximate the water surface. Visually, it appears as shard-like sections of lighter and darker colour in the DTM where different points along the river bank are chosen as sample points for interpolation on a TIN (see Figure 20). While it is most severe in areas further from the border and does not significantly affect the quality of the analysis, it does show that further work is needed to fully integrate the water mask.



Satellite imagery. From Airbus, Maxar Technologies (2026).

Non-continuous water mask.

TIN artefacts observed on DTM.

Figure 20: River mask failure due to changes in the water surface elevation caused by TIN interpolation of the surface.

## 5.4 Analysis

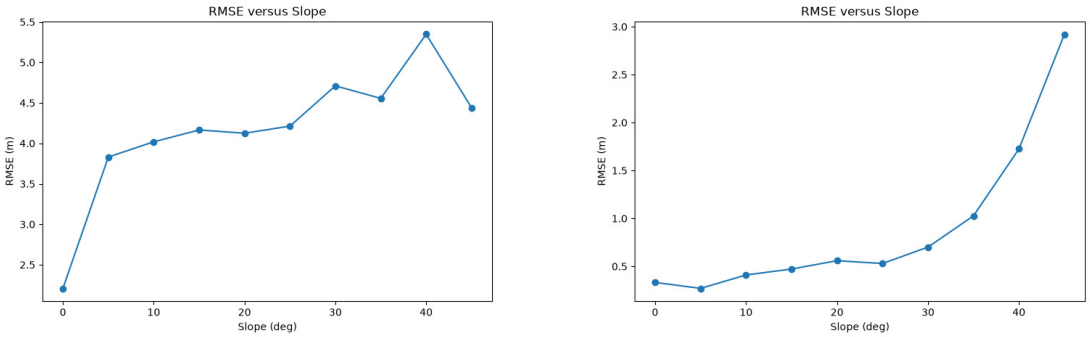
Within this section, we evaluate differences on the border between two neighbouring regions. We automatically generate various metrics the DTMs are evaluated on (Section 4.3). Of those, we find mean, median, RMSE and LE90; the generated raster; RMSE/slope plot and difference histogram to be among the most important as they give information about the general cases. Based on this data, a pattern was found regarding the error sources at the borders, described further below.

Table 5 shows the key statistics for each pair of the datasets. We note that high RMSE and high LE90 correspond to a high systematic error (as is the case for FR/LU border) while high RMSE but low LE90 correspond to an outlier (as is the case for DE-BW/DE-BY border).

| Ref | Comp | mean | median | rmse | nmad | le90 | min | max | Δ z (m) |
|---|---|---|---|---|---|---|---|---|---|
| FR | LU | 0.658 | 0.584 | 3.129 | 0.521 | 3.827 | −23.190 | 39.686 | 0.587 |
| FR | CH | 0.095 | 0.101 | 0.727 | 0.130 | 0.405 | −24.450 | 15.926 | 0.102 |
| DE-BW | FR | −0.479 | −0.458 | 0.873 | 0.161 | 0.779 | −26.461 | 46.219 | −0.458 |
| DE-BW | DE-BY | −0.053 | 0.039 | 5.361 | 0.052 | 0.114 | −961.151 | 29.982 | 0.039 |
| DE-BW | CH | 0.047 | 0.048 | 0.406 | 0.051 | 0.124 | −21.185 | 34.454 | 0.048 |
| DE-HE | DE-BY | 0.003 | 0.002 | 0.178 | 0.051 | 0.098 | −18.354 | 19.075 | 0.002 |
| DE-HE | DE-BW | 0.011 | 0.009 | 0.112 | 0.030 | 0.064 | −10.628 | 14.753 | 0.009 |
| DE-NW | DE-HE | 0.010 | 0.011 | 0.184 | 0.041 | 0.075 | −9.371 | 22.737 | 0.011 |
| DE-NW | DE-RP | −0.028 | −0.023 | 0.195 | 0.110 | 0.196 | −22.004 | 31.373 | −0.021 |
| DE-RP | FR | −0.559 | −0.559 | 0.616 | 0.114 | 0.715 | −30.898 | 34.020 | −0.555 |
| DE-RP | DE-BW | −0.009 | −0.013 | 0.210 | 0.044 | 0.073 | −19.532 | 16.480 | −0.013 |
| DE-RP | DE-HE | 0.013 | 0.045 | 1.297 | 0.073 | 0.186 | −61.556 | 30.440 | 0.053 |
| DE-RP | DE-SL | 0.008 | 0.011 | 0.374 | 0.072 | 0.148 | −38.308 | 26.917 | 0.018 |
| DE-RP | LU | 0.032 | 0.027 | 0.353 | 0.096 | 0.228 | −25.563 | 17.085 | 0.025 |
| DE-SL | FR | −0.514 | −0.563 | 1.317 | 0.234 | 1.161 | −22.604 | 32.059 | −0.564 |
| DE-SL | LU | −0.016 | −0.003 | 0.426 | 0.048 | 0.166 | −10.565 | 10.442 | −0.002 |
| NL | DE-NW | −0.065 | −0.063 | 0.244 | 0.038 | 0.134 | −11.543 | 13.759 | −0.063 |

Table 5: Overview of error statistics between neighbouring datasets.

The majority of the datasets demonstrate high agreement with each other, with 15 out of 17 pairs meeting the INSPIRE requirements for RMSE. The first exception is FR/LU border where the errors are caused by the consistent differences between the French — acquired with radar or photogrammetry — and Luxembourgish — acquired with Aerial Laser Scanning (ALS) — DTMs. In Figure 21, we can observe that France displays consistently high RMSE for all slope angles which even exceeds 5 m which is much higher compared to a more typical graph. The second is a fill value which does not match the NoData value specified in the metadata for DE-BW, described in Section 5.4.6.




Slope/RMSE graph for FR/LU border.

Slope/RMSE graph for DE-RP/DE-SL borders.

Figure 21: Comparison of slope/RMSE graphs.

We also observe that the pairs involving France perform consistently worse. Again, this is due to large sections of the DTM which are not acquired with ALS. France also has a median and mean of about 0.5 m which matches the offset due to co-registration, suggesting a systematic error. We have not been able to determine whether this error is intrinsic to the dataset or a result of the missing CRS transform.

### 5.4.1 Error Sources

These error sources can be divided into six categories representing overarching themes: interpolation, ground filtering, temporal, water mask, pre-processing, and post-processing artefacts. To further specify the type of artefact, each category is divided into subcategories which each represent an example of the artefact. The only error source which will not be discussed is the artefacts caused by shortcomings in the river mask, which have been discussed above in Section 5.3.1.

Table 6 gives an overview of these categories and their subcategories. The percentage of border pairs describes in how many of the 17 evaluated border cases an artefact occurs. We exclude the CH/LI border from the comparison as the countries share the dataset. We calculate what percentage of all border pairs, at least one of the artefacts within the category occurs — most borders have more than one type of artefact, and are therefore counted several times. As can be observed, the interpolation is the most frequently occurring artefact. We also evaluate the artefacts by severity, i.e. the magnitude and perceived effect of the error type. The overview can be found in Table 6.

The full overview of each border pair and their corresponding artefacts can be found in Table 7. Furthermore, a pairwise analysis of all border pairs can be found in Appendix G, which contains additional visuals and a case-by-case evaluation of the differences.

| Category / subcategory | Border pairs | Percentage of border pairs | Rank occurrence | Rank severity |
|---|---|---|---|---|
| **Interpolation** | **17** | **100%** | **1** | **5** |
| Interpolation differences | 17 | 100% | | |
| **Ground filtering** | **15** | **88%** | **2** | **2** |
| Overpass | 5 | 29% | | |
| Slopes | 13 | 76% | | |
| **Temporal** | **14** | **82%** | **3** | **3** |
| Snow | 1 | 6% | | |
| Bulk storage | 5 | 29% | | |
| Quarry | 10 | 59% | | |
| Construction | 11 | 65% | | |
| **Water mask** | **13** | **76%** | **4** | **4** |
| Water handling | 5 | 29% | | |
| River mask failure | 13 | 76% | | |
| **Pre-processing** | **7** | **41%** | **5** | **6** |
| ALS striping | 4 | 24% | | |
| Tile seam | 4 | 24% | | |
| **Post-processing** | **7** | **41%** | **5** | **1** |
| Extrapolation | 5 | 29% | | |
| Edge resampling | 5 | 29% | | |

Table 6: Identified artefact categories and their prevalences.

| Region pair | Interpolation | Ground filtering | | Temporal | | | | Water mask | | Pre-processing | | Post-processing | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Interpolation | Overpass | Slopes | Snow | Bulk storage | Quarry | Construction | Water handling | River mask failure | ALS striping | Tile seam | Extrapolation | Edge resampling |
| NL/DE-NW | ● | | | | ● | | | | ● | | | ● | ● |
| DE-NW/DE-RP | ● | ● | ● | | | ● | ● | | ● | ● | ● | ● | |
| DE-NW/DE-HE | ● | ● | ● | ● | | ● | ● | | | | | | |
| DE-RP/DE-SL | ● | ● | ● | | | ● | ● | | ● | | | ● | ● |
| DE-RP/DE-BW | ● | ● | | | ● | | ● | | ● | | ● | | |
| DE-RP/DE-HE | ● | | | | | | | | | | ● | ● | |
| DE-RP/FR | ● | ● | ● | | | | | ● | ● | ● | | | |
| DE-RP/LU | ● | | ● | | ● | | ● | | ● | | | | ● |
| DE-SL/FR | ● | | ● | | | ● | ● | ● | ● | | | | |
| DE-SL/LU | ● | | ● | | ● | ● | | | ● | | | | |
| DE-HE/DE-BW | ● | | ● | | | ● | ● | | ● | | | | ● |
| DE-HE/DE-BY | ● | | ● | | | ● | ● | ● | ● | | | | |
| DE-BW/DE-BY | ● | ● | | | | ● | ● | | ● | | ● | ● | ● |
| DE-BW/FR | ● | | ● | | ● | ● | | ● | ● | ● | | | |
| DE-BW/CH | ● | | ● | | | ● | ● | ● | ● | | | | |
| FR/LU | ● | | ● | | | ● | ● | | ● | | | | |
| FR/CH | ● | | ● | | | | | | | ● | | | |

Table 7: Overview of border region artefacts. Cells marked with ● imply the subcategory artefact occurs at the border pair.

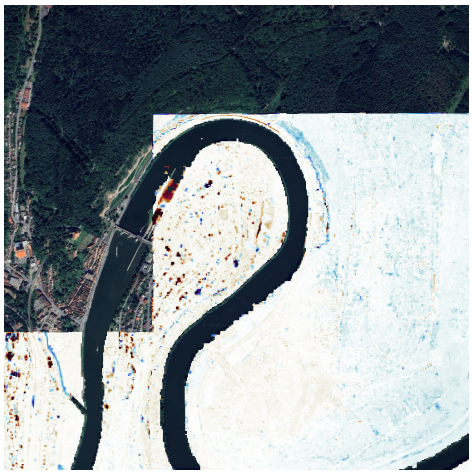
### 5.4.2 Interpolation Artefacts

Interpolation disagreements occur at all borders. This is because a DTM does not include buildings or vegetation within the data, it only includes terrain which means that some data has to be excluded from the source data. There are various methods to filter out these features. This results in gaps in the dataset which need to be filled, typically via interpolation. The exact point within the pipeline, source data, and interpolation algorithm all affect the elevation values of the resulting DTM.

Most data providers do not describe the method through which this filtering and/or interpolation is performed. Therefore, it is hard to definitively say whether differences occur as a result of differences in interpolation strategy, or due to other reasons such as temporal changes without evaluating the underlying point cloud.

The magnitude of the errors differs between region pairs. Some regions show little disagreement, while others can show elevation differences of 1–2 metres, with isolated cases going beyond that. We suggest that the errors can be better assessed if an accurate interpolation mask can be obtained. We can also use point cloud data to harmonise the interpolation — this has not been performed within the project.

Satellite imagery. From Airbus, Maxar Technologies (2026).



Elevation difference map.

Figure 22: Example of elevation differences caused by interpolation disagreements. Dark red values represent large positive difference while dark blue shows negative difference.

One important note is that interpolation is not always at fault for differences observed for buildings. Especially for large area buildings, ground filtering can also cause disagreements. DE-NW uses synthetic ground points from the CSF instead of interpolation to extract the DTM — this causes artefacts seen in Figure 23.

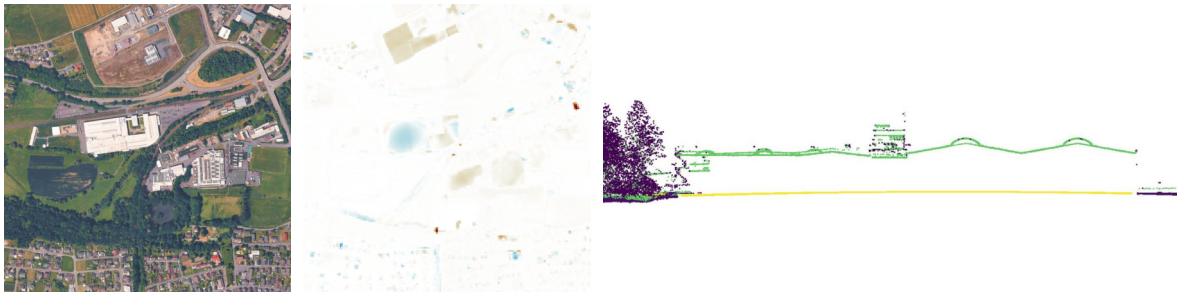
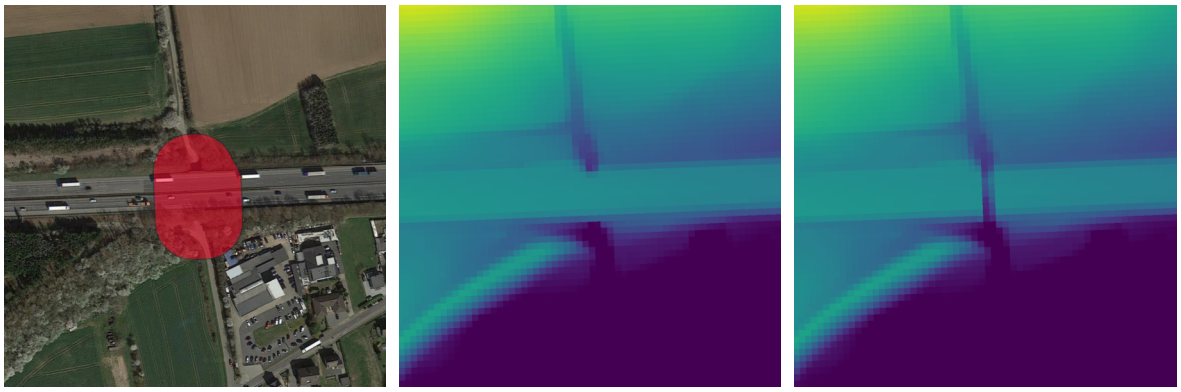


Figure 23: Ground and building filtering differences between DE-NW and DE-HE.

### 5.4.3 Ground Filtering Artefacts

Ground filtering is the second most common artefact encountered for the border pairs. For this category we consider steep slopes (cliffs, valleys), and disagreements caused by overpass handling. Out of these subcategories, steep slopes are both the most common and most severe, while overpasses only create small, local disagreements.

Errors in overpasses can only be found in roughly a quarter of all border pairs — they may also be present in other DTMs but are processed similarly there. The severity of the difference largely depends on the height of the overpass and generally exceeds the 1.67 m threshold. Figure 24 shows an example of different overpass processing.



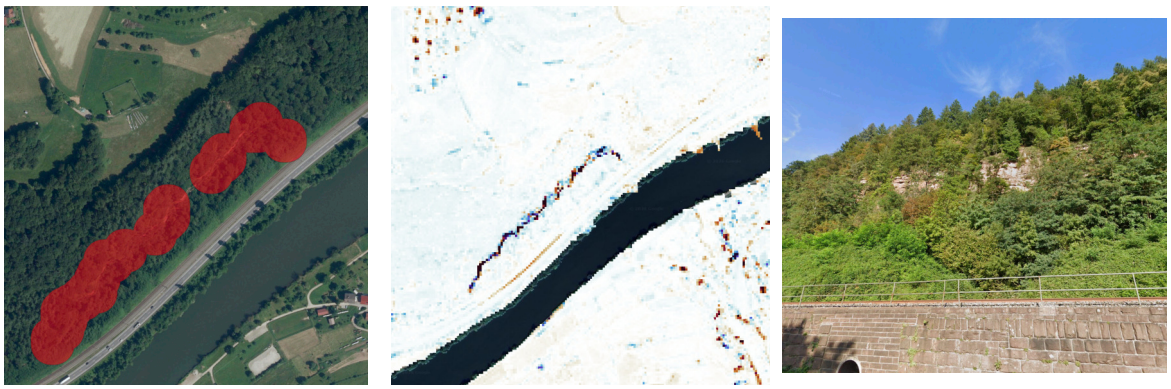
Satellite imagery. From AeroWest, Airbus, Maxar Technologies (2026).

DTM of DE-NW.

DTM of DE-RP.

Figure 24: Example of an overpass being treated differently between DE-NW and DE-RP due to differences in ground filtering.

Errors due to slope processing occur at cliffs and in valleys/hills. A cliff typically shows up as a thin and long artefact on the DEM difference raster, as seen in Figure 25. Due to the steepness of the cliff small horizontal differences between sample points in the DTM can create large vertical differences if one dataset measures the elevation at the bottom of the cliff and the other at the top. Alternatively, this can be a byproduct of differences in ground filtering methods.



Satellite imagery. From AeroWest, Airbus, Geobasis-DE/BKG. (2026)

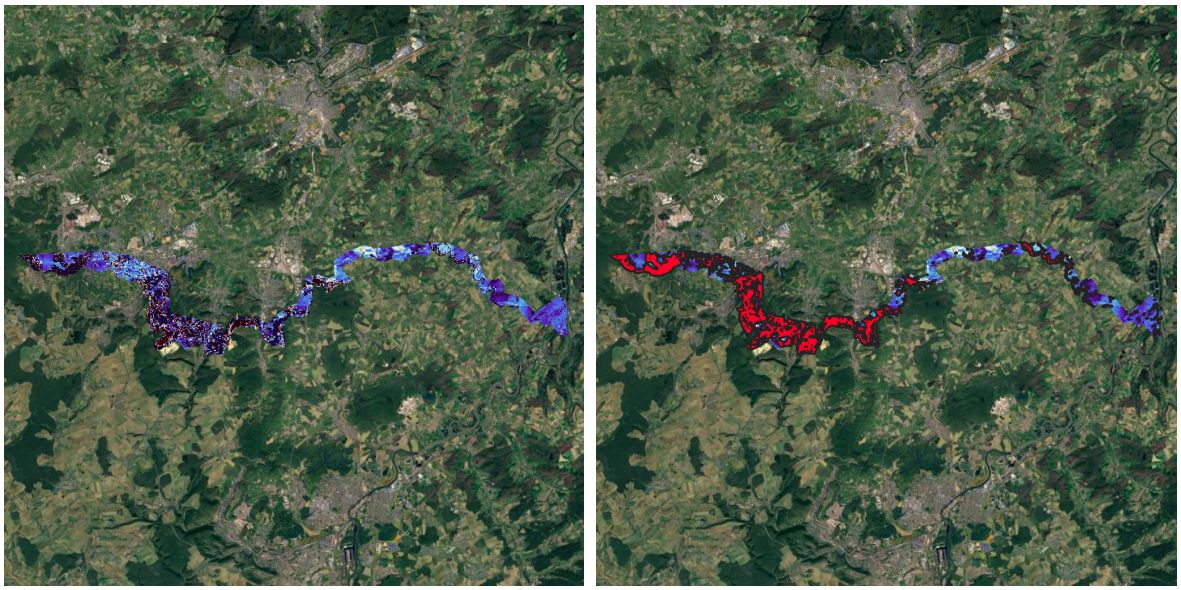
Elevation difference map.

Image from Google Street View (49.4539° N, 8.9474° E)

Figure 25: Example of a cliff artefact flagged as a resolution threshold exceeding disagreement.

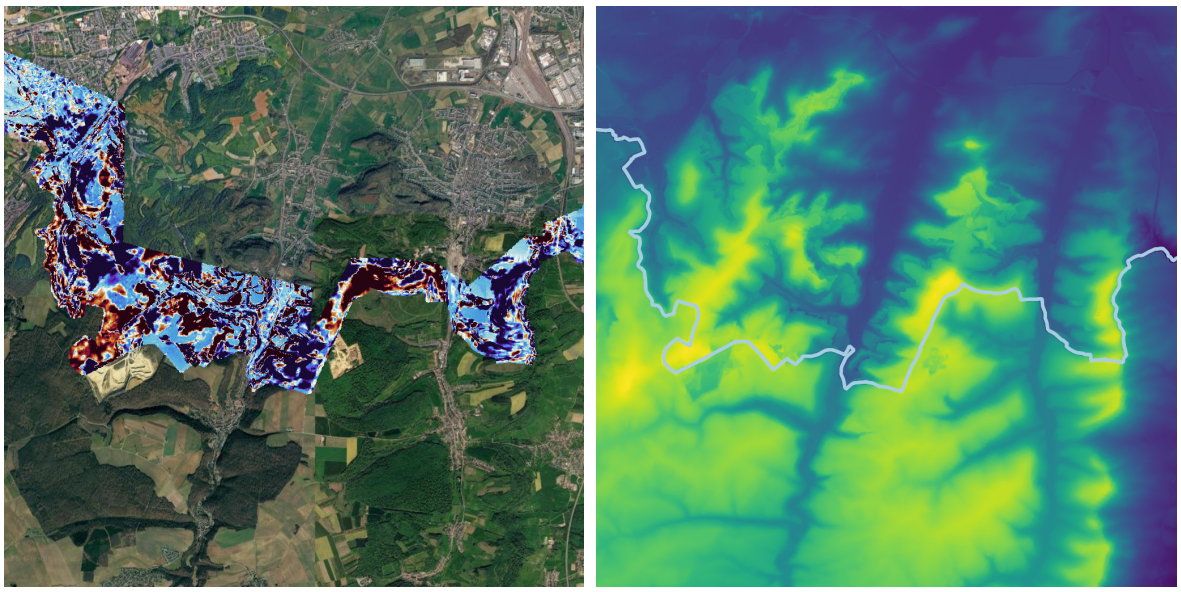
The second type of slope disagreement can be found primarily in regions which contain deep valleys such as the FR/LU border. The underlying causes are similar: differences in either data acquisition or

ground filtering. However, unlike cliffs, this type of artefact can cause large sections of disagreement between DTMs (Figure 26). Figure 27 shows the region around the city of Rumelange, which is located within a valley. The difference in DTM elevations here are up to differences of 25 m in both positive and negative directions. We also observe higher error rates for areas with steep slopes in other datasets, but the magnitude of those is typically under a metre. For France, we expect that a more accurate dataset will mostly resolve these errors; for other regions, standardising the ground filtering algorithm may reduce their magnitude too.



Elevation difference map. Flagged differences in red.

Figure 26: Example of large regions of disagreements. Flagged disagreements in red exceed 1.6 metres of difference between the FR/LU datasets.



Elevation difference map. DTM.

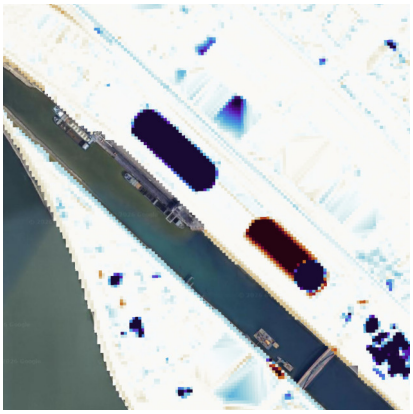
Figure 27: Example of Rumelange in LU, next to the French border. Disagreements of up to 25 m are observed.

### 5.4.4 Temporal Artefacts

Temporal artefacts rank third as the most common cause of artefacts. This is due to the fact that the acquisition time between two regional or national DTMs can differ up to several years. There are only a few typical causes of these differences. Most common are construction sites, which include everything from the construction or demolition of buildings, to the erection of wind turbines, which happens in most German regions. These are followed by excavation within the quarries, frequently found in all regions of Germany.

A more interesting type of temporal difference is found in the area surrounding Luxembourg. These are stockpiles of coal and sand at river ports (Figure 28). Due to their shape, they are classified as ground and therefore included in the DTM. As ships are loaded and the coal is redistributed, a change in elevation is observed.

Satellite imagery. From Airbus, GeoBasis-DE/BKG, Maxar Technologies (2026)

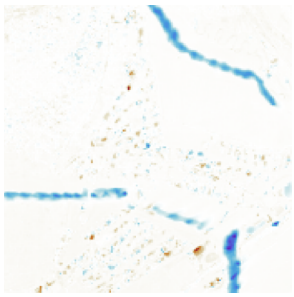


Elevation difference map.

Figure 28: Large elevation differences observed due to changes of a coal storage between two surveys.

Some remnants of snow at a skiing resort are included in the DTM of DE-NW, likely collected in the early spring. It highlights that the elevation models are highly sensitive to the time of acquisition. We have only been able to determine the source of this error from analysing the point clouds (Figure 29).

Satellite imagery. From Airbus, Maxar Technologies, GeoBasis-DE/BKG (2026).



Elevation difference map.

Point cloud of the area.
Yellow areas correspond to snow.

Figure 29: Elevation differences caused by the presence of snow in one of the surveys.

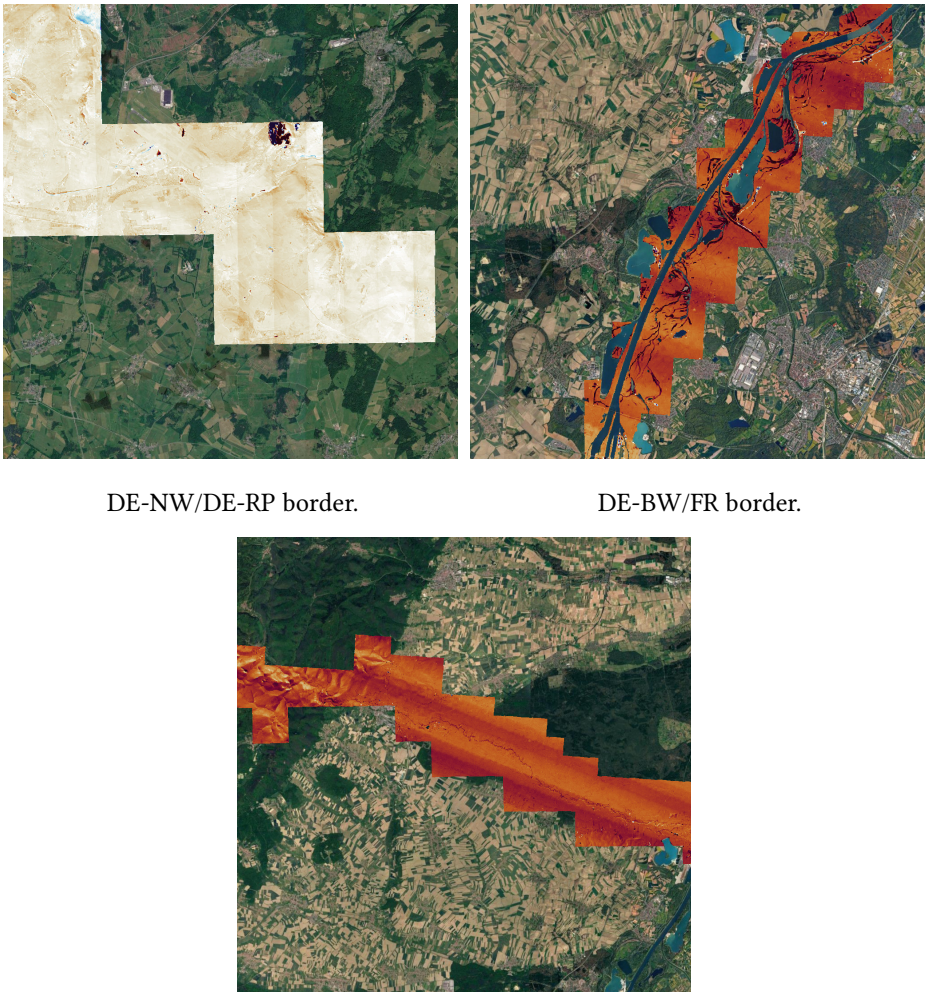
These differences can be only resolved by comparing the point clouds of the neighbouring regions. In the interest of temporal accuracy, it is most favourable to select data acquired most recently. However, this may cause a noticeable "seam" if the change happens at the edge of the dataset.

### 5.4.5 Pre-Processing Artefacts

Pre-processing artefacts are one of the least common types of artefacts. However, these artefacts may have a sizeable effect on the RMSE or LE90 statistics for a given border pair. These differences can be divided into two subcategories.

The first artefact in this category is striping. One common trend among the datasets within the Rhine catchment area is that most elevation data is derived from the point clouds of the region. These are primarily aerial laser scans, which are performed by flying a plane in parallel straight lines. The width of the bands that this creates is determined by the scanner's field of view.

Small deviations of several centimetres in the elevation of each scanning pass can create a striping effect if not corrected for. The severity of this artefact is minimal but clearly noticeable when observing differences in DEMs. It shows up as bands of consistent elevation along-track difference and sharp changes across the tracks, as shown in Figure 30. This artefact cannot be corrected based on gridded datasets alone.

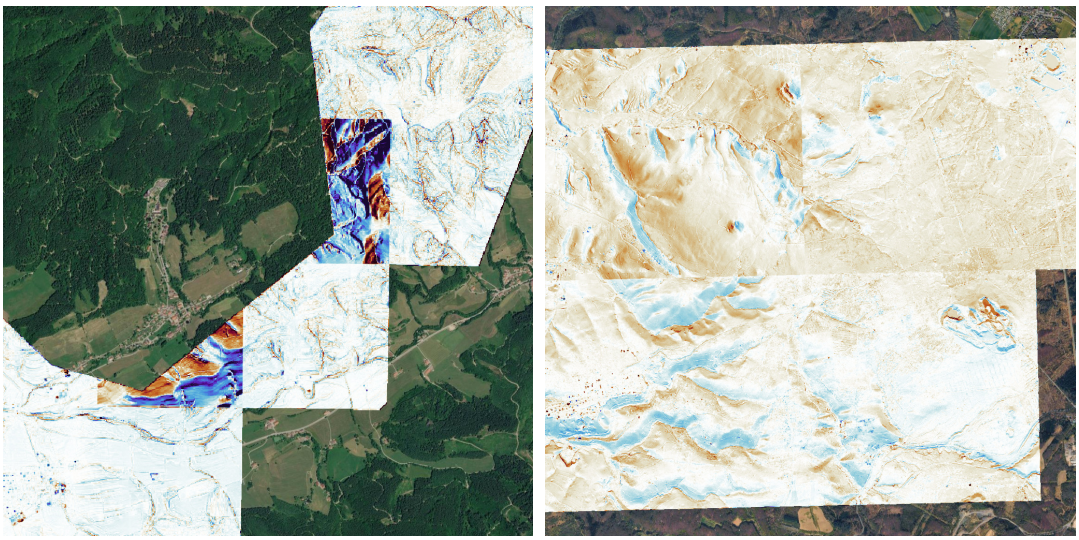


DE-NW/DE-RP border. DE-BW/FR border.

DE-RP/FR border.

Figure 30: Examples of lidar band-like striping artefacts in elevation difference maps. For all examples, the displayed elevation differences were restricted to values between −1 and 1 m to make the striping patterns more visible.

The second artefact in this category is singular tiles, or sections of tiles that have different vertical offsets compared to the rest of a national DTM. This appears as visible seam lines on the DEM difference raster. These differences, however, become less clear when looking at the national DTM alone. This happens for example at the DE-NW/DE-RP border, and more prominently at the DE-BW/DE-BY border (Figure 31). We speculate that this artefact is caused by (mis-)alignment of data between different surveys. Resolving this artefact is thus challenging without having access to source point clouds. The magnitude of this artefact has been inconsistent: in Figure 31 (a), the maximum difference is 2.6 metres; in Figure 31 (b) the difference between tile elevation steps across the seam is measured at 40 cm.



DE-BW/DE-BY border. DE-NW/DE-RP border.

Figure 31: Examples of visible tile seams in elevation difference maps caused by vertical elevation offsets between adjacent DTM tiles.

### 5.4.6 Post-Processing Artefacts

Post-processing artefacts refer to errors due to resampling DTM edges and providers' extrapolation of data to fill in missing data or use of fill values.

When a DTM is resampled in a different CRS and at different resolution, the cells at the edge may have limited number of valid sample points within them. As these errors are introduced by the pipeline itself, a possible solution would be to check the candidate grid points for resampling, and resample only if a certain condition is met, e.g. proximity of the candidate points to the new grid point. Within the current implementation of the pipeline, border harmonisation is achieved by clipping the DTMs to the national or regional extent. Therefore, these artefacts do not make it into the catchment area DTM as it is currently.

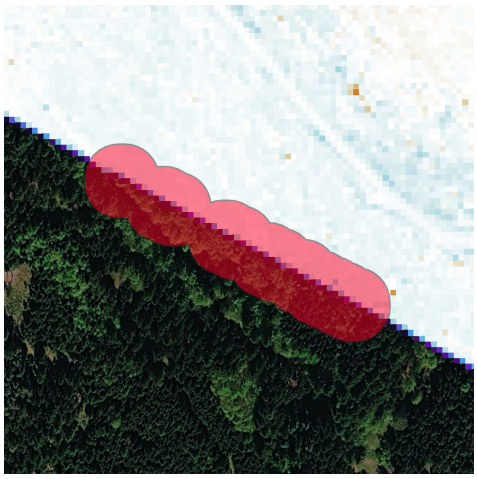
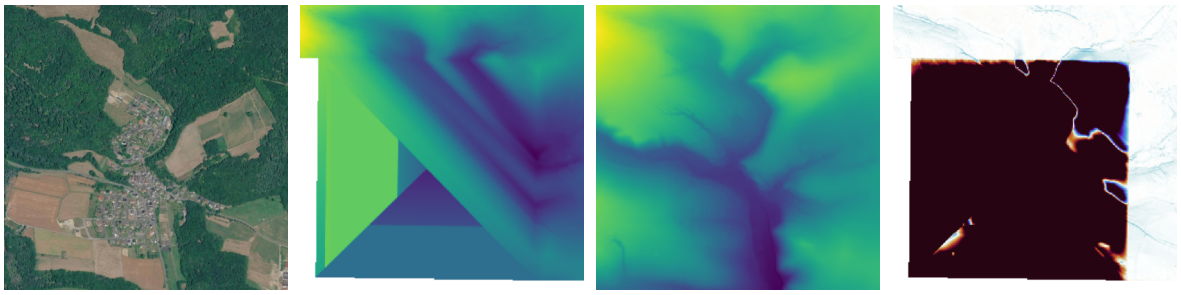


Figure 32: Example where a DTM edge difference exceeds the resolution threshold.

The second type of artefact are border regions where some type of interpolation has occurred to complete or modify the DTM. The severity of this artefact is greater in both magnitude and the area it covers. It is also difficult to distinguish the artefact from valid terrain. The first example can be seen at the DE-HE/DE-RP border, though its exact source is unknown. It appears in DE-HE's DTM and partly presents itself as linear interpolation on a TIN and partly contains flat fill values where the cells fall outside of the sample points' convex hull.



Satellite imagery. From Airbus, Maxar Technologies. (2026)

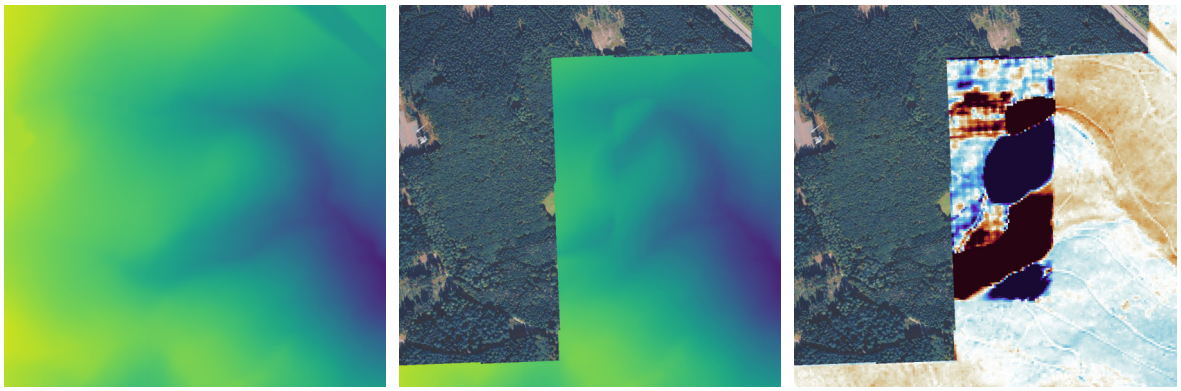
DTM of DE-HE.

DTM of DE-RP.

Elevation difference map.

Figure 33: Interpolation artefact found over an entire tile for the DE-HE DTM. The DE-RP DTM in the same location is also shown.

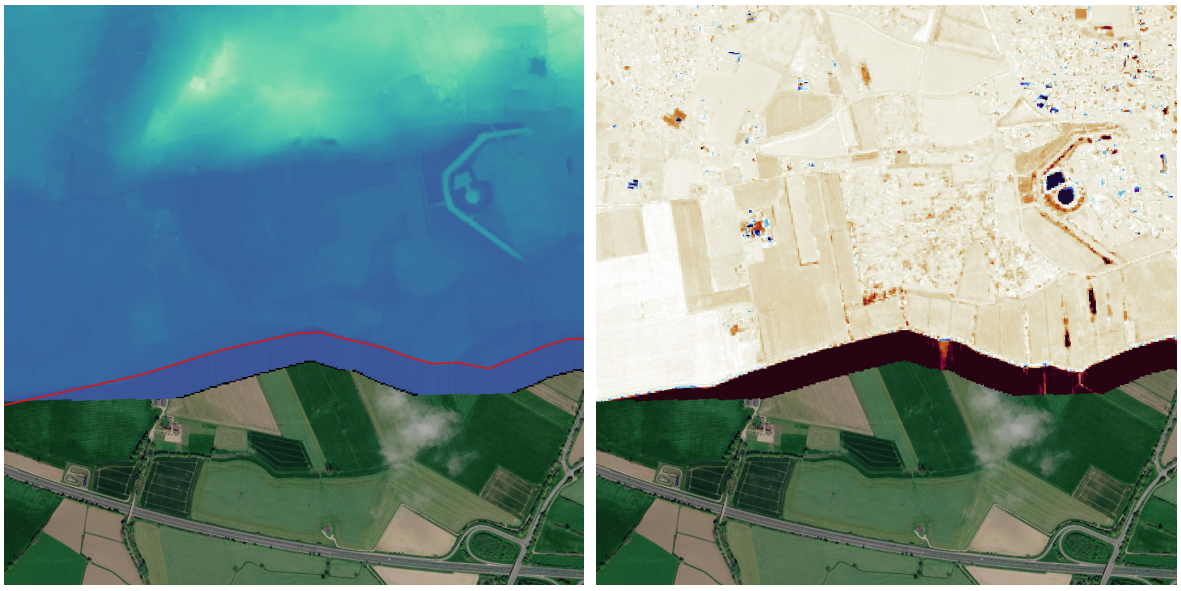
A different case occurs at the DE-RP/DE-SL border. As can be observed in Figure 34, the valley shape (in blue) does not line up between the two DTMs. Furthermore, the visuals appear "smeared" in the difference raster, pointing towards values interpolated e.g. using IDW. In contrast to the previous case, only part of the tile appears to be interpolated. However, the postprocessing at the border section has caused a disagreement of up to 16 m in this area.



DTM of DE-RP. DTM of DE-SL. Elevation difference map.

Figure 34: Large elevation differences caused by extrapolation artefacts. The artefacts are visible in the DE-SL DTM.

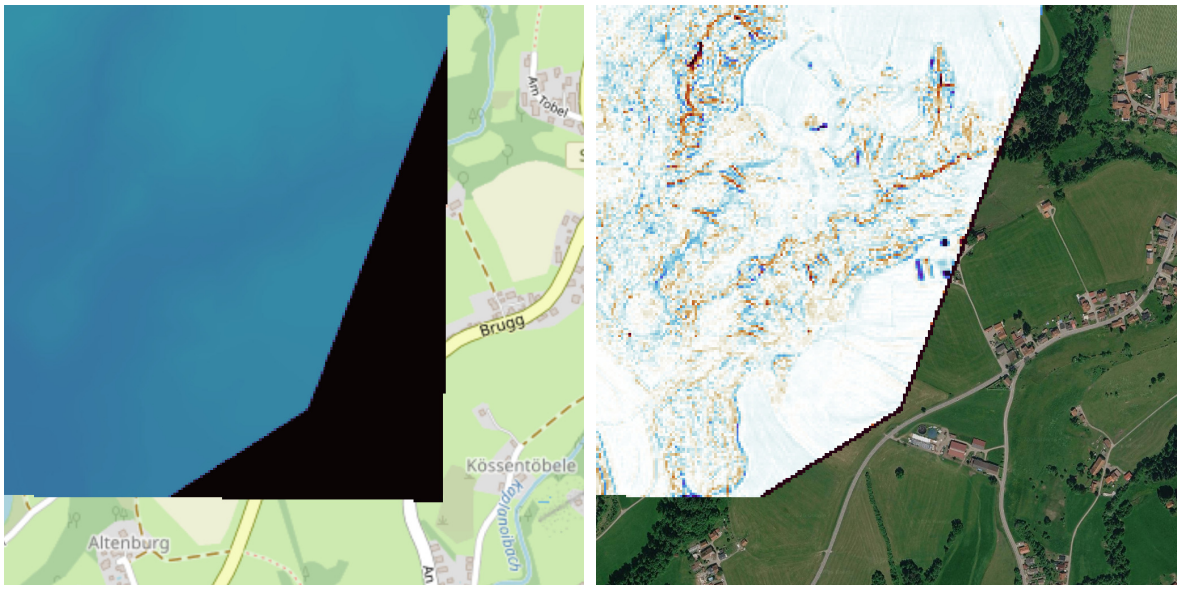
The third example of this type of artefact is one which was introduced due to our own interpolation. In an effort to interpolate the Dutch DTM `NoData` values we introduce extrapolation errors (Figure 35) where the border dataset we use does not match the edge of the Dutch DTM. We later solve this error by using a more accurate dataset for the borders. Furthermore, the availability of Actueel Hoogtebestand Nederland (AHN) version 6 will eliminate this issue as it no longer crops the data at the border.



DTM of NL's border with DE-NW. Elevation difference map.

Figure 35: Elevation differences introduced along NL/DE-NW border caused by interpolation by the authors.

Lastly, DE-BW fills an area at the edge of the dataset which is different from the `NoData` they use elsewhere. This results in a flat region of near-zero elevation and an elevation difference of up to 961 m. This again does not affect the quality of the final product as the error falls outside of the region's extent, but is another error on the part of the data provider.




DTM of DE-BW. Elevation difference map.

Figure 36: Edge resampling artefact caused by DE-BW using both a 0 and −9999 `NoData` value, while only declaring the −9999 value as `NoData`. The difference raster was set to −1 to 1 m to highlight the differences.

# 6 Discussion

This chapter contains the conclusions of the report and the recommendations. First, the limitations of the project are acknowledged. It continues by answering the main question of the project and by giving answers to the subquestions based on the contents of the report. It concludes by suggesting future work and giving recommendations for the use of the results of this report.

## 6.1 Limitations

The project has resulted in our successfully producing a harmonised DTM of the Rhine catchment area. Furthermore, the automated pipeline for processing and analysis has been completed and tested, resulting in the evaluation of data quality between bordering regions to determine the quality of the product.

However despite this success, there are limitations to the project. The first limitation is the scope of the project itself. Originally designed for a group of 4–6, it was completed by only three people over the course of 10 weeks. Within this time frame, they performed acquisition, harmonisation, and evaluation; carried out project management; and presented their results in this report and in talks. The small number of people essentially prevented multitasking — for instance, data acquisition and processing were paused at one point to allow the team to focus on presenting the results. As a result, several aspects of the project were excluded from its scope: these include re-extracting the DTM from raw point cloud data, automated artefact correction, and more extensive validation of the final product.

Several issues were anticipated within the risk analysis carried out at the start of the project (Appendix E.3). During week 6 of the project, the processing server was unavailable due to planned but unannounced maintenance. While coding and documentation continued, no testing or processing was possible. Furthermore, personal circumstances affected the availability of the team members during weeks 8 and 9. These events occurred during the final stages of analysis and reporting, leaving little opportunity to mitigate them. However, the impact of these events was not significant enough to completely stop the project.

As described above, the scope of the project was reduced to focus primarily on DTMs instead of point clouds. While it significantly reduced the total workload, it did prevent evaluating raw data instead of processed elevation data. The raw data should in theory contain fewer processing differences such as ground filtering or water handling. Furthermore, using a unified processing method would enable a better extraction of the water surface.

Another limitation is the reliance on external raster datasets. The quality of the end product is determined by the sum of its parts. Throughout the analysis it became clear that most artefacts are the result of temporal, spatial and processing differences between datasets, and not necessarily the processing pipeline. In several cases, large artefacts such as extrapolation data or misaligned tiles caused large differences between the elevation data. In other cases, the exact cause could not be established due to the lack of data about processing methodology, or raw point cloud data to evaluate the raw data for clues.

The water mask itself also introduces limitations. While not an originally intended element of the pipeline, the analysis proved that water was a consistent source of large disagreements and would need to be filtered out for any meaningful analysis to be carried out. However, it makes assumptions about water such as having small elevation changes, and requires a coarse, incomplete existing mask to function. So while the analysis benefits from the feature, it still needs refinement. It is currently also not used to correct the water surface in the final product.

Several limitations related to the border analysis itself also exist. The comparisons evaluate agreement based on the quality of the overlap between datasets instead of their absolute accuracy. Accounting for the dataset accuracy and combining it with the border differences would give a better idea of the accuracy of the DTM, however due to the lack of such metadata, it is not possible to determine this accuracy.

Finally, a number of artefact categories are based on qualitative rather than quantitative evidence. While in many cases, interpretation allows the cause of a difference to be determined based on facts and data, we haven't always been able to determine it exactly. Verifying whether the interpretations are correct would require processing of metadata, and classified point cloud data for all regions. Such data was not widely available, or within the scope and temporal constraints of the project.

## 6.2 Conclusion

During this project, an automated pipeline for producing a harmonised DTM of the Rhine catchment area was developed. The DTM of most of the Rhine catchment area was successfully produced by executing the pipeline. The DTM is at 5 m resolution, its CRS is ETRS89-LAEA + EVRF2007; and it has an omission rate of 2.67%. The DTM is near-seamless, except for water surface and small parts of land border.

Within the EU, the INSPIRE directive informs the requirements to several themes of geographical information, including elevation. For land-elevation, INSPIRE Specification on Elevation requires that gridded (i.e. raster) data is available in either GeoTIFF or GML format. It requires that ETRS89-derived CRS is used, recommending geodetic ETRS89. The specification further establishes the requirements for data quality: the ones we consider expressly relevant are commission rate, omission rate, and the relationship between GSD and vertical RMSE, where $3 \leq \frac{\text{GSD}}{\text{RMSE}} \leq 20$.

Several definitions of the catchment area were considered. Within the project, we are using the catchment as defined by HydroSHEDS through performing runoff modelling on a global elevation dataset. This catchment definition includes the main body of the Rhine but excludes the Maas and parts of the delta such as IJssel. It covers parts of Italy, Austria, Liechtenstein, Switzerland, Germany, France, Luxembourg, Belgium, and the Netherlands.

Existing approaches to creating cross-border DEMs were also reviewed. Mapterhorn combines heterogeneous — global, national, and regional — sources into a single DEM. However, Mapterhorn's current harmonisation algorithm prefers smoothness to accuracy. Alternative cross-border products are global elevation models like Copernicus DEM which is only available at low (30 m) resolution.

Alongside global elevation models, national and regional datasets were investigated. Switzerland and Liechtenstein (jointly), France, Luxembourg, and the Netherlands each collect and publish elevation data. Within Germany, each region collects and publishes their own elevation data. To our knowledge, there is no elevation product for regions of Italy and Austria falling within the catchment area. Similarly, we did not succeed at acquiring the elevation data for Belgium.

The datasets are available at a range of resolutions from 0.5 m to 5 m several providers publish both a high- and a low-resolution version. For different products, the data has a different horizontal and vertical CRS; different `NoData` values; different approaches to terrain filtering and interpolation are used. Several providers do not disclose, or incorrectly disclose, the `NoData` values and CRSs of their datasets. Parts of the datasets have also been acquired at different times, resulting in differences in elevation due to changes in terrain, such as earthworks or construction. Each of these results in an extra need to perform data discovery and either an additional processing step — such as CRS transformation or harmonising `NoData` values — or a difference in elevation data at the border.

We find that the pipeline is sensitive to the specifics of the input (meta-)data, such as correct CRS or `NoData` values. Several of the datasets feature incorrect or incomplete information which warrants additional data discovery and manually specifying this information for each dataset. The pipeline currently does not correct disagreements between the datasets due to the lack of a robust way to evaluate which of the datasets is "correct" in each given case. It also has been designed assuming all data within a given region has the same CRS and `NoData` values. While this assumption holds for the regions we have investigated, the opposite would require further pre-processing of the data.

To test the accuracy of the DEM, we analyse the errors between the overlapping DTM products. We observe several kinds of errors when comparing bordering datasets. These can be categorised into interpolation errors, ground filtering errors, errors due to temporal differences, water mask errors, and errors due to pre- and post-processing (here listed from most to least common). The majority of the datasets demonstrate good agreement as illustrated by low RMSE and LE90, with errors largely caused by changes in elevation over time, often observed in quarries. Of the 17 evaluated border pairs, 15 meet the INSPIRE guidance for vertical accuracy, with FR/LU and DE-BW/DE-BY exceeding it. DE-HE/DE-RP and DE-BY/DE-BW border pairs have higher RMSE due to errors with interpolation and `NoData` values — however, these areas fall outside of the extent of their corresponding region and therefore are not present in the product. The French national elevation product appears to be using a global interpolation function on some of the terrain which also results in high error rate and does have an effect on the quality of the product. Based on border region comparison alone, it is impossible to evaluate the exact magnitude of these errors.

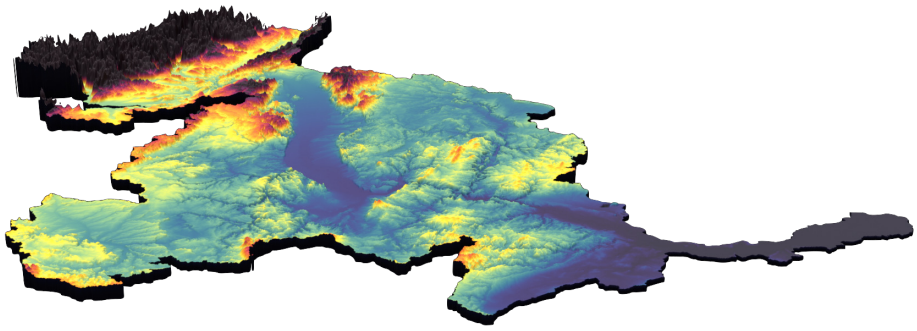


Figure 37: 3D Render of the Rhine Catchment DTM. Elevation is exaggerated by a factor of 10.

## 6.3 Recommendations

### 6.3.1 Recommendations on Standardisation

Currently, INSPIRE specification on elevation is a 260-page document, encompassing everything from the basic definition of a terrain to implementations of the GML schema, with a particular focus on the latter. We find that the specification is quite limited, focusing on the interoperability only on a very basic level (essentially, that all data follows a schema which can be read in an automated way) without much regard for the congruity between the datasets themselves. Currently, the specification is not fully followed: while most elevation datasets appear to be INSPIRE-compliant, France publishes its data as ASCII grid which is expressly disallowed (TWG-EL, 2024, p. 166).

However, this guidance is limited to one line within the specification, presented as a note rather than requirement, which makes it easy to ignore. We would encourage that the specification is reviewed and rewritten in a way that enables easier implementation, e.g. by including a catalogue of all requirements, recommendations, and their scope.

To further ensure better cross-border interoperability, we recommend the following:

- Data providers to adopt a unified way to access and download the data. We find WCS (as adopted by the Netherlands and DE-NW) particularly convenient and allows for automatic server-side querying of a particular geographic extent. Alternatively, we can also recommend providing tiled GeoTIFFs with a spatial catalogue of tiles or COGs.
- Data providers to adopt a unified `NoData` value, and expressly include it in the elevation data.
- Data providers to adopt a unified way to extract a DTM from a surface model.
  - Based on the work by Cai *et al.* (2023), we suggest ICSF will likely perform best for ground filtering but further research is needed.
  - A standardised interpolation method for filling gaps in the ground points is also needed, as we see artefacts introduced by linear interpolation on a TIN or extracting data from CSF points. We believe that RBF would produce the best results deterministically, but further comparison of the different interpolation methods is needed.
  - We also recommend that the river surface is extracted with a hydro-flattening approach based on the elevation of points classified as water given the errors introduced by interpolation on a TIN or the need for post-processing if the water surface is excluded.
- Supplementary data on dataset quality to be provided. This data can include masks showing the interpolated areas, the water surface, the time of acquisition of each given elevation sample, or the density of (ground) points within each grid cell. This data can support integrating elevation data from different providers.

### 6.3.2 Further Work

There are several areas of the project which warrant improvement.

The evaluation of the results within this report focused on the agreement between national datasets due to absence of robust ground truth. We recommend that such a ground truth be obtained and results be compared against it. One such approach, as outlined in Section 3.3.1, is evaluating the DTM against the national geodetic control networks.

Although runoff modelling is the key use case of the DTM produced within the project, the elevation raster is currently not optimised for it, and no runoff modelling was performed. We suggest that such optimisation and modelling can be performed for the given cross-border DTM.

The pipeline still struggles with some `NoData` values as seen with DE-BW elevation dataset. While a small change, a more robust way to handle `NoData` even where it is not encoded into the dataset is needed. The river mask is rather basic in its functionality and currently misses sections of the river; work can continue to develop a more robust water detection algorithm which can better handle the different methods used to extract the water surface.

Within this project, the differences between the different DTMs are only analysed but not resolved. We recommend that ways to detect and automatically fix those differences are investigated, based either on the raster data or a combination of raster and point cloud data.

A more ambitious but worthwhile version of this project would involve a re-extraction of the DTM from the national and regional point cloud data, including re-classification and performing hydro-flattening. The outcomes of such a project can more specifically inform the recommendations listed in Section 6.3.1.

As an alternative to hydro-flattening, the bathymetric data can be introduced into the rivers. Such data can either be synthetic, as proposed by Dysarz (2018), or integrate existing river- and lakebed surveys with aerial lidar data.

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# A Work Allocation & AI Statement

RV is the primary author of the abstract. AB is the primary author of the introduction. AB, AK, and RV have collaborated on writing the project definition.

AB, AK, and RV have each performed literature review. In the report, AK describes the INSPIRE specification on elevation (reading which was one of the hardest — and surprisingly pointless — things he's ever done), global elevation models, and the Rhine catchment area definition. RV and AK both contribute to describing point cloud processing, interpolation, and river elevation.

AB and AK have both performed data discovery. AB, AK, and RV have collectively designed the processing pipeline. AB has implemented data acquisition for all national and regional datasets alongside initial data harmonisation. RV has implemented the river mask generation and the analysis. AK has integrated all parts of the pipeline to produce the final DTM. AK describes the pipeline in the report; RV and AK both contribute to the description of the river mask. AB and RV describe the evaluation and visualisation.

AB describes the results of the data acquisition. AK describes the results of running the pipeline and the resulting aggregate DEM. AB, AK, and RV have all performed the analysis of border regions, each focusing on a different set of border pairs. AB, AK, and RV present the aggregated description of issues found between bordering DTMs.

AK presents the conclusions and recommendations. RV describes limitations. AB ensured the consistency and correctness of all figure captions in the report.

AK has led a (highly opinionated) edit of the report prior to its publication. AK has also written this section.

AB, AK, and RV have all used generative AI over the course of the project. AB has used ChatGPT 6 Astra to aid with production of the code for data acquisition; help understand code produced by other group members; and as a proof-reading/editing aid. AK has used Claude Opus 5.5 to integrate different aspects of the pipeline — AK is not sure he wrote a single line of code by hand during this project. Reluctantly, he also used AI for proof-reading and editing — tightly controlling the scope of those edits. RV has used Claude Opus 5.5. to perform bug-fixing and implement a small set of features in his code. He has also used Microsoft Copilot as a proof-reading/editing aid.

# B Data and Code

The code is available on GitHub at https://github.com/a-kurski/synthesis2026-eudem.

The data is available from national providers, and can be acquired by executing corresponding parts of the code. The DTM, river mask, and analysis can be fully reproduced by executing the processing and analysis parts of the pipeline.

# C Pipeline README

Merges digital terrain models from several regions (countries, states, provinces, …) into one seamless DTM in a common horizontal and vertical reference system, and compares the regions where they overlap.

Each region can come with its own tile format, CRS, height system, `NoData` convention and gaps. All of these differences are described in one JSON file; none of them is hardcoded. To process a different set of regions you write a new JSON, not new code.

Downloading data and point-cloud processing are out of scope: the pipeline starts from raster tiles that are already on disk.

## C.1 Contents


- How it works
- Installation
- Quick start
- Configuration
- Outputs
- Comparison
- Running large jobs
- Tests
- Code layout
- Limitations


## C.2 How it works

`main.py` runs five stages in order. Each stage reads the JSON, does its work and writes what it produced (file paths) back into the same JSON. A rerun skips everything that's already recorded and exists on disk, so an interrupted run resumes where it stopped.

1. **`vectors`**: buffers every region outline by `buffer_m` and records which regions intersect.
2. **`rasters`**: for each region, builds a virtual mosaic of its tiles and warps it in one step to the target CRS, height system and resolution. Each region's DEM keeps a `buffer_m` margin beyond its outline, so neighbouring DEMs overlap. Optional extras per region: subtracting a geoid for ellipsoidal heights, gap filling (with a mask of the filled cells) and cropping to the outline.
3. **`rivers`**: for each region, grows a water mask over the DEM, starting from seed cells taken from a water raster. The comparison uses these masks to leave water surfaces out of the statistics.
4. **`overlaps`**: where two regions overlap and one region's cell was interpolated (gap-filled) while the other has a measured value, the measured value replaces it. Nothing else is changed: no averaging, no feathering.
5. **`assemble`**: writes each region's DEM inside its own outline, applies the overrides from the previous stage, masks everything outside the overall extent and writes the result as a Cloud-Optimized GeoTIFF. It also stitches the regions' river masks the same way, each inside its own outline, into one binary river mask on the same grid (1 = water, 0 = land, 255 = `NoData`).

The comparison (`python -m dtm.comparison`) is a separate step that runs on the region DEMs after the pipeline.

## C.3 Installation

You need Python 3.10 or newer and GDAL 3.8 or newer with its Python bindings. Install GDAL from conda-forge or your system packages; `pip install gdal` usually fails or gives a GDAL without the

drivers you need. If your tiles use LERC compression, the GDAL build must support LERC; conda-forge's does.

With micromamba or conda:

```
micromamba create -n dtm -c conda-forge python gdal numpy scipy tqdm \
    xdem geoutils rasterio numba geopandas pandas matplotlib
micromamba activate dtm
pip install scikit-gstat        # used by xdem's uncertainty estimate; not on
conda-forge
```

The `vectors`, `rasters`, `overlaps` and `assemble` stages only need GDAL, numpy, scipy and tqdm. The `rivers` stage and the comparison also need the xdem stack (see `requirements.txt`).

Check your install by running the tests (see Tests).

### C.3.1 Height transformations and PROJ grids

Height system changes are done by PROJ, and many of them need grid files. Either install the grids you need (`projsync --source-id <agency>`) or let PROJ download them on demand:

```
export PROJ_NETWORK=ON
```

If PROJ only knows a "ballpark" transformation between two height systems, it leaves heights unchanged. The pipeline then logs a warning for that region (`PROJ has no EPSG:a -> EPSG:b transformation here; heights are NOT shifted`). You can see which transformations PROJ would use with:

```
projinfo -s EPSG:<source> -t EPSG:<target> --spatial-test intersects
```

## C.4 Quick start

1. Put each region's outline (a polygon in any vector format GDAL reads) and its DEM tiles somewhere on disk.
2. Write a JSON file. A minimal example with two regions:

```
{
  "settings": {
    "crs": 3035,
    "vertical_crs": 5621,
    "resolution": 5,
    "extent": null,
    "output_dir": "output"
  },
  "regions": {
    "region_a": {
      "vector": "outlines/region_a.gpkg",
      "raster": "data/region_a/**/*.tif",
      "crs": 25832,
      "vertical_crs": 7837,
      "interpolate": false
    },
    "region_b": {
      "vector": "outlines/region_b.gpkg",
      "raster": "data/region_b/*.asc",
      "crs": 2154,
      "vertical_crs": 5720,
      "interpolate": true
    }
```

```
    }
  }
```

3. Run it:

```
python main.py my_config.json
```

The merged DTM is written to `output/merged_dtm.tif`. Paths in the JSON are relative to the JSON file itself, not to the folder you run from.

Other ways to run it:

```
python main.py my_config.json --stages rasters overlaps assemble   # only some
stages
python -m dtm.river_mask my_config.json                            # river masks
only
python -m dtm.comparison my_config.json                            # comparison,
after the pipeline
python main.py my_config.json && python -m dtm.comparison my_config.json
```

## C.5 Configuration

The JSON has two parts you write, `settings` and `regions`. The pipeline adds more entries as it runs (see State written by the pipeline).

### C.5.1 `settings`

| Key | Meaning | Default |
|---|---|---|
| `crs` | EPSG code of the output horizontal CRS. | required |
| `vertical_crs` | EPSG code of the output height system. `null` turns off all height transformations. | `null` |
| `resolution` | Output cell size, in units of `crs`. | required |
| `extent` | Polygon of the area to cover; the final DTM is masked to it. `null` uses the union of all region outlines. | `null` |
| `output_dir` | Folder for all outputs. | `data/pipeline` |
| `product_name` | File name of the merged DTM. | `merged_dtm.tif` |
| `buffer_m` | Buffer around each outline: used for the intersection test and as the margin of each region's DEM. | `1000` |
| `NoData` | `NoData` value of all outputs. | `-9999` |
| `max_vertical_shift_m` | The pipeline logs the height shift PROJ applies at each region's centre, and warns if it is larger than this. A shift of tens of metres usually means ellipsoidal and physical heights are being mixed. | `10` |
| `interpolation_mask` | Optional polygon; gap filling is limited to the part of each region inside it. | `null` |
| `borders` | Optional border lines, used for the distance-to-border analyses in the comparison. | none |
| `skip_missing` | `true`: regions with a missing outline, missing tiles or a missing geoid are skipped with a warning. `false`: the run stops. | `false` |
| `threads` | Total number of threads, or `"ALL_CPUS"`. | `"ALL_CPUS"` |

| Key | Meaning | Default |
|---|---|---|
| `region_workers` | Number of regions warped at the same time; `threads` is split evenly between them. | `1` |
| `warp_memory_mb` | Memory per warp, in MB. | `2048` |
| `hole_filling` | Gap-filling settings, see below. | see below |
| `river_growing` | River-mask settings, see below. Each region can override these. | see below |
| `comparison` | Comparison settings, see Comparison. | see there |
| `qgis_root` | Windows only: path to a QGIS/OSGeo4W installation whose Python and GDAL should be used. Leave `null` otherwise. | `null` |

`hole_filling` (used for regions with `"interpolate": true`):

| Key | Meaning | Default |
|---|---|---|
| `method` | `"gdal_idw"`, inverse-distance weighting with GDAL's FillNodata. The only method available. | `"gdal_idw"` |
| `selection` | `"all_aoi_NoData"`: every `NoData` cell inside the region outline is a candidate. The only option available. | `"all_aoi_NoData"` |
| `max_distance_m` | How far to search for valid cells. Gaps wider than about twice this distance stay `NoData`. | `100` |
| `nearest_fallback` | Fill whatever is left with the nearest valid value. Very slow on large regions with big empty areas. | `true` |
| `smoothing_iterations` | Must be `0`. | `0` |

`river_growing`:

| Key | Meaning | Default |
|---|---|---|
| `seed_values` | Values in the region's water raster (`river_seeds`) that mark water. | `[2, 3]` |
| `tolerance_m` | Largest height step between neighbouring cells of the same water body. Growth stops at larger steps. | `0.02` |
| `seed_buffer_cells` | Dilate the seed cells by this many DEM cells before growing. | `0` |

### C.5.2 `regions`

One entry per region; the key is the region's name, used in file names and logs.

| Key | Meaning | Default |
|---|---|---|
| `vector` | Region outline (polygon). The region's data is used inside it. | required |
| `raster` | Path or glob pattern for the source tiles (** searches subfolders). Any format GDAL reads, e.g. GeoTIFF or ASCII grid. | required |
| `crs` | EPSG code of the tiles' horizontal CRS. It's set explicitly because tiles often lack a CRS, or spell the same one differently. | required |
| `vertical_crs` | EPSG code of the tiles' height system. If `settings.vertical_crs` is set as well, heights are transformed by PROJ. | none |

| Key | Meaning | Default |
|---|---|---|
| `geoid` | Geoid raster: geoid heights N in metres, any CRS. When set, the source heights are taken as ellipsoidal and converted as H = h − N after the warp; `vertical_crs` is then informational only (e.g. `4937` for ETRS89). | none |
| `interpolate` | Fill gaps in this region's DEM and record which cells were filled. | `false` |
| `crop` | Cut the DEM to the outline. If `false`, the DEM keeps its buffer and neighbours overlap; the merge still uses each region only inside its outline. | `false` |
| `resampling` | GDAL resampling method used by the warp. | `"average"` |
| `source_NoData` | A value the tiles use for missing data without declaring it as `NoData` (e.g. `0`). These cells are then ignored by the warp, including its averaging. Overrides any `NoData` value the tiles declare. | none |
| `river_seeds` | Water raster for this region (any CRS). Cells with one of the `seed_values` are seeds. Without it, the region gets no river mask. | none |
| `river_growing` | Overrides for `settings.river_growing`, for this region only. | none |
| `point_clouds` | Ignored (point clouds are out of scope). | ~— |

How the source data is handled:

- **Tiles without a `NoData` value**: gaps between tiles are given `settings.NoData`, so they're treated as gaps rather than read as 0 m. If the tiles themselves fill missing areas with a value such as 0, set `source_NoData`. Otherwise those cells come out as near-0 heights, shifted by the height transformation, and are averaged into neighbouring cells.
- **Tiles in different but equivalent CRS definitions** are mosaicked together. The region's `crs` is used for all of them.
- **Height systems with a ballpark transformation only** are left unchanged, with a warning (see PROJ grids).
- **Ellipsoidal heights**: give a `geoid`. Any geoid model works, for example the global EGM2008 (`curl -fL -o geoids/egm08.tif https://cdn.proj.org/us_nga_egm08_25.tif`). A regional geoid that matches your target height system is more accurate.

### C.5.3 State written by the pipeline

| Entry | Written by | Content |
|---|---|---|
| `regions.<name>.intersections` | `vectors` | Names of the intersecting regions. |
| `overlaps.<a>__<b>` | `vectors`, `overlaps` | Overlap polygon and override raster. |
| `regions.<name>.dem` | `rasters` | Processed DEM of the region. |
| `regions.<name>.fill_mask` | `rasters` | Mask of the gap-filled cells, or `null`. |
| `regions.<name>.river_mask` | `rivers` | River mask of the region. |
| `product` | `assemble` | Merged DTM. |
| `river_product` | `assemble` | Merged binary river mask. |
| `comparisons.<a>__<b>` | comparison | Statistics CSV of a finished pair. |

Don't edit these by hand. To redo a region, delete its folder in `output_dir` (or its `dem` entry) and rerun. To redo a comparison pair, delete its `comparisons` entry or its statistics CSV.

## C.6 Outputs

Everything is written under `settings.output_dir`:

```
<output_dir>/
├── <product_name>                      merged DTM (Cloud-Optimized GeoTIFF)
├── <product name>_river_mask.tif       merged river mask, Byte COG: 1 = water, 0 =
land, 255 = `NoData`
├── extent.gpkg                         area covered by the merged DTM
├── pipeline.log                        log of all runs, appended
├── intersections/<a>__<b>.gpkg         buffered overlap of two regions
├── <region>/
│   ├── tiles.vrt                       virtual mosaic of the source tiles
│   ├── dem.tif                         region DEM in the target CRS/height system,
with buffer
│   ├── hole_filling/                   filled DEM and fill mask (regions with
interpolate: true)
│   └── *_river_mask.tif                0 = no water, 1..n = water body id
├── overlaps/<a>__<b>/merged.tif        measured-over-interpolated overrides (NaN
elsewhere)
└── comparison/                         see Comparison
```

## C.7 Comparison

`python -m dtm.comparison my_config.json` compares every pair of region DEMs whose extents overlap. Finished pairs are recorded in the JSON (`comparisons`) and skipped on the next run, as long as their statistics file exists and is newer than both DEMs, so an interrupted comparison resumes and a reprocessed region is compared again. A pair is recognised in either order (`a__b` or `b__a`). The summary table always lists all finished pairs. With `comparison.workers` above 1, several pairs run at once, each in its own process. Their log lines are prefixed with the pair name, e.g. `[a__b]`. It uses the region DEMs with their buffers, not the merged DTM, so it sees both datasets on either side of a border. For each pair it:

- excludes water cells (both regions' river masks, dilated by `river_buffer_cells`);
- computes the raw difference `dh = first − second`, and the difference after Nuth and Kääb coregistration (a horizontal and vertical shift only);
- computes statistics (mean, median, standard deviation, RMSE, MAE, NMAD, LE90, min, max, cells above the threshold), with and without the gap-filled cells;
- flags cells where |dh| exceeds `resolution × threshold_fraction` and groups them into problem regions. Gap-filled cells are left out of both, since measured data replaces them where the neighbour has it;
- if `settings.borders` is set, plots the error against distance to the border;
- estimates the random error with xdem's uncertainty workflow, on stable terrain (no water, no gap-filled cells) after coregistration:
  1. **Heteroscedasticity:** the spread (NMAD) of the differences is binned by terrain attributes (slope and maximum curvature by default), which gives an error map for every cell.
  2. **Spatial correlation:** a variogram of the standardized differences, fitted with a sum of models; each model gives one correlation range.
  3. **Propagation:** the error of the mean difference is the mean error divided by √(number of effective samples). It's reported in each statistics row as `mean_dh_sigma_m`, together with `n_effective`, `mean_sigma_dh_m` and `error_correlation_1km`. The overlap area is treated as a disk, which is conservative for thin border strips (fewer effective samples, so a larger error).

`settings.comparison`:

| Key | Meaning | Default |
|---|---|---|
| `threshold_fraction` | Threshold for problem cells, as a fraction of the resolution. The default, resolution / 3, follows the INSPIRE recommendation. | `0.333…` |
| `river_buffer_cells` | Dilation of the river masks before excluding them. | `1` |
| `max_distance_m` | Range of the distance plots (error correlation, distance to border). | `2000` |
| `border_strip_m` | Width of the distance-to-border strips. | `50` |
| `local_rmse_tile_m` | Tile size of the local RMSE raster. | `50` |
| `problem_min_area_m2` | Smallest problem region kept. | `200` |
| `problem_merge_distance_m` | Problem cells closer together than this are merged into one region. | `60` |
| `uncertainty` | Run the uncertainty analysis. It's the slowest part, so `false` skips it. | `true` |
| `precision_of_other` | `"same"`: both DEMs are equally precise, so each DEM's error is the difference's error / √2. `"finer"`: the first DEM is much more precise, so all of the error is assigned to the second. | `"same"` |
| `uncertainty_variables` | Terrain attributes used to explain how the error varies (any xdem terrain attribute). | `["slope", "max_curvature"]` |
| `variogram_models` | Variogram models summed to model the spatial correlation. | `["Gaussian", "Spherical"]` |
| `workers` | Number of pairs compared at the same time, each in its own process. Memory is the limit, see Running large jobs. | `1` |

Outputs, in `<output_dir>/comparison/`:

```
comparison/
├── dem_difference_statistics.csv          all pairs in one table
└── <a>__<b>/
    ├── statistics_<a>_<b>.csv             before/after coregistration, with/
without filled cells
    ├── dh_<a>_<b>.tif                     raw difference
    ├── exceeds_threshold_<a>_<b>.tif      1 = above threshold, 0 = within, 255 =
no data, water or gap-filled
    ├── problem_regions_<a>_<b>.gpkg
    ├── border_strips_<a>_<b>.csv, border_effect_<a>_<b>.png
```

```
    ├── histogram_*.png, cdf_*.png, slope_rmse_*.png
    ├── uncertainty_<a>_<b>.tif/.png         error map (1σ per DEM)
    ├── heteroscedasticity_<var>_<a>_<b>.png  error vs terrain attribute
    ├── variogram_<a>_<b>.png, Correlation_<a>_<b>.png   spatial correlation of
the error
    └── local_inspire_rmse_*.tif, dh_std_*.tif, aspect_difference_*.tif
```

The Nuth and Kääb shift in the log is a quick diagnostic. A few centimetres is normal. A large vertical shift points to a wrong height system, for example ellipsoidal heights declared as orthometric. A consistent horizontal shift points to a georeferencing problem in one of the datasets.

## C.8 Running large jobs

- **Memory:** each warp uses up to `warp_memory_mb`, and there are `region_workers` warps at a time. River masks are built one region at a time but hold the whole region DEM in memory: roughly 25–30 bytes per cell at peak, so 75–90 GB for a 3-billion-cell region.
- **Comparison:** each pair loads the overlap of the two DEMs' bounding boxes and holds several copies of it in memory (several GB to a few tens of GB for large regions). Raise `comparison.workers` only as far as memory allows. Each pair uses about one core, so more workers mostly cost memory, not CPU.
- **Threads:** `threads` is the total budget. For many medium-sized regions, several `region_workers` with fewer threads each usually finish sooner than one worker with all threads.
- **Long runs:** use `screen` or `tmux`. Keep a copy of the output with `2>&1 | tee run.log`. With `set -o pipefail`, the exit code still reflects failures.
- **Missing data:** with `"skip_missing": true`, incomplete regions are skipped and retried on the next run, so you can start before all data is in.

## C.9 Tests

```
python -m unittest discover tests
```

The tests run offline on small synthetic rasters. They cover the merge rules, a two-region run end to end, skipping and retrying missing regions, geoid conversion, river masks, the merged river mask and the comparison statistics. The river-mask and comparison tests are skipped if xdem isn't installed.

## C.10 Code layout

```
main.py                  entry point (python main.py config.json)
dtm/
├── pipeline.py          the five stages and the JSON handling
├── river_mask.py        river masks (runnable on its own)
├── comparison.py        pairwise comparison (runnable on its own)
├── hole_filling.py      gap filling
├── process.py           raster helpers, COG writer
├── geo.py               vector/raster helpers
├── raster_io.py         xdem/rasterio I/O for comparison and river masks
├── progress.py          progress logging
└── runtime.py           optional QGIS environment on Windows
tests/test_pipeline.py
```

## C.11 Limitations

- Downloads and point clouds aren't handled.
- Height transformations are only as good as what PROJ provides. For a missing grid, PROJ either fails or makes no change at all (with a warning in the log). Check the warnings in `pipeline.log`.

- Overlaps are resolved only where one side is interpolated. Disagreements between two measured datasets are reported by the comparison, but not corrected.
- River growing follows flat water surfaces using `tolerance_m`. Steep or stepped water bodies, such as weirs and rapids, may be captured only in part.
- See KNOWN_ISSUES.md for findings and open issues from the Rhine-basin runs.

# D Contributors

This section provides an overview of the student members of the team and their roles, as well as the clients and the supervision team.

## D.1 Team Members

**Arda Baysal**
a.baysal@student.tudelft.nl

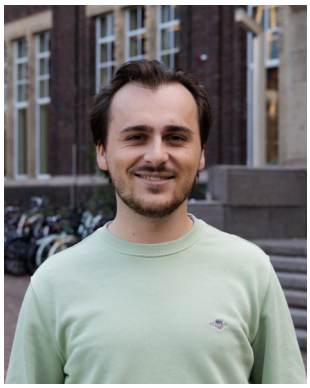
Background:
Turkey
BSc in Electrical Engineering, Eindhoven University of Technology, The Netherlands

Interests: Point clouds, geospatial data visualisation, digital terrain modelling

**Artemi Kurski**
a.kurski@student.tudelft.nl

Background:
Estonia
BSc in Architecture, University of Bath, United Kingdom

Interests: Computational modelling of terrains and the built environment



**Ruben Vons**
r.m.b.vons@student.tudelft.nl

Background:
The Netherlands
BSc in Aeronautical Engineering, Inholland University of Applied Sciences, The Netherlands

Interests: Solving programming challenges and creating nice visuals.

## D.2 Supervision Team

The supervision team for this project consists of Gina Stavropoulou, representing Delft University of Technology's 3D Geoinformation Group, Maarten Pronk, representing Deltares, and Daan van der Heide, representing Rijkswaterstaat; Maarten Pronk and Daan van der Heide are also working toward obtaining their PhD in the 3D Geoinformation Group. The supervisors provide feedback on the team's work, help define the scope and desired outcomes, and share their experience with existing approaches to the challenge.

## D.3 Clients

Rijkswaterstaat is the Netherlands' government agency for infrastructure and water management. Originally created in 1798 for flood prevention, it now also oversees the construction and maintenance of national infrastructure (Rijkswaterstaat, no date). Some of the notable projects completed by the agency include the Afsluitdijk, built in the interwar period, and the Delta Works, the construction of which finished in 1997. Flooding remains a key priority for Rijkswaterstaat.

Deltares is an independent research institute specialising in water and subsurface research (Deltares, no date). Its key mission is 'Enabling Delta Life', which encompasses five key areas focusing on health, safety, and sustainability of life in river deltas. Creating a more accurate river model contributes to several of these areas, in particular "Safer from flooding" and "Healthy water systems".

# E Project Planning

## E.1 Prioritisation

| Req ID | Description | MoSCoW | Status |
|---|---|---|---|
| DT-01 | Global / EU DEM | Must | Done |
| DT-02 | Rhine Watershed Mask | Must | Done |
| DT-03 | Rhine Bathymetry | Will not have | Not done |
| DT-04 | National/regional PC | Must | Not done |
| DT-05 | National/regional DTM | Must | Done |
| DT-06 | National/regional DSM | Should | Not done |
| DT-07 | National/regional topographic map | Could | Not done |
| CT-01 | NL is included | Must | Done |
| CT-02 | DE is included | Must | Done |
| CT-03 | Belgium is included | Should | Not done |
| CT-04 | CH is included | Must | Done |
| CT-05 | FR is included | Must | Done |
| CT-06 | LU is included | Must | Done |
| CT-07 | Austria is included | Should | Not done |
| CT-08 | LI is included | Must | Done |
| CT-09 | Italy is included | Will not have | Not done |
| MP-01 | DTM raster | Must | Done |
| MP-02 | DSM raster | Should | Not done |
| MP-04 | Auxiliary masks and maps | Should | Done |
| TC-01 | Workflow is fully automated | Must | Done |
| TC-02 | Workflow is efficient | Should | Done |
| TC-03 | Workflow is reproducible | Must | Done |
| RP-01 | Report on findings | Must | Done |
| RP-02 | Report details the workflow | Must | Done |
| RP-03 | Report details the issues with cross-border data | Must | Done |
| RP-04 | Report details the limitations of the workflow | Must | Done |

Table 8: Prioritisation of data requirements using MoSCoW.

## E.2 Timeline

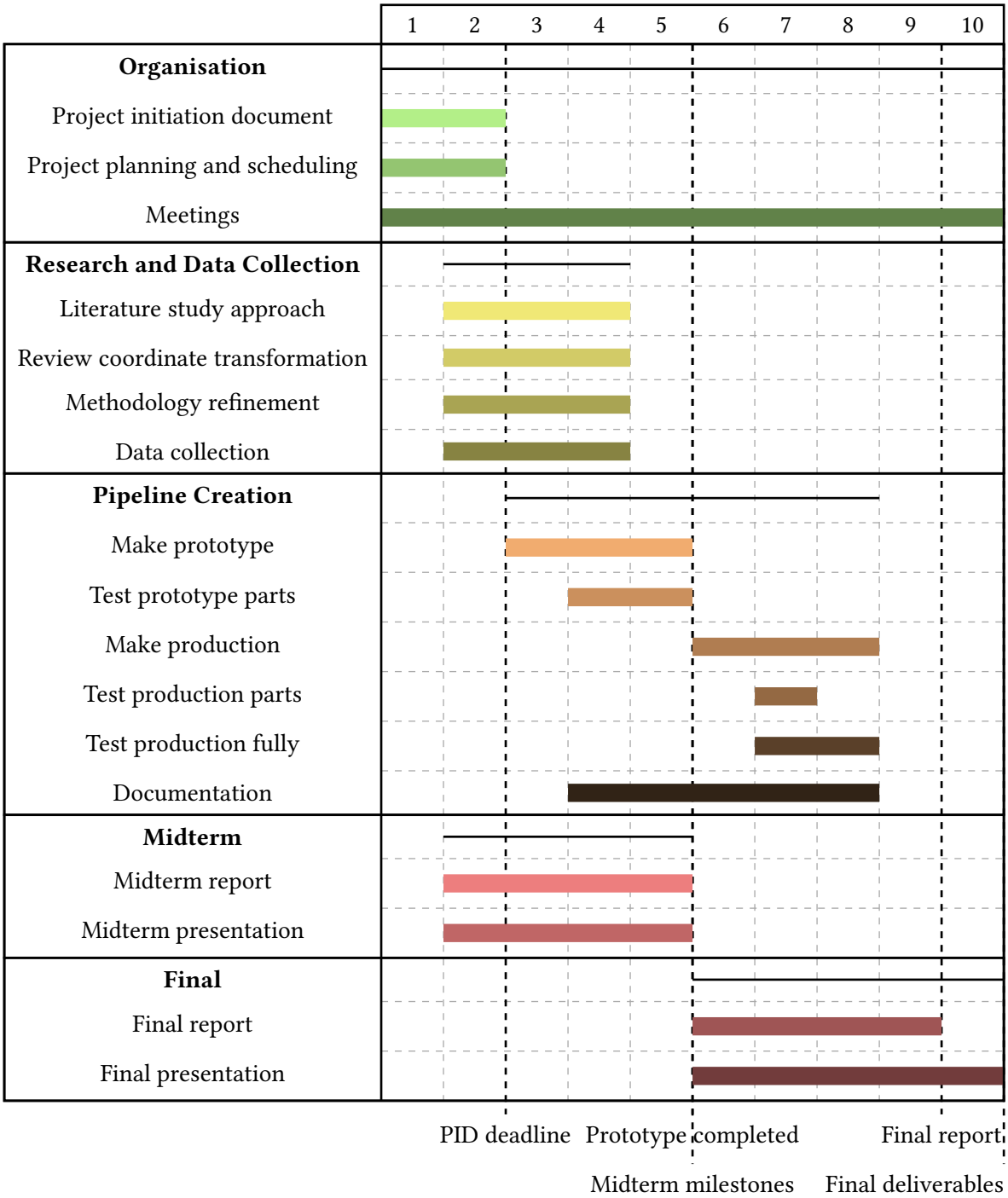



Figure 38: Gantt chart.

## E.3 Risk Analysis

| Likelihood | Harm severity | | | |
|---|---|---|---|---|
| | Minor | Marginal | Critical | Catastrophic |
| Certain | High | High | Very high | Very high |
| Likely | Medium | High | High | Very high |
| Possible | Low | Medium | High | Very high |
| Unlikely | Low | Medium | Medium | High |
| Rare | Low | Low | Medium | Medium |
| Eliminated | Eliminated | | | |

Figure 39: Risk Assessment Matrix.

| Risk ID | Description | Impact | Likelihood | Mitigation |
|---|---|---|---|---|
| 1. | Pipeline too computationally taxing or insufficient computational resources. | Critical | Possible | Test with small dataset and adjust spatial extent if necessary. |
| 2. | Insufficient quality/ availability of point cloud data. | Marginal | Rare | Shift focus to regions with available high-quality data. |
| 3. | CRS transformations are inaccurate. | Critical | Unlikely | Test CRS alignment on select border regions. |
| 4. | Missing/conflicting data on border areas. | Critical | Likely | Check spatial overlap of datasets and prioritise one dataset. |
| 5. | Task is too ambitious given timeframe/team size. | Catastrophic | Unlikely | Check if internal deadlines are met and adjust scope if necessary. |
| 6. | Edge cases not properly assessed due to data quantity and variability. | Minor | Possible | Not ideal but acceptable within the scope of the project. |
| 7. | Task is insufficiently constrained/defined. | Critical | Unlikely | Regular meetings with team and stakeholders to assess and adjust scope and requirements if necessary. |
| 8. | Lacking/inadequate communication within the team and/ or with stakeholders. | Critical | Possible | Schedule regular meetings and maintain open communication channels. |
| 9. | Discovery of unexpected issues. | Marginal | Possible | Regular meetings with team and stakeholders to assess impact and devise mitigation strategies if necessary. |
| 10. | Temporary or permanent absence of team members. | Critical | Rare | Set internal deadlines early to allow for adjustments in case of absence. |
| 11. | Data loss. | Catastrophic | Possible | Implement regular backup procedures. |

Table 9: Risk Assessment Table.

## F Data Source List

| Dataset | URL | Notes |
|---|---|---|
| NL: DTM | see https://www.ahn.nl/dataroom | Map tiles — COG, 0.5 m resolution, not interpolated; also available at 5 m resolution |
| NL: DSM | see https://www.ahn.nl/dataroom | Map tiles — COG, 0.5 m resolution, not interpolated; also available at 5 m resolution |
| NL: PC | | Map tiles — LASzip-compressed point-cloud format (LAZ), 10–14 pt/cm² |
| CH: DTM | https://www.swisstopo.admin.ch/en/height-model-swissalti3d | Map tiles — COG, 0.5 m and 2 m resolution, interpolated |
| CH: DSM | https://www.swisstopo.admin.ch/en/height-model-swisssurface3d-raster | Map tiles — COG, 0.5 m resolution only, interpolated |
| CH: PC | https://www.swisstopo.admin.ch/en/height-model-swisssurface3d | Map tiles — LAZ or Cloud-Optimized Point Cloud (COPC), point density 15–40 pt/cm² (region-dependent) |
| DE-NW: DTM | https://www.opengeodata.nrw.de/produkte/geobasis/hm/dgm1_tiff/dgm1_tiff/ | Map tiles — GeoTIFF, 1 m resolution |
| DE-NW: DSM | https://www.opengeodata.nrw.de/produkte/geobasis/hm/dom1_tiff/dom1_tiff/ | Map tiles — GeoTIFF, 1 m resolution |
| DE-NW: PC | https://www.opengeodata.nrw.de/produkte/geobasis/hm/3dm_l_las/3dm_l_las/ | LAZ |
| DE-RP: DTM | https://geoshop.rlp.de/digitale_gelaendemodelle/digitale_gelaendemodelle_dgm.html | Map tiles — GeoTIFF, 1 m resolution |
| DE-RP: DSM | https://geoshop.rlp.de/digitale_oberflaechenmodelle/digitales_oberflaechenmodell_domb.html | Map tiles — GeoTIFF, 0.2 m resolution |
| DE-RP: PC — surface | https://geoshop.rlp.de/digitale_oberflaechenmodelle/laserpunkte_objekte_lpo.html | LAZ |
| DE-RP: PC — terrain | https://geoshop.rlp.de/digitale_gelaendemodelle/laserpunkte_gelaende_lpg.html | LAZ |
| DE-SL: DTM | https://www.shop.lvgl.saarland.de/index.php?option=com_virtuemart&view=category& | GeoTIFF, 1 m resolution |

| Dataset | URL | Notes |
|---|---|---|
| | virtuemart_category_id=1060& Itemid=475 | |
| DE-SL: DSM | https://www.shop.lvgl.saarland.de/index.php?option=com_virtuemart&view=category&virtuemart_category_id=1066&Itemid=475 | GeoTIFF, 1 m resolution |
| DE-SL: PC | https://www.shop.lvgl.saarland.de/index.php?option=com_virtuemart&view=category&virtuemart_category_id=1067&Itemid=475 | LAZ |
| DE-BW: DTM | https://www.lgl-bw.de/Produkte/3D-Produkte/Digitale-Gelaendemodelle/ | Map tiles — GeoTIFF, 0.25 m resolution |
| DE-BW: DSM | https://www.lgl-bw.de/Produkte/3D-Produkte/Digitale-Oberflaechenmodelle/DOM1/ | Map tiles — GeoTIFF (1 m resolution) |
| DE-BW: PC | https://www.lgl-bw.de/Produkte/3D-Produkte/Laserscandaten/ | not open data, available for a fee |
| DE-HE: DTM | https://hvbg.hessen.de/landesvermessung/geotopographie/3d-daten/digitale-gelaendemodelle | Map tiles — GeoTIFF (1 m resolution) |
| DE-HE: DSM | https://hvbg.hessen.de/landesvermessung/geotopographie/3d-daten/digitale-oberflaechenmodelle | Map tiles — GeoTIFF (1 m resolution) |
| DE-HE: PC | https://hvbg.hessen.de/landesvermessung/geotopographie/3d-daten/airborne-laserscanning | not open data, available for a fee |
| DE-BY: DTM | https://geodaten.bayern.de/opengeodata/OpenDataDetail.html?pn=dgm1 | Map tiles — GeoTIFF, 1 m resolution |
| DE-BY: DSM | https://geodaten.bayern.de/opengeodata/OpenDataDetail.html?pn=dom20 | Map tiles — GeoTIFF, 0.2 m resolution |
| DE-BY: PC | https://geodaten.bayern.de/opengeodata/OpenDataDetail.html?pn=laserdaten | Map tiles — LAZ |
| LI: DTM, DSM, PC | | Included in Swiss datasets |
| FR: DTM | https://cartes.gouv.fr/rechercher-une-donnee/dataset/IGNF_MNT-LIDAR-HD | Map tiles — GeoTIFF, 0.5 m resolution |

| Dataset | URL | Notes |
|---|---|---|
| FR: DSM | https://cartes.gouv.fr/rechercher-une-donnee/dataset/IGNF_MNS-LIDAR-HD | Map tiles — GeoTIFF, 0.5 m resolution |
| FR: PC | https://cartes.gouv.fr/rechercher-une-donnee/dataset/IGNF_NUAGES-DE-POINTS-LIDAR-HD?redirected_from=geoservices.ign.fr | Map tiles — LAZ |
| LU: DTM | https://data.public.lu/fr/datasets/lidar-2024-releve-3d-du-territoire-luxembourgeois/ | Map tiles — GeoTIFF, 0.5 m resolution |
| LU: DSM | https://data.public.lu/fr/datasets/lidar-2024-releve-3d-du-territoire-luxembourgeois/ | Map tiles — GeoTIFF, 0.5 m resolution |
| LU: PC | https://data.public.lu/fr/datasets/lidar-2024-releve-3d-du-territoire-luxembourgeois/ | Map tiles — LAZ |

Table 10: Dataset availability.

# G Border Pair Analysis

## G.1 Regions of Germany

### G.1.1 North Rhine-Westphalia / Rhineland-Palatinate

North Rhine-Westphalia, Germany (DE-NW), and DE-RP both fall within the extent of Germany. The overlapping border area is hilly with few water bodies spread throughout the landscape. As with all Germany DTM data, the tiles are aligned to the same tile grid, and overlap by one or two tiles.

Loading in the `dh_germany_nrw_germany_rlp.tif` raster shows several interesting features. Figure 40 shows North-South or South-North scanning tracks created by aerial Light Detection and Ranging (lidar) scanning. The elevation jump in the order of 3 cm, which while minimal, shows that some differences are caused by the misalignment of the scan passes within tiles.

Figure 40: Lidar flight path artefact on the elevation difference raster, magnitude of differences up to 3 cm.

Secondly, there appears to be misalignment between tiles within the DTM. This is illustrated in Figure 41 as the sharp transition between the four tiles. This transition is however more drastic at a difference of 30 cm in the difference DTM. On close inspection of the DE-NW and DE-RP DTMs, it appears that cells on the edge between tiles, the jump between neighbouring cell values increased with 40 cm more on the DE-NW DTM, compared to the DE-RP DTM. However, it is possible this effect occurs on both the DE-NW and DE-RP DTMs.

`statistics_germany_nrw_germany_rlp.csv` corroborates these findings. It lists the similarity of errors within one kilometre to be 99.9%. High correlation which is consistent throughout the correlation graph either points towards a tiling or vertical transform artefact. The median difference between the two DTMs is −2cm, which also corresponds with the z-shift for the coregistration.

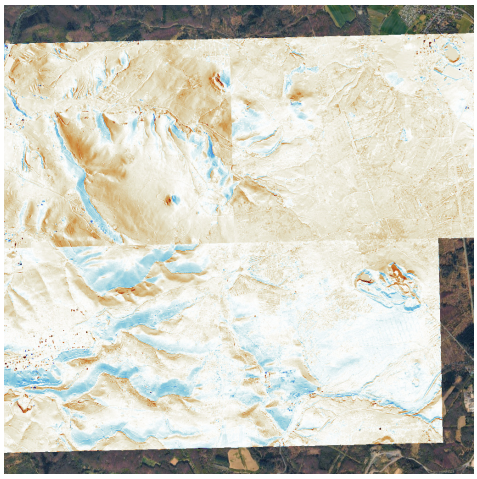
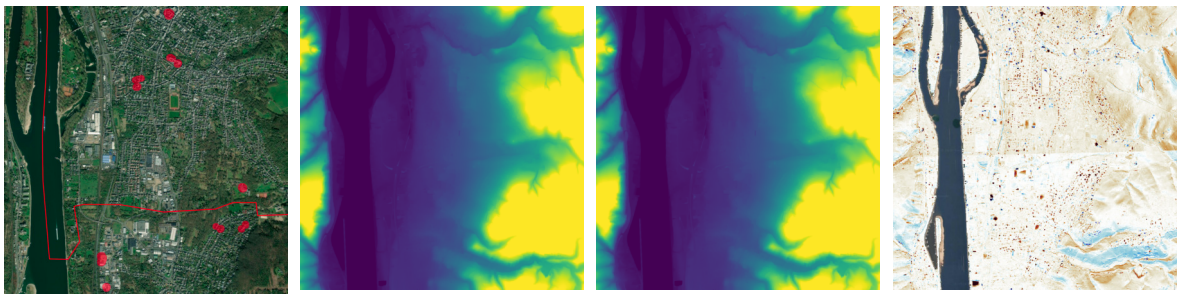


Figure 41: Tiling artefacts on the elevation difference map over the DE-NW/DE-RP border.

Lastly, as described in Section 4.2, small water bodies such as narrow rivers or small lakes are not filtered out and are therefore counted towards the differences in the comparison. This however, a small contribution and does not appear disproportionately represented in the calculated problem areas.

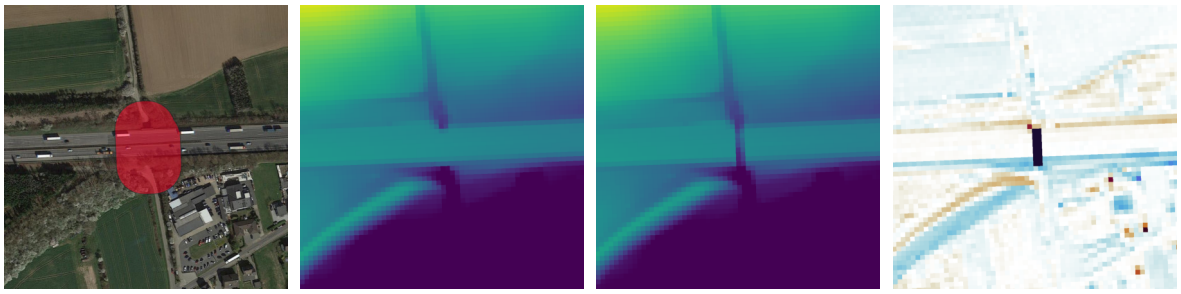
There do appear to be several differences which are likely related to lidar sensing limitations or subsequent processing steps. Examples include interpolation errors for buildings (Figure 42), ground filtering discrepancies (Figure 43), temporal changes (Figure 44)(verified using Google Earth), and steep slopes (Figure 45).

Nevertheless, the overall alignment of the DE-NW and DE-RP DTMs does comply with the INSPIRE target described in Section 3.2.4. Furthermore, only 0.05% of points within the overlap fall outside of the INSPIRE derived threshold described in Section 4.3. Therefore, the quality of the resulting DTM does comply with the guidelines we have followed for the full Rhine DTM.



DTM of DE-BW. DTM of DE-NW. DTM of DE-RP. Elevation difference map.

Figure 42: Example of elevation differences caused by roof interpolations. Red polygons in the leftmost figure indicate areas where the elevation difference exceeds 1.6 m.



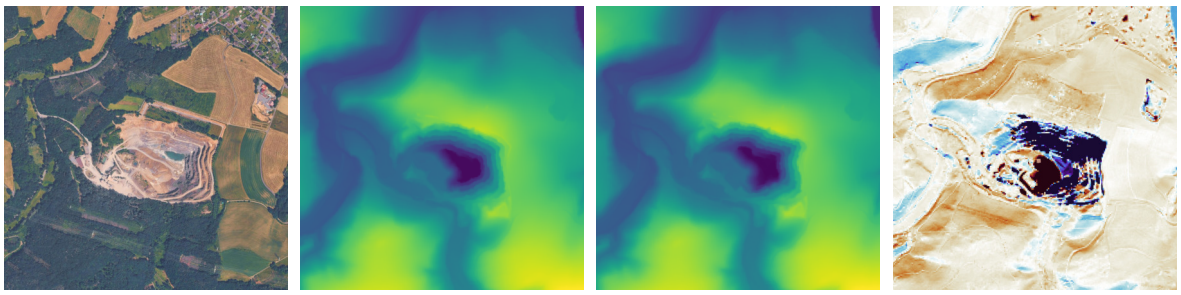
Satellite imagery. From Airbus, Maxar Technologies. (2026).

DTM of DE-NW.

DTM of DE-RP.

Elevation difference map.

Figure 43: Example of a highway overpass, where DE-NW prioritizes the highway as ground, while DE-RP prioritizes the road underneath. The DE-NW and DE-RP rasters were set to 195–215 metres. The color scale was set to −1 to 1 m to highlight the differences.



Satellite imagery. From Airbus, Maxar Technologies. (2026).

DTM of DE-NW.

DTM of DE-RP.

Elevation difference map.

Figure 44: Example of differences between the DE-NW and DE-RP DTMs, as result of temporal quarry changes. The DE-NW and DE-RP rasters were set to 115–235 metres. The color scale was set to −1 to 1 m to highlight the differences.

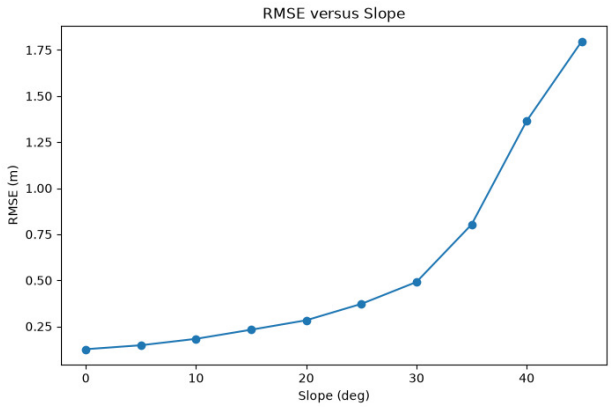



Figure 45: RMSE of the elevation difference plotted against terrain slope.

### G.1.2 Hessen / Baden-Württemberg

The border between DE-HE and DE-BW is one of the shorter borders, coming in at roughly 33 km. It does however contain some interesting elements such as an exclave of DE-HE inside of DE-BW. This exclave does however not appear to contain any notable differences. Furthermore, the

elevations in this region range from 90 metres in the large and relatively flat valley, to nearly 550 metres in the hills (Figure 46).

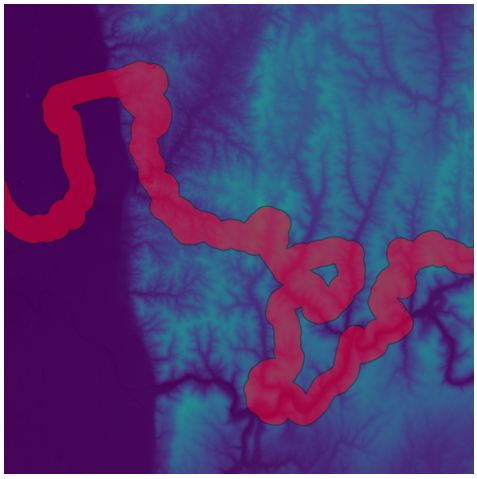


Figure 46: DE-HE/DE-BW border. The red polygon shows the overlapping area of the two DTMs.

The first obvious difference is a bright highlight at the border of the DTM (Figure 47). This artefact is a byproduct of the DE-BW dataset being cut differently compared to the DE-HE dataset. Therefore, this difference is an artefact caused by the clipped dataset and not a real physical difference. This effect occurs on all difference analysis DTMs, though it is not always as pronounced as within this comparison.

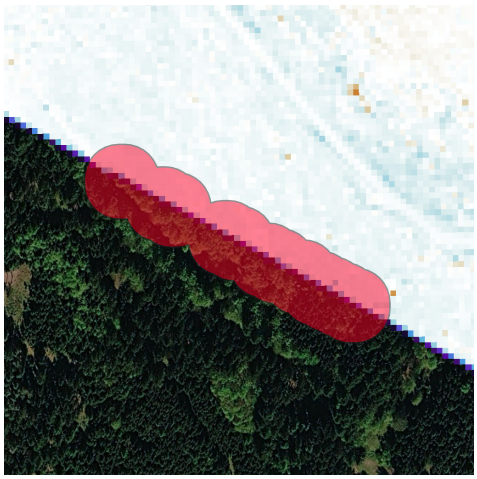
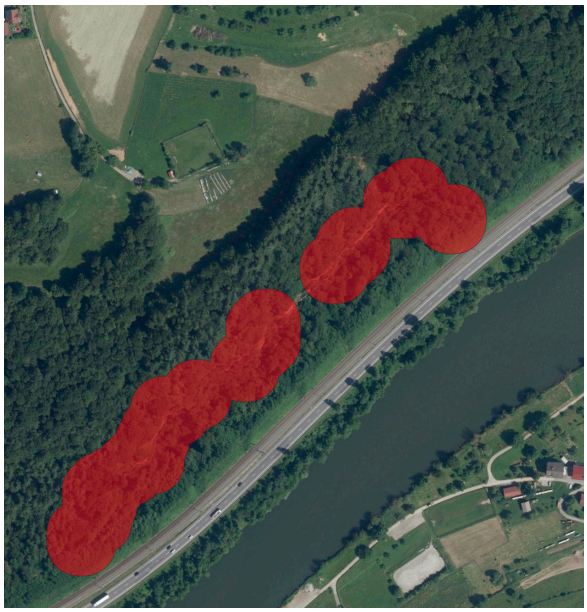
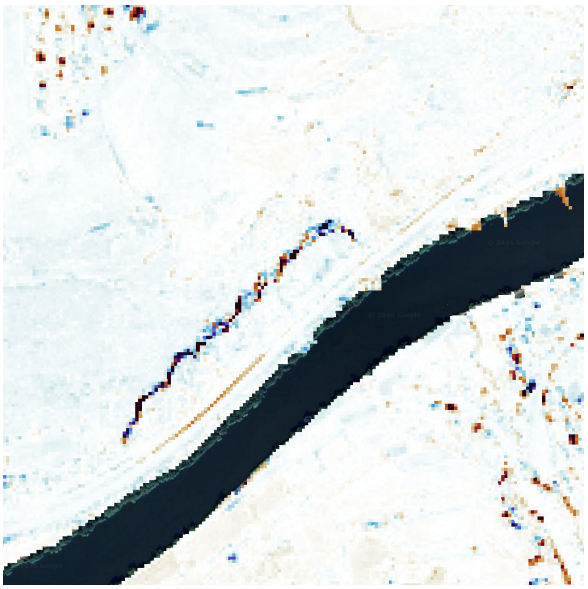


Figure 47: DE-HE/DE-BW border artefact. The color scale was set to −1 to 1 m to highlight the differences. The red polygon shows the area exceeding the 1.6 m threshold.

A secondary source of differences appears to be the narrow clearings within wooded areas (Figure 48). This occurs in several locations. It is unclear exactly why this happens. The uncertainty DTM representing random error shows a very constant, yet small error of around 3 cm. This implies that the observed error is not simply consisting of noise. Observing the DE-BW DTM, we see a steep drop in elevation. Streetview furthermore confirms that it is indeed a cliff face. The elevation difference is likely caused by ground filtering or different flight paths during lidar scanning.

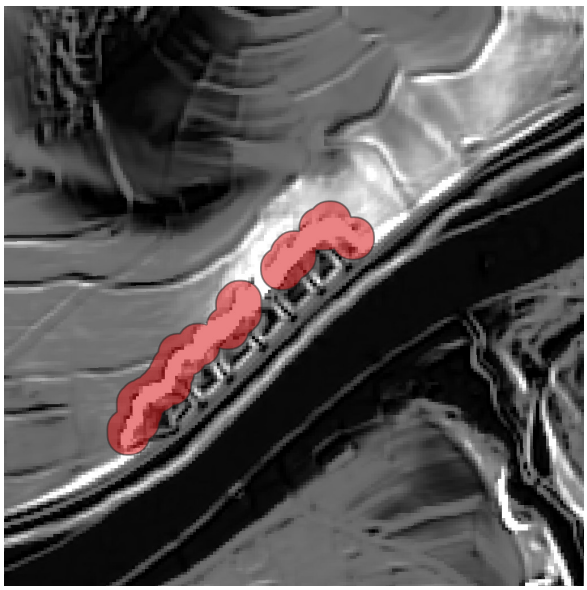


Satellite imagery. From AeroWest, Airbus Geobasis-DE/BKG. (2026)



Elevation difference map.

Figure 48: Cliff face example, the red polygons show a difference above 1.6 m on the ESRI aerial image. The color scale was set to −1 to 1 m to highlight the differences.



Uncertainty map.

Google Street View (49.4762° N, 8.4208° E).

Figure 49: Cliff face example, random error inspection shows random errors in 3 cm range which imply this is not the main cause, Google Street View confirms cliff hypothesis.

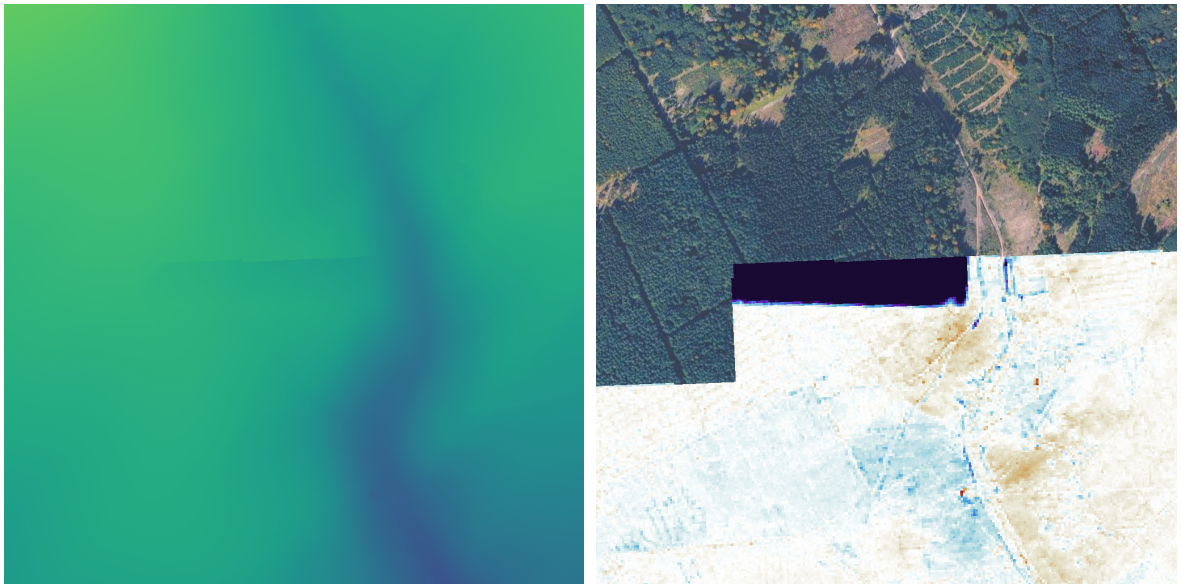
Further artefacts include common elements throughout the catchment DTM. These include building interpolation differences, quarries/construction sites, and lakes/rivers not caught by the river mask. These errors are expected and can likely only be fixed through the re-extraction of the DTM, by using both the DE-HE and DE-BW point clouds. Nevertheless, the DE-HE/DE-BW border falls well within the INSPIRE threshold, with only 0.06% of points being non-compliant. The alignment of the two DTMs also appears to be accurate, with the coregistration recommending a vertical transform of only 1 cm for the best possible alignment.

### G.1.3 Saarland / Rhineland-Palatinate

The DE-SL/DE-RP border is roughly 207 km long and consists of mostly hilly terrain. The amount of non-compliant pixels for this area appears slightly higher than that for other regions. However, at 0.15% it is still not a significant amount.

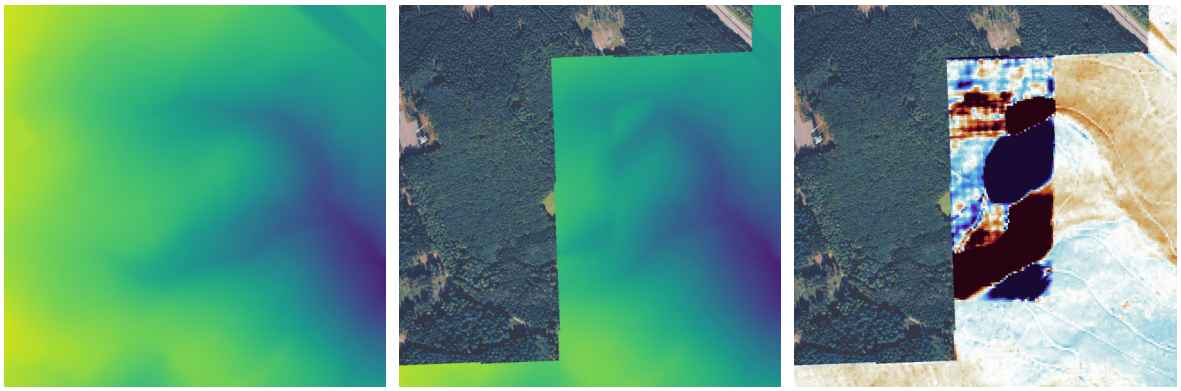
Most errors within this region can be attributed to interpolation errors, temporal changes such as wind turbine construction or quarries, steep slopes, water bodies missed by the mask, and border artefacts. The only notable difference between the DE-SL and DE-RP DTMs, appears to be the inclusion of extrapolation artefacts at the borders, as seen in Figure 50 and Figure 51. These are only two out of several occurrences. These large areas of similar errors do explain the secondary peaks seen within the difference histogram (Figure 52). Upon further inspection, there do also appear to be occurrences of overpasses as discussed in DE-NW/DE-RP, and cliffs as discussed in DE-HE/DE-BW.

All in all, despite issues with the source data, the quality of the DTM is not affected, due to the region cutoff at the border for the catchment DTM.



DTM of DE-RP and DE-SL. Elevation difference map.

Figure 50: Extrapolation example 1, large negative difference accompanied by feathering of elevation values due to extrapolation. The DE-RP and DE-SL rasters were set to 400–700 metres. The color scale was set to −1 to 1 m to highlight the differences.



DTM of DE-RP. DTM of DE-SL. Elevation difference map.

Figure 51: Extrapolation example 2, large positive and negative differences accompanied by feathering of elevation values due to extrapolation. The DE-RP and DE-SL rasters were set to 400–575 metres. The color scale was set to −1 to 1 m to highlight the differences.

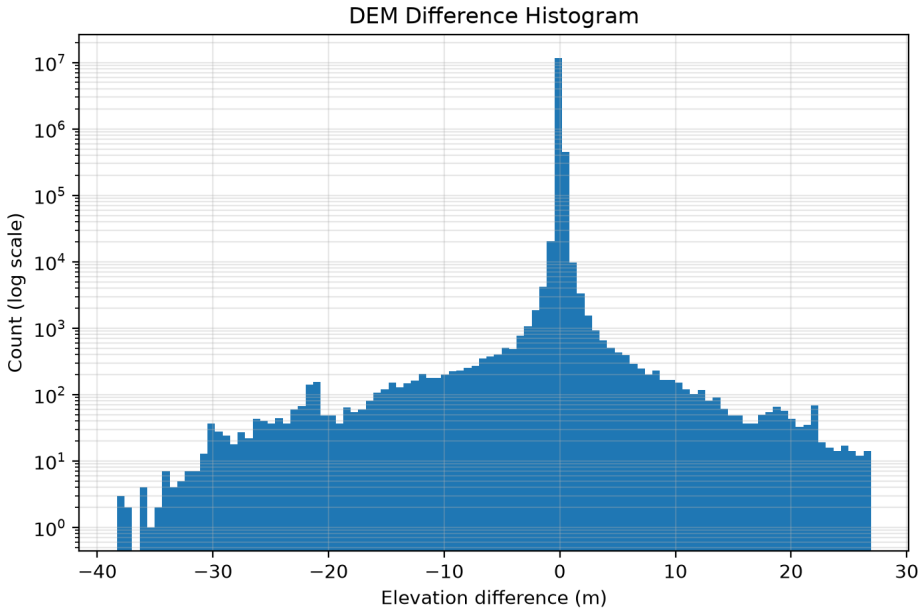


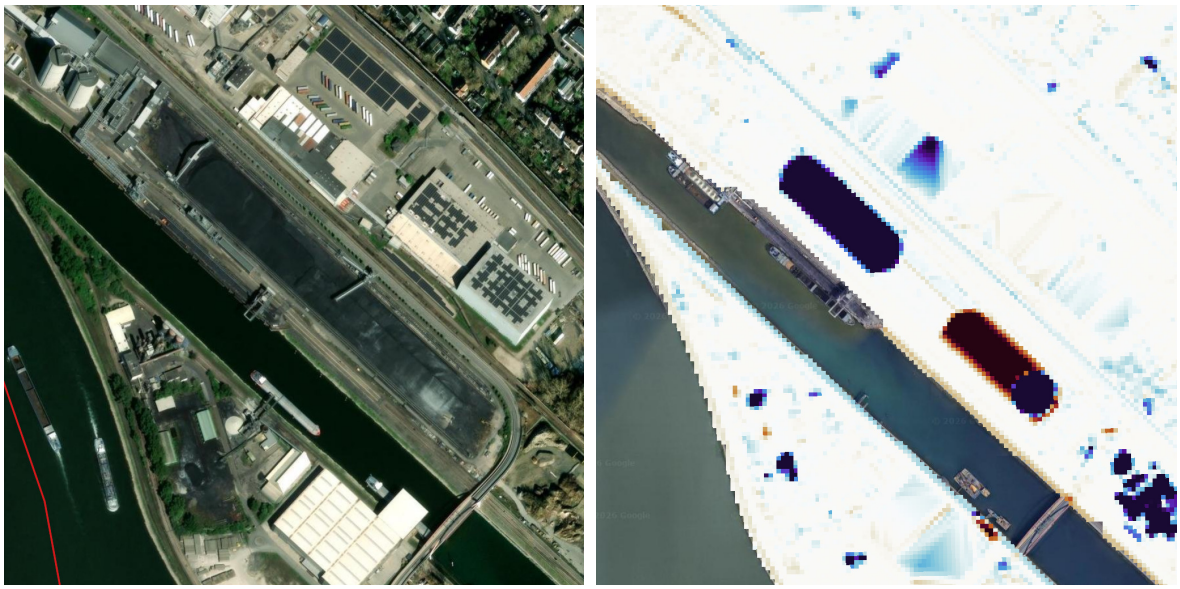

Figure 52: DE-SL/DE-RP histogram of elevation differences. Secondary peaks around the -30, -20, and 20 metre marks mostly represent extrapolation artefacts.

### G.1.4 Rhineland-Palatinate / Baden-Württemberg

The DE-RP/DE-BW border is a very flat region, which is entirely bisected by the Rhine river. The border region is home to several ports, which also created the greatest source of discrepancy. These ports prominently contain piles of sand and coal which are moved around during the loading and unloading of ships (Figure 53).

Another source of differences are interpolation errors. These are mostly found around the city of Mannheim. These correspond to building roofs. Furthermore, the tile offset found in several of the other datasets also occurs at the DE-RP/DE-BW border, as shown in Figure 54. More common differences are related to water bodies, overpasses, and construction sites.

The only notable outlier within the data is the convergence of error similarity at 60%, which is explained by the tile offset. The reason why the percentage is so high is due to the small overall range of elevation differences. Overall, the comparison falls well within the allowed threshold and causes no noticeable issues for the full DTM.



Satellite imagery. From Airbus, GeoBasis-DE/BKG, Maxar Technologies (2026)

Elevation difference map.

Figure 53: Example of elevation differences encountered on a coal storage. The temporal changes in the coal mine cause this difference. The color scale was set to −1 to 1 m to highlight the differences.

Figure 54: Elevation differences caused by tiling artefacts. The color scale was set to −0.25 to 0.25 m to highlight the differences.

### G.1.5 North Rhine-Westphalia / Hessen

Only 59 km of DE-NW/DE-HE border and 127 1km×1km tiles fall into the study area. Due to the shape of the catchment area, the border is split in two — a longer section in the South and a shorter, approximately 3-kilometre section in the North. Notably, DE-NW openly publishes point cloud data which enables further analysis, whereas DE-HE does not.

The errors in this region are relatively low: RMSE is 18 cm, LE90 is 7.5 cm, and only 0.07% of cells exceed the 1.67 m difference threshold, suggesting a high agreement between the DTMs. Still, the biggest differences are −9 m and 22 m in the negative and positive direction respectively.

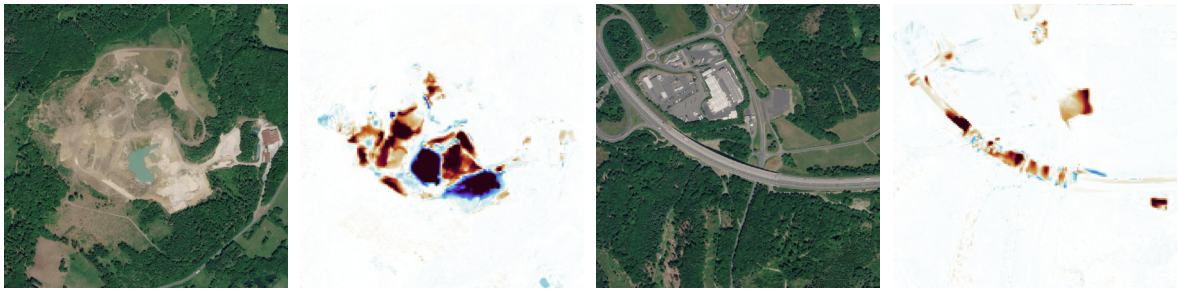


Figure 55: Elevation difference maps showing areas on the DE-HE/DE-NW border with high elevation differences due to temporal changes. Satellite imagery from Airbus, GeoBasis-DE/BKG, Maxar Technologies.

Most of the errors which exceed threshold are best explained by the temporal differences in the datasets. There are two large quarries in the overlapping region, and the earthworks at both sites contribute to the largest errors. Additionally, there are several places where judging by the historical Google Earth imagery, the error is likely to be due to new or ongoing construction, such as the bridge near the Haiger service station. Additionally at the time of acquisition, there was snow on the ground in Willingen ski resort which was classified as ground by DE-NW (Figure 56, snow shown in yellow).

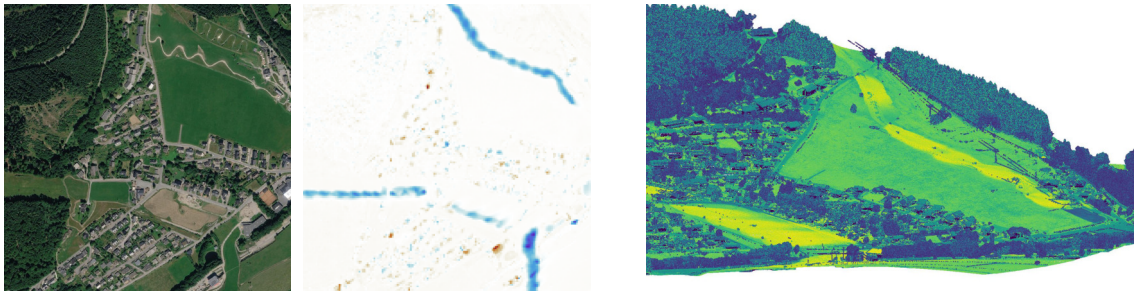


Figure 56: Figure 29: Elevation differences caused by the presence of snow in one of the DE-NW surveys. Satellite imagery from Airbus, GeoBasis-DE/BKG, Maxar Technologies.

There are several smaller issues which we believe are best attributed to the differences in processing. The built-up areas display a high number of scattered differences around 0.5 m which are likely caused by either the states' approach to ground filtering or interpolation algorithms used to create the original DTMs. An interesting artefact is a 0.6m-tall "mound" between Wallau and Bad Laasphe. DE-NW appears to be using CSF and extract the heights of synthetic cloth points instead of interpolating authentic ground points — we believe the mound is most likely caused by the "sagging" of the simulated cloth (see Figure 57, simulated cloth in yellow), which we verify by examining the point clouds.

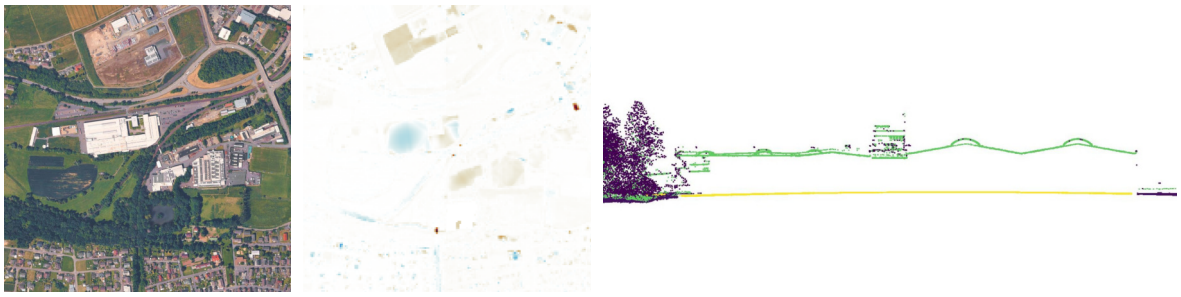


Figure 57: Ground and building filtering differences between DE-NW and DE-HE.

### G.1.6 Hessen / Bavaria

The statistics of the errors along DE-HE/DE-BY border are similar to those of DE-HE/DE-NW border, but their sources appear to be somewhat different. Between DE-HE and DE-BY, RMSE is 0.18 m and the LE90 is 0.10 m the minimum and the maximum errors are −18.35 m and 19.07 m respectively.

The highest absolute errors can all be traced to temporal changes, specifically the excavation taking place at a single quarry outside Ringheim. There are additionally some construction works taking place along the border, but there, the magnitude of the errors there is lower.

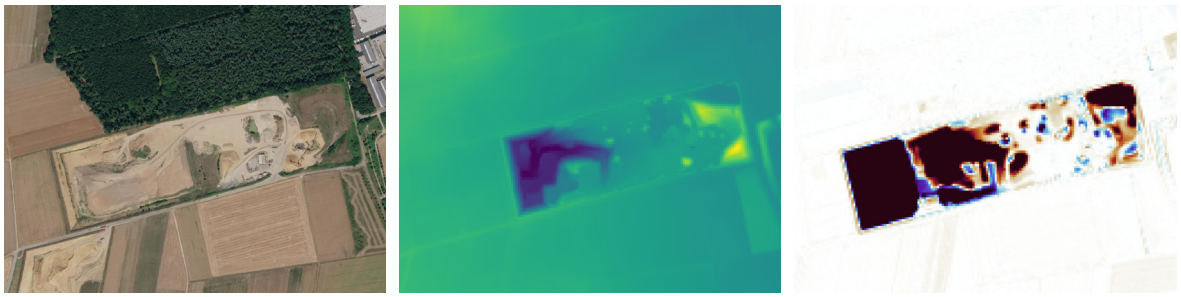


Figure 58: Elevation differences caused by temporal artefacts between DE-HE and DE-BY.

More noteworthy are the small (typically around 0.3–0.5 m) differences in rural, especially forested, areas. These are areas of difference which appear to correspond to linear features in the landscape such as roads, streams, or edges between a forest and an open field. These are likely consequences of the approaches to ground and vegetation classification employed by the different data providers.

Examining the DE-BY point cloud reveals that all points in the relevant tile are strictly classified into ground (2) and "objects" (20); as well as scan patterns and non-uniform point distribution, largely due to shadowing from trees. Due to lack of open point cloud data in DE-HE, the comparison between two states is impossible.

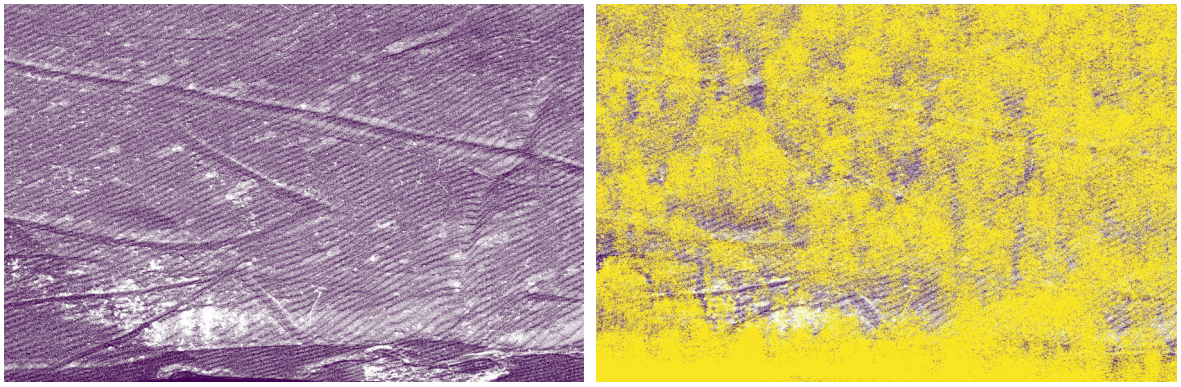
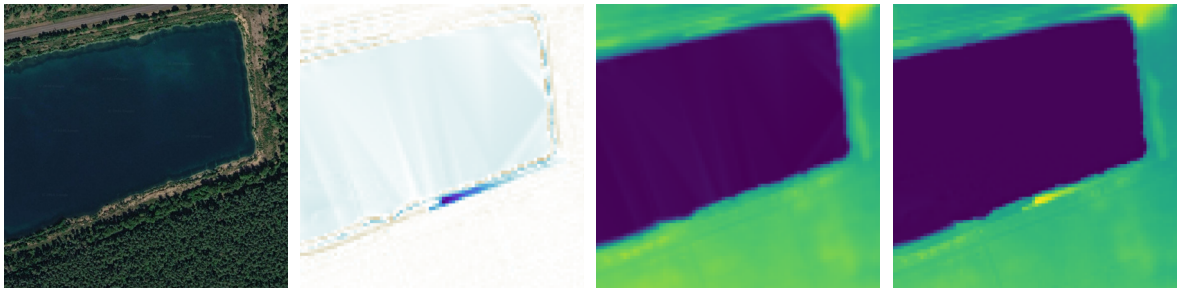


Figure 59: DE-BY point cloud data. Classified ground points in purple.

Another detail evident from the comparison is the difference in approach to estimating the elevation of the water surface. DE-HE appears to be relying on the data about the water surface obtained from the point cloud — the surface does not appear to be perfectly smooth and seems to contain small artefacts. DE-BY instead uses Delaunay triangulation, seemingly based on the lowest point along the shoreline of the water body.



Satellite imagery. From AeroWest, Airbus, Geobasis-DE/BKG. (2026)

Elevation difference map.

DTM of DE-BY.

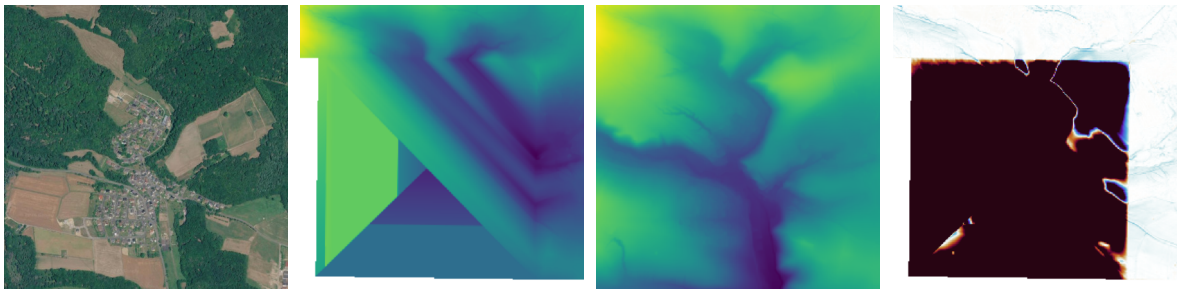
DTM of DE-HE.

Figure 60: Differences in water processing between DE-BY and DE-HE.

### G.1.7 Hessen / Rhineland-Palatinate

The border between DE-HE and DE-RP largely follows the Rhine. Compared to the borders between German states, it has the second-highest RMSE at 1.30 m and the second-highest LE90 at 0.19 m. It also has the second-highest absolute error at −61.56 m. High RMSE can largely be attributed to errors due to processing in one particular tile; high LE90 is a symptom of systematic errors.

One of the tiles DE-HE includes in its elevation dataset is almost entirely interpolated — seemingly using two interpolation methods: linear interpolation on a TIN where sample points are available; and another method where data was not available. DE-HE does not publish the point cloud data, yet having observed raster elevation data, we believe that the entire tile was cropped out from the point cloud survey as it only contained partial data. A small part of DE-HE still falls within that tile; and it was re-included by DE-HE during the raster processing phase, interpolated based on the ground points from the neighbouring tiles. It is unclear what the elevation of the southwest extreme point is based on.



Satellite imagery. From Airbus, Maxar Technologies (2026).

DTM of DE-HE.

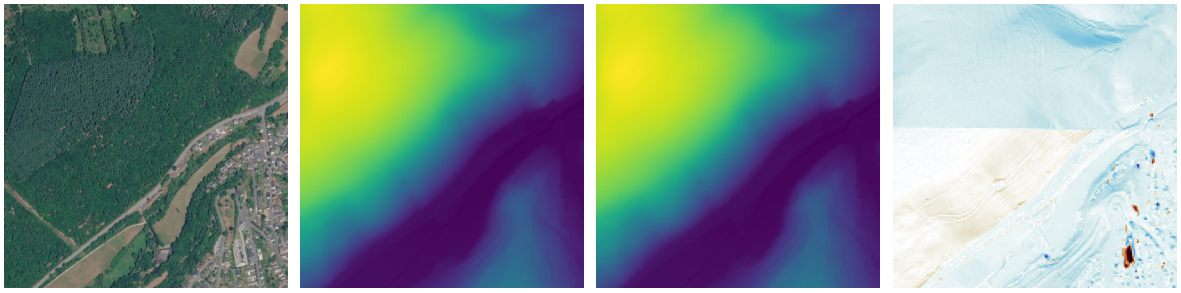
DTM of DE-RP.

Elevation difference map.

Figure 61: Interpolation artefact by DE-HE.

Systematic errors can be observed throughout the region where, especially on hilly terrain, there is a relatively consistent 5–20 cm difference between the two DTMs. There are visible seams in

the difference map where two tiles meet, which likely point at either a different ground filtering algorithm or a misalignment between two different surveys carried out by the same provider. The hilly terrain makes it impossible to confidently conclude which of the data providers is responsible for this error based on raster data alone.



Satellite imagery. From AeroWest, Airbus, Geobasis-DE/BKG. (2026)

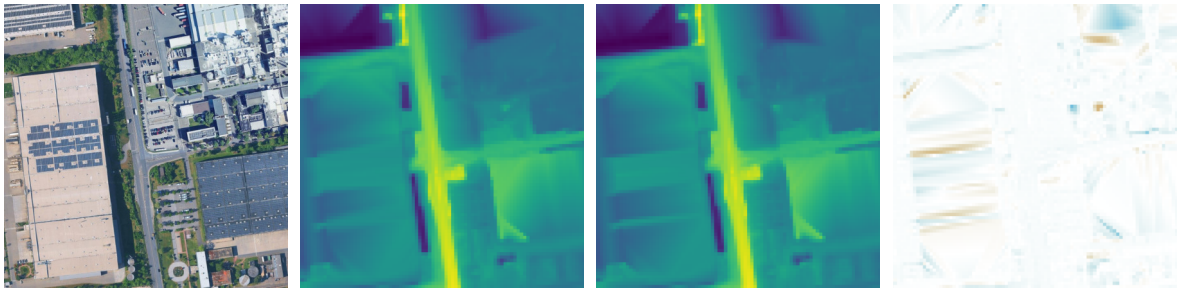
DTM of DE-HE.

DTM of DE-RP.

Elevation difference map.

Figure 62: Example of systematic difference between DE-HE and DE-RP.

Smaller errors are also visible in building footprints, where both providers are performing linear interpolation on a TIN but based on somewhat different set of points.



Satellite imagery. From AeroWest, Airbus, Geobasis-DE/BKG. (2026)

DTM of DE-HE.

DTM of DE-RP.

Elevation difference map.

Figure 63: Interpolation differences between DE-HE and DE-RP.

### G.1.8 Baden-Württemberg / Bavaria

The DE-BW/DE-BY border is similar to the DE-NW/DE-HE border, as it is split in two sections due to the catchment area shape. For this region DE-BY provides point cloud data, whereas DE-BW does not. This means that point cloud comparisons will not be possible for this border region.

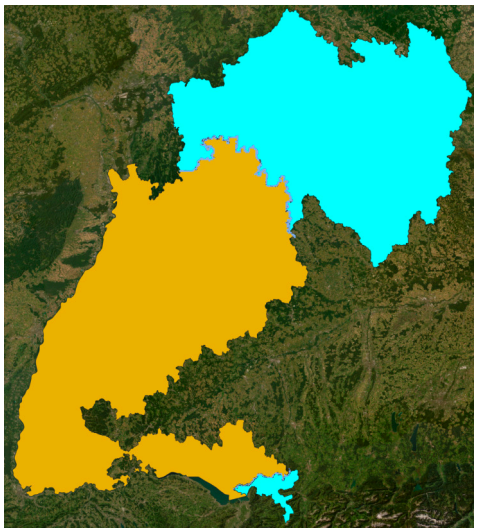
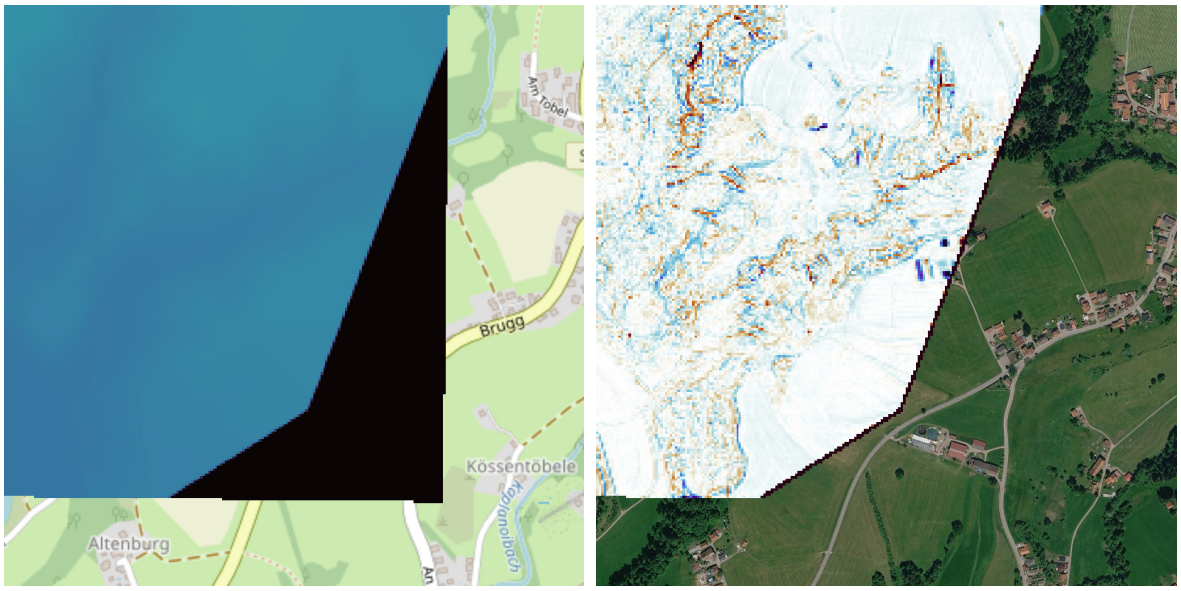


Figure 64: DE-BW/DE-BY border is divided into a northern and southern section in the catchment area.

The border area appears to display two main artefacts. The first of these being a pronounced difference DTM edge (Figure 65). However, this effect only appears at the southern border, and appears absent from the northern border. The cause of this artefact is that the DE-BW DTM appears to have two `NoData` values. The metadata states that the `NoData` value is at −9999, while the value in the black area is 0 before the vertical transform. The code currently corrects for this artefact. However, the border edge still has traces of this effect due to the rotation applied during the horizontal transform, and the interpolation that occurs at that stage. These errors make up almost 25% of all designated problem areas. Figure 66 illustrates the effect this has on the total error distribution for the border comparison. Nearly all values below −5 metres can be attributed to this effect.




DTM of DE-BW. Elevation difference map.

Figure 65: Border artefact caused by DE-BW using both a 0 and −9999 `NoData` value, while only declaring the −9999 value as `NoData`. The color scale was set to −1 to 1 m to highlight the differences.

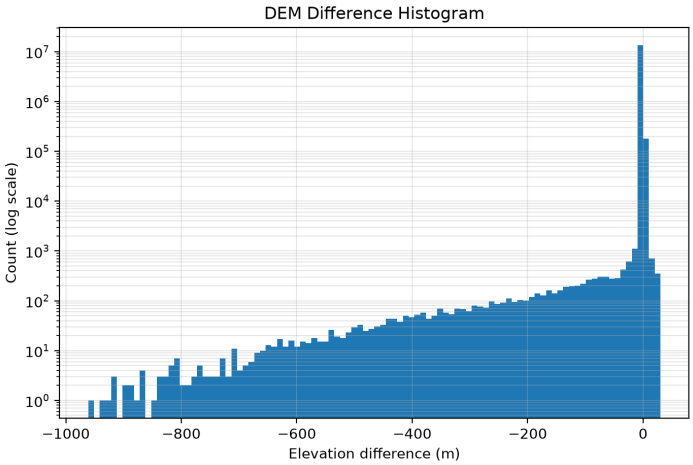



Figure 66: DE-BW/DE-BY border elevation differences histogram. This region appears to be strongly affected by the edge artefacts caused by the effect of 0 as `NoData` value and bilinear interpolation during CRS transformation, with elevation differences of up to 1000 m of magnitude.

The second artefact within this border comparison is similar to the one encountered in DE-NW/DE-RP section. In this instance, the effect clearly occurs in the DE-BY DTM. Figure 67 shows this effect. It appears as if these specific tiles are separate from the rest of the DTM as it contains both positive and negative differences, meaning it cannot be the result of a vertical shift. Zooming in on the border confirms that these different tiles exist in DE-BY, shown by the faint line at the same line as the elevation difference map. The cause for this can be that the tiles have either been collected at a different time, or that the acquisition was done through another method.

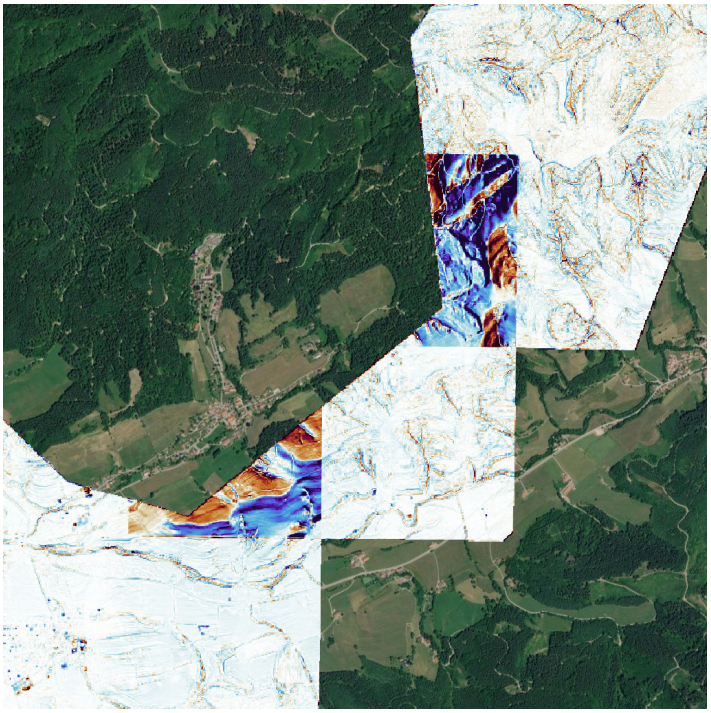
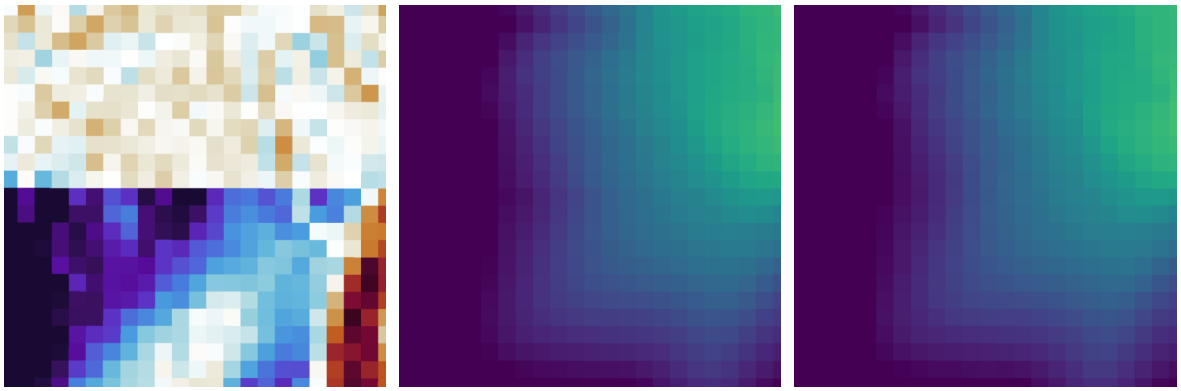


Figure 67: Elevation differences appearing like tiling artefacts. Pronounced red and blue regions show larger differences.



Elevation difference map. DTM of DE-BY. DTM of DE-BW.

Figure 68: Discrete tile step zoomed, shows DE-BY mirrors the tile step observed in the difference DEM.

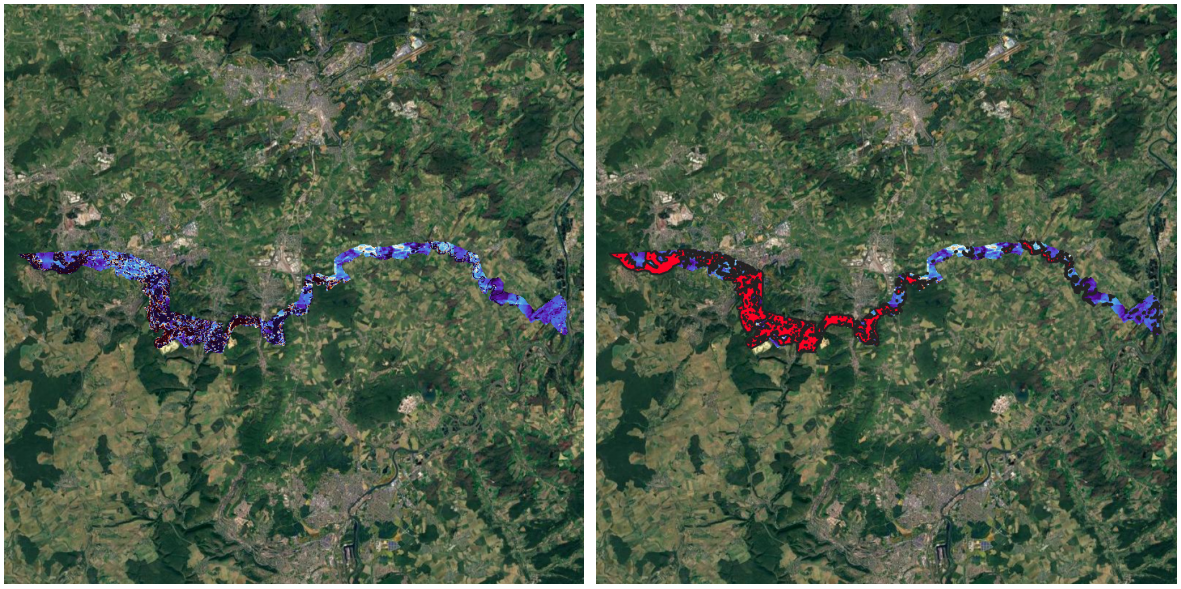
Several other errors exist within the border comparison. These are in line with the errors encountered at other borders. These include interpolation errors, differences in ground filtering (differences in overpass handling), temporal changes (mostly due to quarries or construction areas), differences due to watermask shortcomings (including small pools, small lakes, creeks), steep slopes

For this border comparison, the INSPIRE target is not met. The amount of points above the threshold is only 0.14%. However, the total RMSE is 5.36 m — a factor 3.2 greater than what it is allowed to be for the 5 metre resolution. These statistics suggest this border pair is of unacceptable quality; however, the error regions get overwritten and thus do not introduce any conflicts within the final product. Solving this problem does not fall within the scope or time constraints of this project, and is primarily a challenge created due to available input data and not by the processing pipeline.

## G.2 Luxembourg

### G.2.1 Luxembourg / France

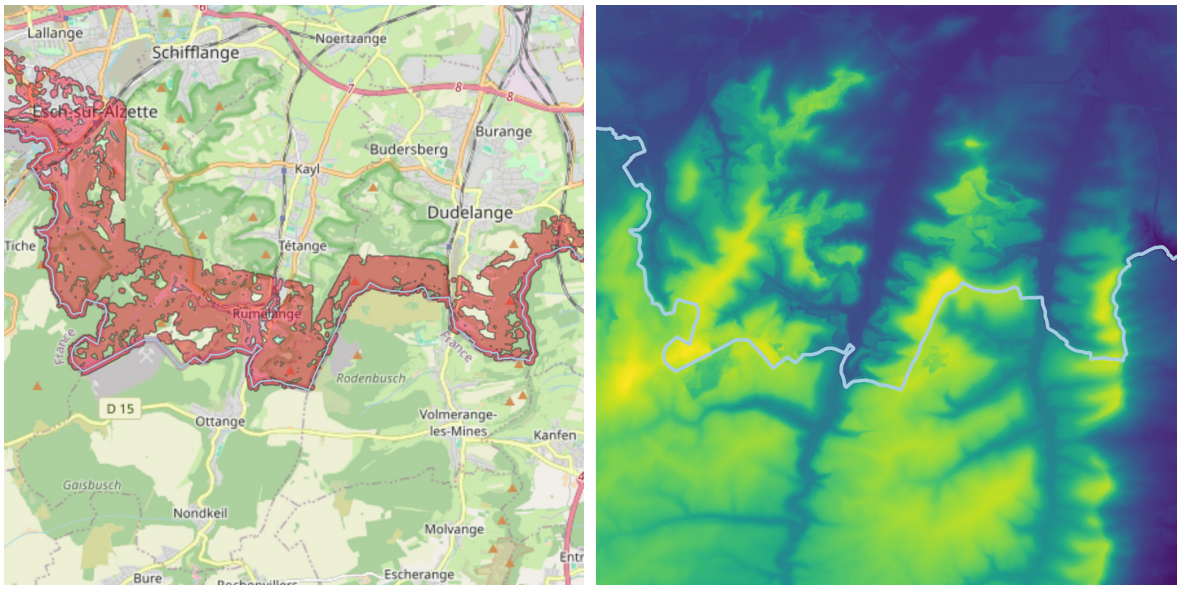
The border area between Luxembourg and France is roughly 50 km long and consists of more even terrain in the east and increasingly hilly terrain in the west with deviations of over 100 m at relatively short distances. As can be seen in Figure 69, most of the western region of the border is flagged as having a deviation greater than the resolution-RMSE ratio.



Elevation difference map. Flagged Differences.

Figure 69: Overlaping border DTM region for LU/FR. The western section of the border almost entirely exceeds the 1.6 metre resolution threshold, shown by the red polygons (right).

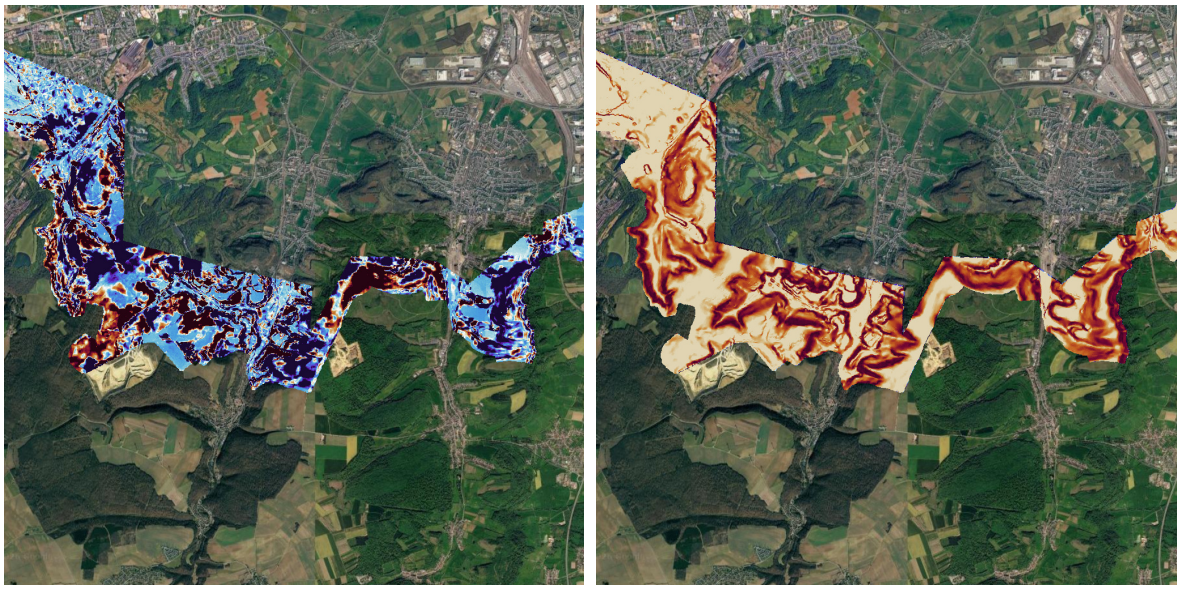
As a primary example of this phenomenon the area around Rumelange was more closely evaluated. Unlike most border regions, the differences are not limited to isolated artefacts, but affect most of the border area. A prominent example is the area around Rumelange. Inspecting the DTM shows that the area is populated with steep slopes, valley sides, ridges and cliff faces (Figure 70).




Flagged Differences. DTM.

Figure 70: Elevation difference in Rumelange, Luxembourg. The DTM shows a hilly terrain, which corresponds to primary areas of disagreement.

Further examining the uncertainty pattern of the border region further supports these observations (Figure 71). Areas with high uncertainty (shown in red), represent elevation differences due to random errors of 1.6 metres, while low uncertainty regions (in blue), correspond with the regions of small differences in the elevation difference map. Due to the respective national DTMs being generated from lidar data, small differences in sample point locations can create large differences in the elevation of a point. This effect becomes more pronounced due to the 5 m resolution of the French DTM and the 0.5 m resolution of the Luxembourg DTM.



Elevation difference map. Uncertainty map.

Figure 71: Random elevation differences due to sloped terrain appear (in red) up to values of 5 metres. The uncertainty map shows random errors with a colour scale of −1.5 to 1.5 m.

Influence of the slope on the RMSE can also be observed directly from the RMSE analysis (Figure 72). Areas with a slope above 5 degrees have an RMSE surpassing 4 m. This behaviour deviates from

most other border regions where the RMSE gradually increases with slope. The result suggests that the steep terrain is one of the dominant causes of disagreement between the two DTMs.

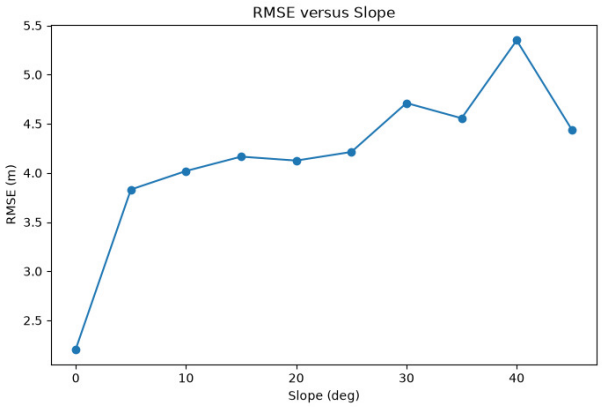



Figure 72: RMSE of elevation differences against slope of the terrain. RMSE grows strongly with higher slope, up to 5.5 m at 40 degree slopes.

Further artefacts within the region do more closely correspond with the rest of Luxembourg and the border regions in general. These include building interpolation differences, narrow streams not captured by the river mask, and isolated quarry or construction areas. This region does not comply with the thresholds set within Section 4, with 23% of all points surpassing the threshold. Furthermore, the resolution would need to be increased by over 80% to nearly 10 m for compliance with the INSPIRE target. As alluded to earlier, this difference may be a byproduct of the low-resolution DTM compared to the fine resolution of the Luxembourg DTM, which has introduced resampling differences. While outside of the time constraint of the project, evaluating the results of the comparison against the higher resolution French DTM could result in better alignment. Alternatively, re-extraction from the national point clouds could also be a way to rectify these differences.

### G.2.2 Luxembourg / Rhineland-Palatinate

The LU/DE-RP border is roughly 80 km long. On the east side of the overlapping area there is the Our river. Observing the overlap map shows that the rivermask was not successful at filtering out the water bodies entirely for the region. This is a known artefact caused by narrow water bodies. Furthermore, these river areas have been flagged as noticeable differences and count towards the difference at the border.

Other differences in this region are in line with the other borders surrounding Luxembourg, such as river ports with piled resources, steep cliffs, construction areas and building interpolation differences. The only real artefact that shows up on the histogram, is a singular secondary peak. Observing the difference DTM, shows that this is caused by a lake, which was missed by the river mask. This is illustrated in Figure 73 as a peak starting at 700 m. The border effect graph also clearly illustrates the difference at the border river.

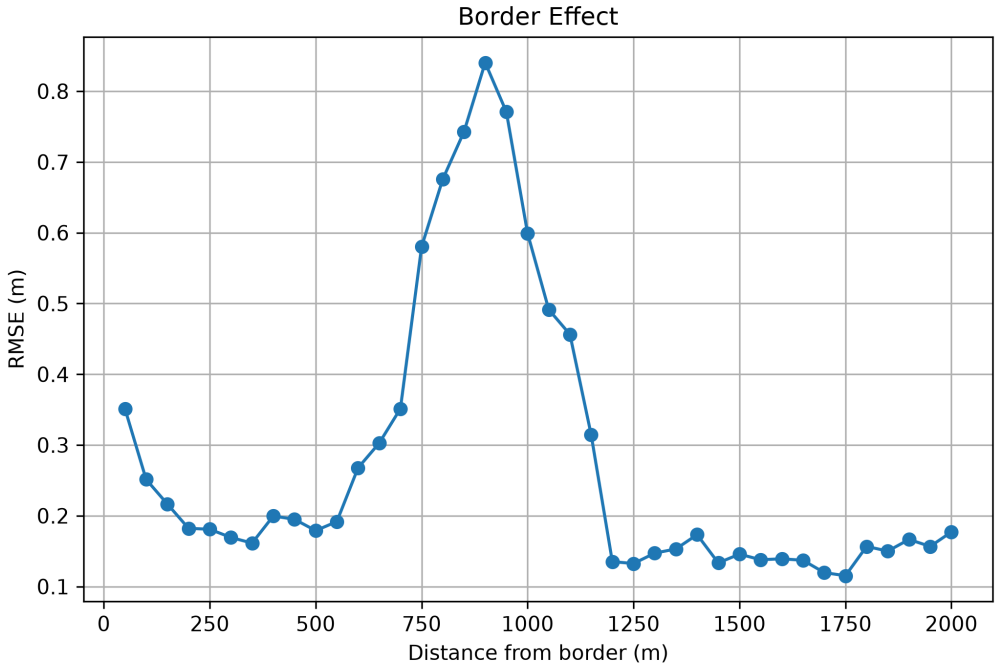


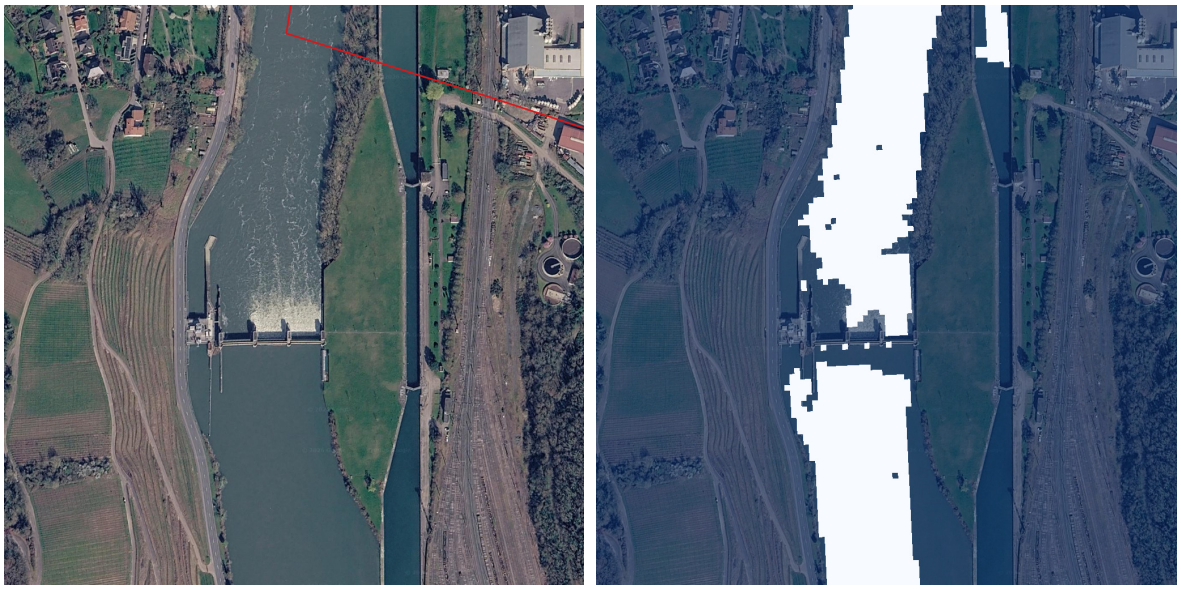

Figure 73: Elevation differences plotted against distance from border. River differences (0 m), lake river mask failure (700 – 1200 m).

### G.2.3 Luxembourg / Saarland

The border area between Luxembourg and DE-SL is roughly 11 km and like DE-RP/DE-BW, is entirely bisected by a river, that being the Mosel in this particular case. The border area consists of flat areas around the river and consists of hills and cliffs on the west side.

This region contains several of the more common differences encountered in other regions can also be found here. This includes harbour loading areas for sand, steep cliffs, construction sites, water bodies not entirely caught by the river mask, and roof interpolation differences.

The only unique error found around this area is the inclusion of hydroelectric power plants. While this difference can really be attributed to the limits of the river mask, it is an interesting feature within the water which could pose a challenge to solve should improving the river mask be attempted as the water here does not follow the smooth water hypothesis. Furthermore, the decision would need to be made if the plant should be removed entirely. This depends on whether all countries model this feature in their DTM, or simply classify it as part of the river, removing it from the DTM entirely.



Satellite imagery. From Airbus, Maxar Technologies. (2026).

Non-continuous river mask.

Figure 74: River mask failure where the mask is not continuous due to a hydroelectric power plant.

For this region, the percentage of points above the INSPIRE threshold, is around 1%. While above average, this is caused by the fact that it is a small region, where the spatial dimensions of the differences remain consistent with differences found in other regions. Nevertheless, the difference DTM falls well within the INSPIRE target threshold, and contains no significant errors such as border extrapolation, misaligned tiles or inexplicable differences.

## G.3 Germany / France

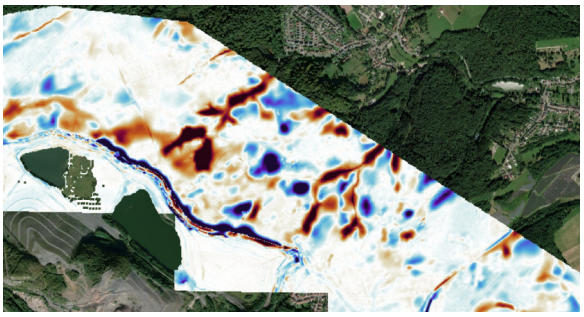
### G.3.1 Saarland / France

The approximately 157 km border between DE-SL and France crosses rolling hills, forested slopes and river valleys, with sections following the Saar and Blies rivers. Across the compared overlap, the two DTMs generally agree closely: the mean elevation difference is +0.035 m and the median is −0.020 m, where positive differences indicate higher elevations in DE-SL. Half of the absolute differences are below 0.163 m, and 90% are below 0.981 m. Nevertheless, the RMSE reaches 1.213 m, reflecting substantial discrepancies at particular locations.

These discrepancies are concentrated rather than evenly distributed. Only 5.81% of the compared area exceeds the 1.667 m comparison threshold, yet this portion contributes approximately 90% of the total squared elevation difference. Differences greater than 5 m occupy just 1.27% of the area but contribute about 63% of that total. The distribution therefore has a narrow centre and long tails, with extremes from approximately −22 m to +33 m. Neighbouring positive and negative differences partly cancel, so the small mean understates the local disagreement.

The Google Earth inspections and corresponding difference maps place these results in their terrain context. In Figure 75, the long dark blue difference pattern on the right corresponds to a cliff visible in the Google Earth imagery on the left. The pronounced discrepancies follow this abrupt terrain edge, with additional patches within the surrounding woodland. In Figure 76, the largest differences on the right coincide with an apparent valley-like terrain feature visible on the left. These selected examples illustrate recurring settings of disagreement rather than representing every location along the border.

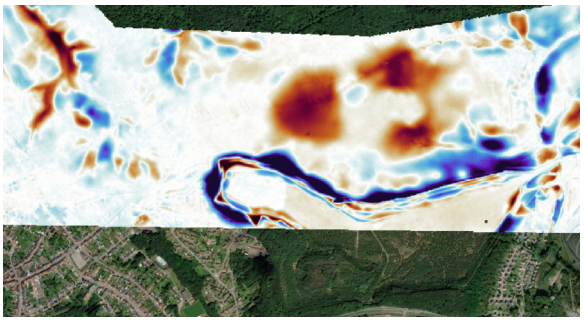
Google Earth Imagery. From Airbus, Landsat/Copernicus.



Elevation difference map.

Figure 75: Elevation differences in forested area with small cliff faces.

Google Earth Imagery. From Airbus, Landsat/Copernicus.



Elevation difference map.

Figure 76: Elevation differences on terrain with abrupt elevation changes.

Terrain steepness is the strongest quantified association. RMSE rises from approximately 0.8 m on slopes of 0–5° to 2.7 m at 20–25° and 5.6 m at 45–50°. This supports the visual observation that steep slopes and rapidly changing relief are particularly problematic. Forest cover is a recurring characteristic of the inspected hotspots, although its contribution has not been quantified separately from slope. Differences in ground filtering beneath canopy and restricted ground observations in narrow valleys are possible contributors. In Figure 77, the largest differences, shown as dark blue patches, coincide with forested areas identified during the visual inspection. Nearby patches of contrasting differences further illustrate how substantial local mismatches can cancel in the regional mean.

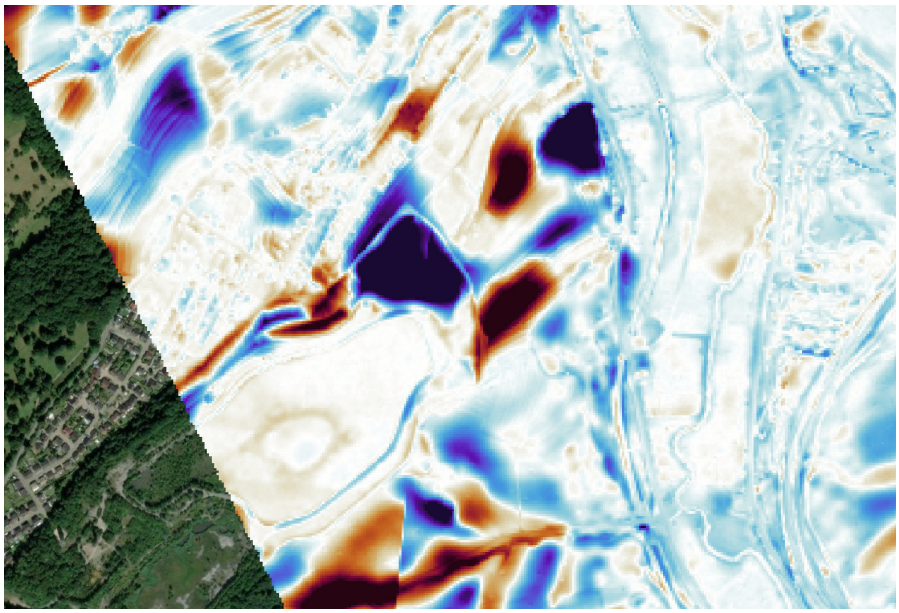
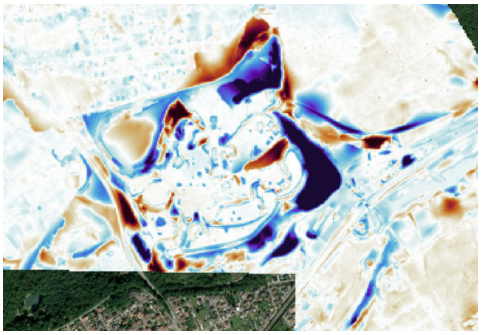


Figure 77: Large patches of high elevation differences.

Differences in DTM acquisition dates may also introduce temporal artefacts. The datasets may capture different stages of excavation, earthworks or vegetation clearance, while seasonal differences may affect the vegetation conditions under which the ground surface was estimated. Recently cleared areas identified during the Google Earth inspection suggest a possible influence of acquisition timing, particularly if an older surface retains residual vegetation effects. Some discrepancies may therefore reflect genuine terrain changes between surveys or changes in residual vegetation influence. The individual contributions of these processes have not been established from the selected examples.

The quarry in Figure 78 provides a clear setting in which acquisition timing may matter. Strong differences trace excavation faces and terraced edges. If the DTMs were acquired at different times, excavation or material deposition between surveys could have changed the height and position of these features. The discrepancies may therefore partly reflect different stages of quarry development, alongside differences in how the DTMs represent steep faces and sharp terrain edges.

Satellite imagery. From Airbus, CNES. (2026)



Elevation difference map.

Figure 78: Elevation differences caused by temporal changes in a quarry.

The regional RMSE is driven primarily by a limited set of pronounced local mismatches. Steep and complex terrain provides the clearest quantitative association, while the inspected images highlight forest cover, exposed escarpments and excavation features as recurring settings. Differences in

acquisition timing are an additional possible contributor where the landscape has changed between surveys. Most of the overlap agrees much more closely than the overall RMSE suggests.

### G.3.2 Rhineland-Palatinate / France

The border between DE-RP and France is approximately 112 km long and crosses predominantly hilly, forested terrain, particularly through the Pfälzerwald, which continues into France as the Northern Vosges. The Rhine does not form the main boundary, only the easternmost section approaches the Upper Rhine, while much of the border runs over land and partly follows smaller watercourses such as the Lauter. Castle ruins and distinctive rock formations are also characteristic features of this region.

Elevation differences are defined as DE-RP minus France, so positive values indicate a higher DE-RP surface. The mean difference is −0.033 m and the median is −0.032 m. Half of the absolute differences are below 0.079 m and 90% are below 0.225 m. The RMSE of 0.260 m, compared with an NMAD of 0.113 m, indicates that a relatively small number of large discrepancies strongly influence the overall result.

Figure 79 shows the narrow central distribution and the longer tails. Only 0.237% of the compared area, equivalent to 0.670 km², exceeds the 1.667 m comparison threshold, yet these cells contribute 60.4% of the total squared elevation difference. Differences greater than 5 m occupy just 0.044% of the area but contribute 36.3% of that total. Most of the overlap therefore agrees closely, while a small number of locations require separate investigation.

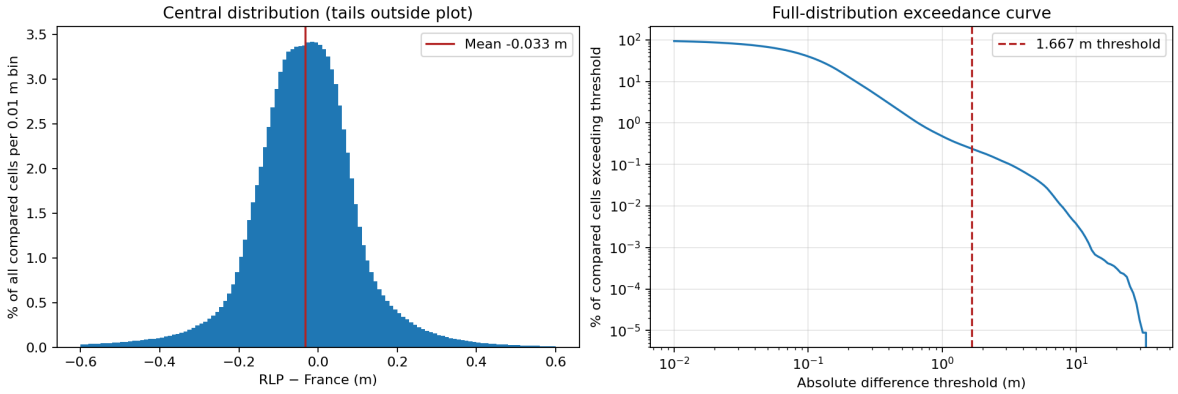



Figure 79: Central distribution and full-distribution exceedance curve. The left figure shows the distribution of the regional differences around the mean. Values outside ±0.6 m are not shown for visual simplicity. The right figure shows the percentage of all compared cells exceeding the absolute-difference threshold.

The spatial overview in Figure 80 distinguishes broad patterns of signed difference from concentrated large discrepancies. The ten largest problem regions lie at the western end of the overlap and contain 73.7% of all area above the threshold. The first five of those regions are shown in Figure 80 and contain 60.0%. However, the strongest individual differences are not in the regions with the biggest concentration of high difference. The locations of the largest individual differences are shown in Figure 81.

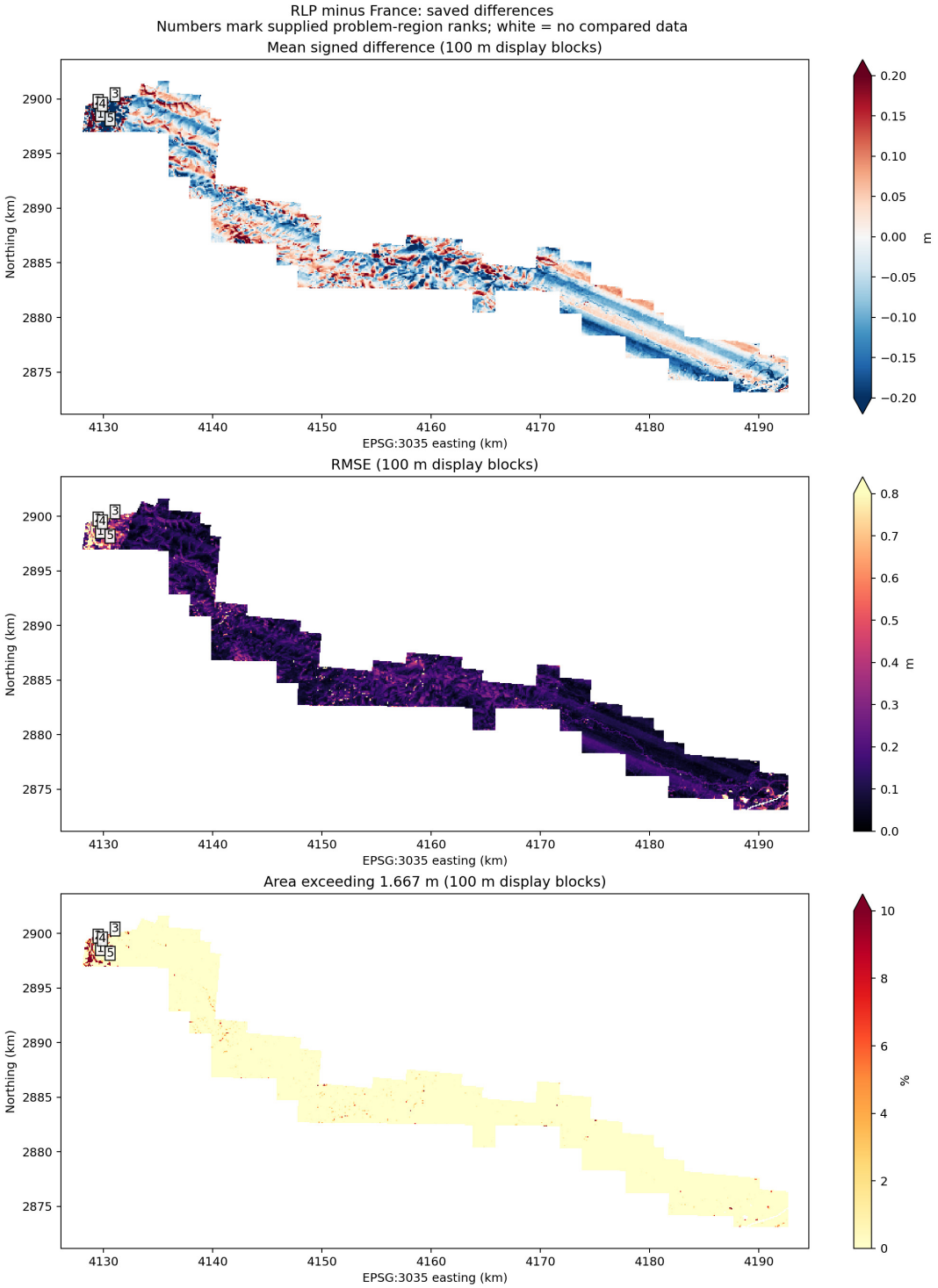
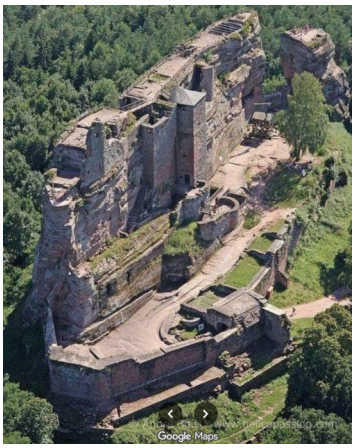


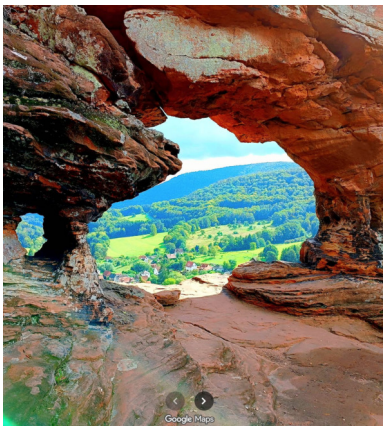

Figure 80: Elevation differences over the entire border between DE-RP and FR. From top to bottom: mean signed difference, RMSE, and percentage of compared area exceeding 1.667 m. Numbers identify the five largest problem regions. Extreme values go over the displayed limits by the colour scheme.

The inspected locations in Figure 81 place the largest differences in their physical setting. The minimum of −30.37 m occurs at Fleckenstein Castle (49.0449° N, 7.7723° E), where the French DTM is higher. The maximum of +34.54 m occurs at the Klein Arsberg ruins (49.0386° N, 7.6904° E), where the DE-RP DTM is higher. Another pronounced positive difference was identified at the nearby rock opening (49.0392° N, 7.6931° E). These photographs were taken from public contributions found in Google Maps, and they provide terrain context rather than a direct measurement of DTM error.




Fleckenstein Castle.

Klein Arsberg ruins.




Nearby rock opening.

Figure 81: Terrain settings of selected large differences. The left and centre photographs show the locations largest negative and positive elevation differences respectively. The photograph on the right shows another location with large positive elevation difference. Photos from Google Maps.

Different treatment of ruins and exposed rock during ground classification, followed by interpolation across removed or sparsely sampled features, could produce these local discrepancies. Forest canopy may further limit ground observations around the sites. A raster DTM stores one elevation per horizontal cell and cannot explicitly represent both the top and underside of an overhang. The 5 m grid also smooths narrow rock edges. Consequently, disagreement at these locations may reflect different representations of complex terrain as well as processing errors. Verification requires the original classified point clouds and source-resolution DTMs, cross-sections through the exact cells, and an independent survey or reliable terrain reference. Acquisition dates and processing specifications are needed to distinguish temporal change from classification and interpolation effects.

Figure 82 shows a different type of discrepancy: broad, approximately parallel bands extending across multiple fields and woodland areas. This pattern suggests an acquisition or processing component rather than terrain features. They resemble artefacts originating from flight line patterns. Survey trajectories and strip identifiers should be compared with the bands, and overlapping point-cloud strips examined separately.

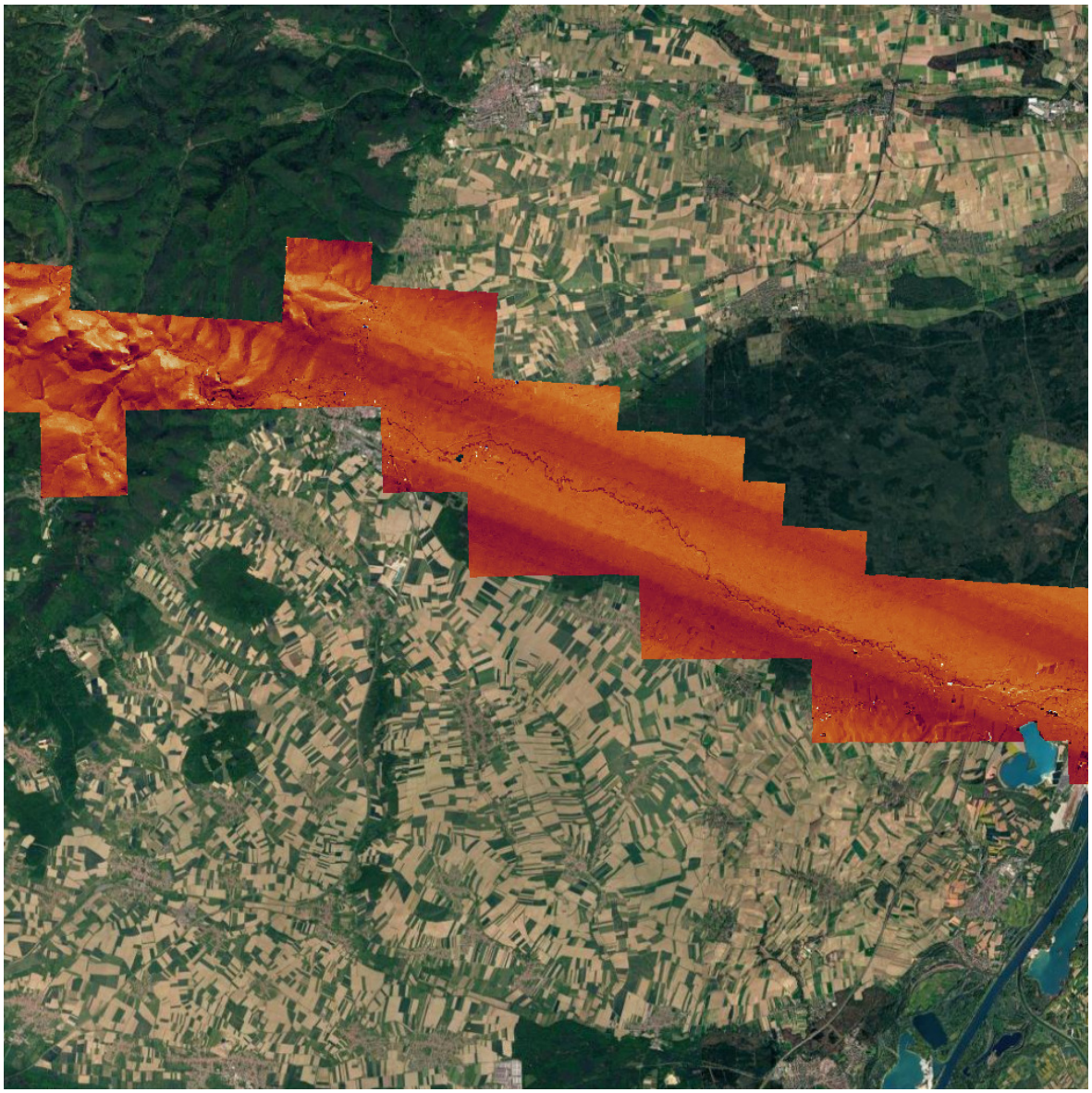


Figure 82: Flight-path artefacts over the border region. The color scale was set to −1 to 1 m to highlight the differences.

### G.3.3 Baden-Württemberg / France

The DE-BW (DE-BW)–France comparison shows close agreement across much of the Rhine corridor, interrupted by pronounced local discrepancies and coherent spatial patterns. Across 166.52 km² of valid overlap at 5 m resolution, the mean elevation difference is +0.034 m and the median is +0.058 m, where positive values mean that DE-BW is higher than France. The RMSE of 0.730 m is considerably larger than the NMAD of 0.153 m, indicating that large local differences have a strong influence on the overall result.

The difference distribution has a narrow centre and long tails. 95% of the absolute differences are below 1.059 m. Only 2.744% of compared cells exceed the configured 1.667 m threshold, but these cells occupy 4.57 km² and contribute 82.73% of the total squared difference. Differences larger than 5 m occupy just 0.397%. The extremes range from −25.96 m to +46.72 m of elevation difference.

Figure 83 shows the strongest regional contrast, between the southern overlap and the remainder of border. The southern segment of the border region contains 89.82% of all threshold-exceeding cells and 85.53% of total squared difference. Its RMSE is 1.680 m, compared with 0.303 m in the northern

segment. Exceedance rates are 15.26% and 0.333% for the two segments, respectively. NMAD also rises from 0.132 m in the north to 0.661 m in the south, showing that the southern disagreement affects the broader distribution as well as its most extreme values.

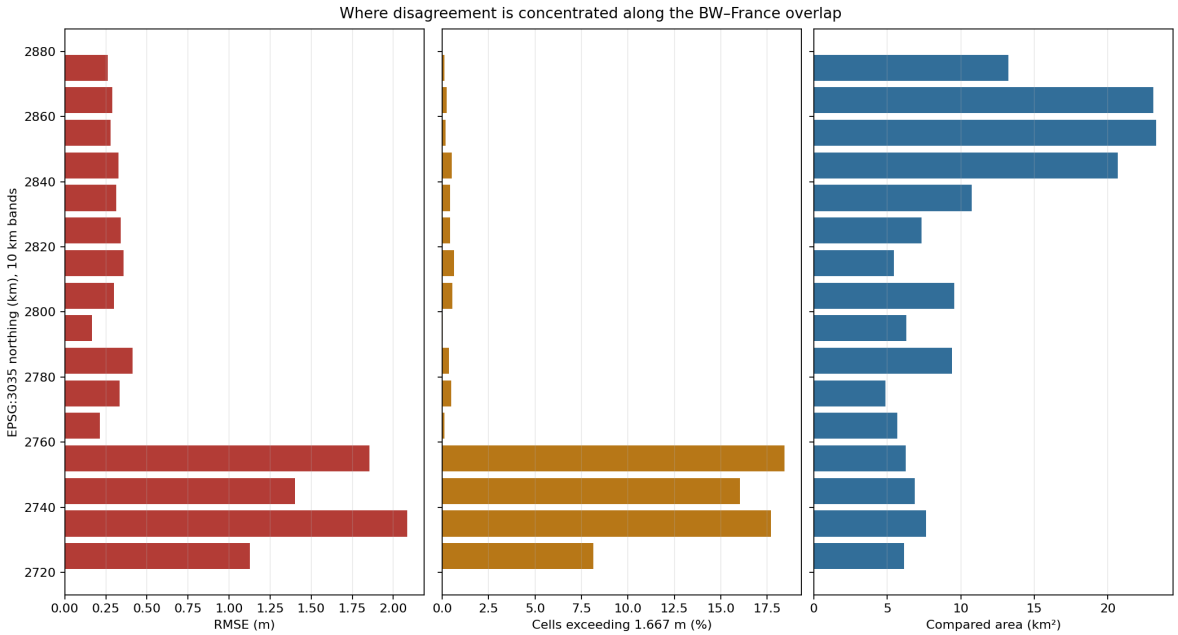



Figure 83: Regional variation of elevation differences in the DE-BW/FR border. RMSE, threshold exceedances and compared area in 10 km northing bands. The southern bands have substantially larger differences despite containing less compared area than several northern bands. Coverage refers to the available overlap.

Alongside these large local differences, Figure 84 shows flight-line-like bands similar to those in the DE-RP/FR comparison (Figure 82). The comparative visual inspection identifies bands running in the same direction and matching exactly with the DE-RP/FR patterns. Because the French DTM is common to both comparisons, this correspondence strongly suggests an origin in the French acquisition and the relative alignment or adjustment of its flight paths. The spatially varying bands are consistent with residual differences between flight strips that persist across the two regional comparisons. Flight-line footprints and adjustment records would allow this interpretation to be tested directly and help distinguish acquisition effects from later processing.

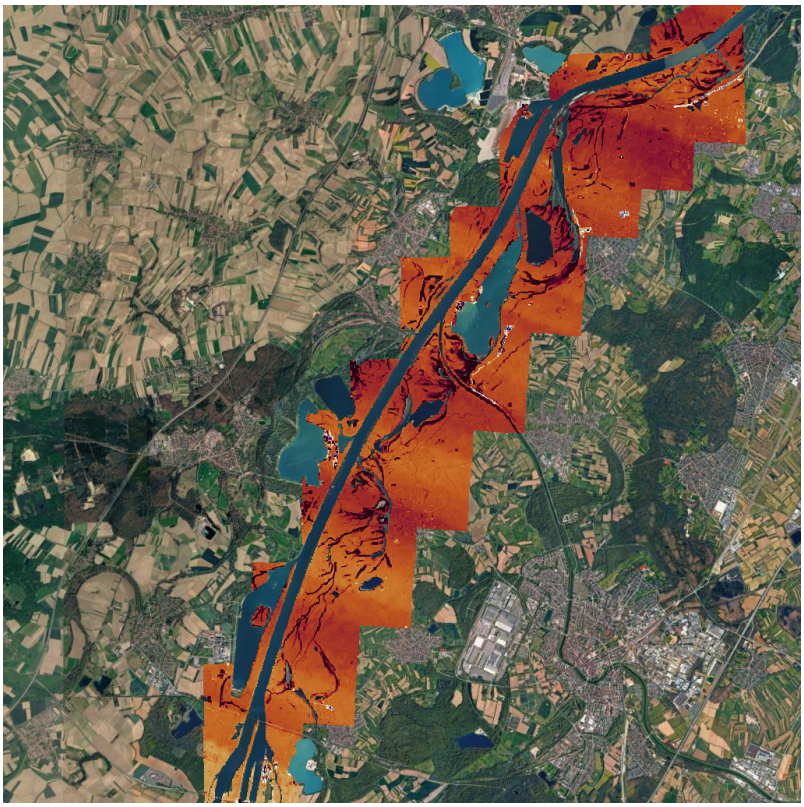


Figure 84: Flight-path artefacts over the border region. The color scale was set to −1 to 1 m to highlight the differences.

## G.4 Switzerland

### G.4.1 Switzerland / France

Overall, the border between France and Switzerland contains errors similar to those found on the French–German border. These include one instance of striping, most visible in the suburbs of Basel. The stripe appears to follow the edge of the French DTM, which further points towards it being caused by poorly aligned flight-paths.

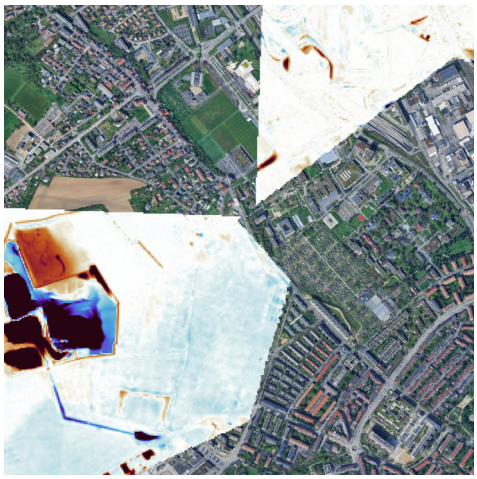
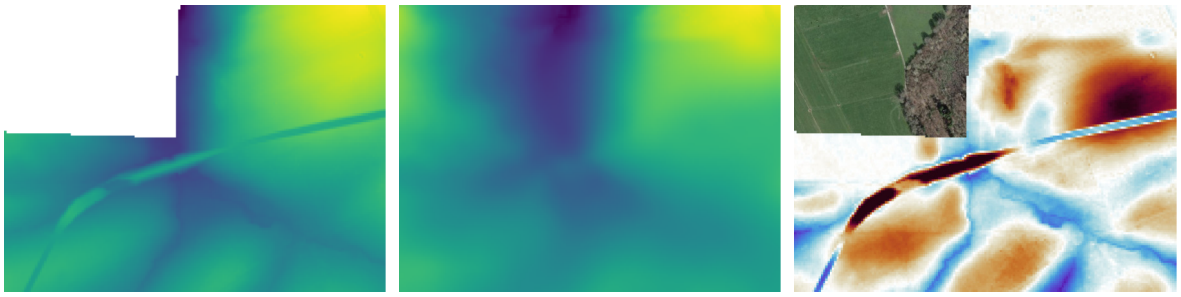


Figure 85: Flight-path artefacts between FR and CH.

There is also a large artefact in the forested region near Pfetterhouse. In the French DTM, the elevation of peaks appears to be underestimated while the elevation of valleys is overestimated — essentially, the DTM is over-smoothed. We speculate that regions like this come from a different

survey with different approach to ground filtering (perhaps ground is estimated with a global interpolation function) but we have not been able to verify or falsify this hypothesis.



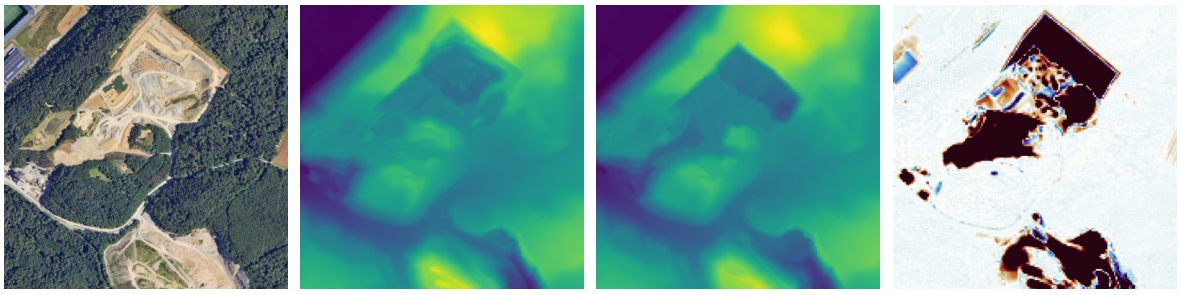
DTM of CH. DTM of FR. Elevation difference map.

Figure 86: Ground filtering differences between CH and FR.

### G.4.2 Switzerland / Baden-Württemberg

The border between Switzerland and DE-BW largely follows the Rhine between Konstanz and Basel and notably includes Büsingen, an exclave of DE-BW within Switzerland. The regions are overall well-aligned with LE90 of only 0.12 m but a high RMSE of 0.4 m.

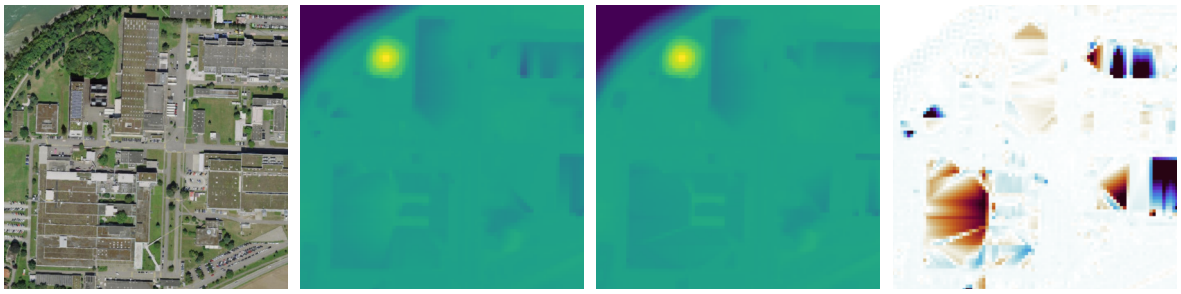
The largest errors are best explained by temporal differences — like in other areas examined above, there are several quarries present near the border. The largest of them is located northeast of Schaffhausen and is responsible for the largest differences in elevation, up to 30 m.



Satellite imagery. From Airbus, CNES 2026. DTM of CH. DTM of DE-BW. Elevation difference map.

Figure 87: Temporal differences between CH and DE-BW.

A more unique error is due to ground filtering, as can be observed with the industrial complex north of Stein, where differences between the different ground filtering/interpolation methods can exceed 3 m. As is evident from the difference mask, both data providers use TIN to interpolate the terrain but a different set of points for constructing said TIN. A likely misclassification of the ground, perhaps due to shadowing or multipath errors, causes the error in the elevation of ground points and therefore the error in the TIN.

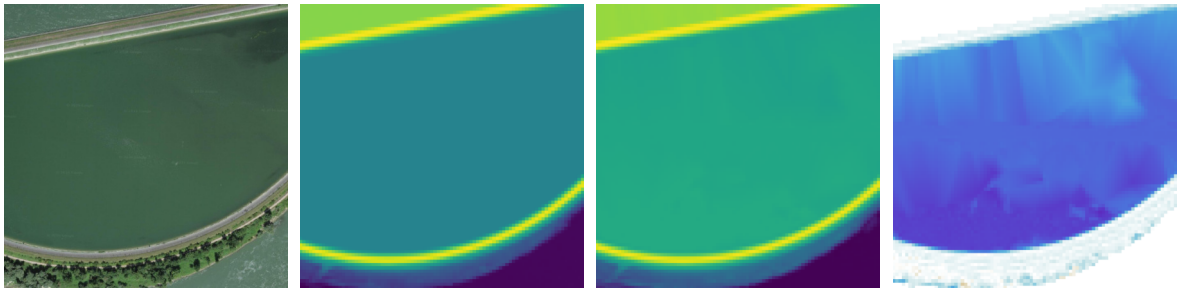


Satellite imagery. From Airbus, CNES 2026. | DTM of CH. | DTM of DE-BW. | Elevation difference map.

Figure 88: Differences in building processing between CH and DE-BW.

Ground and vegetation filtering also have an effect on forested slopes between Hallau and Eggingen where minor striping can be observed. The stripes appear to be running perpendicular to the slope and, we speculate, can be symptomatic of either differences in ground processing by data providers; or they could be a resampling artefact. We recommend performing point cloud analysis to validate either of these hypotheses, which was skipped due to time constraints of the project.

Similarly to the building interpolation, there is also an observable difference in the water surface. Switzerland appears to be using a hydro-flattening approach whereas DE-BW is instead using linear interpolation on a TIN.



Satellite imagery. From Airbus, CNES 2026. | DTM of CH. | DTM of DE-BW. | Elevation difference map.

Figure 89: Differences in water processing between CH and DE-BW.

## G.5 Germany / Netherlands

### G.5.1 Netherlands / North Rhine-Westphalia

The border between the Netherlands and DE-NW is approximately 50 km long, is mainly flat, but also contains large hills. The main error found within this region is unfortunately a byproduct of our attempt to interpolate the non-interpolated regions of the Netherlands (Figure 90). These errors are found primarily in the form of extrapolation errors found at various positions along the border. This is a discrepancy caused by differences in the specified borders for the Netherlands within the interpolation algorithm, and has been later resolved by changing the border dataset

DTM of NL. Elevation difference map.

Figure 90: Border artefact caused by interpolation at the border.

Other errors within this region include both small and large scale earthworks. Examples include the “Steenfabriek de Bijland”, which also includes swampy areas, which are not caught by the river mask (Figure 91). Another example includes storage of manure and dirt at farms scattered throughout the border area (Figure 92).

For this border area in particular, there is a strong effect of water bodies on the difference. The RMSE increases by 10% if the water bodies are included, compared to the case without. This is likely due to the attempt at interpolating the river areas, which will differ from the approach taken in the German region, which most often involves triangulation of the water surface.

Satellite imagery. From Airbus, Landsat / Copernicus, Maxar Technologies (2026). Incomplete water mask.

Figure 91: Example of a region whre the river mask is incomplete/non-continuous.

Satellite imagery. From Airbus, Maxar Technologies. (2026).

Elevation difference map.

Figure 92: Elevation differences caused by small earthworks.